

SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

JINEON BAEK, JEEWON KIM

ABSTRACT. We prove that the regular hexagon of side length one cannot be covered by seven open equilateral triangles of side length one. The center and the six vertices canonically label the seven covering triangles. A midpoint-supplier reduction routes every hypothetical cover to one of three strategies: trace-length deficit, area loss, or a finite equilateral-enclosure obstruction. The finite-enclosure strategy uses fixed four-point supported-rescuer and two-gap witnesses, a type-independent six-point transverse witness, and the explicit zero-gap nine-point theorem. A direct neighboring-ray bound removes V-type splits along nonsupercritical paths. Appendices collect the exact local reach calculations certifying that the relevant finite sets do not fit in a unit C triangle. The paper is organized as one proof, and the two disk-tangent contact bounds are verified in a self-contained exact certificate appendix over $\mathbb{Q}(\sqrt{3})$.

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1. INTRODUCTION AND PROOF OUTLINE

1.1. The theorem and the canonical labeling. Let

$$O = (0, 0), \quad V_i = \left(\cos \frac{i\pi}{3}, \sin \frac{i\pi}{3} \right), \quad H = \text{conv}\{V_0, \dots, V_5\},$$

2020 *Mathematics Subject Classification.* 52C15, 52A40, 51M04.

Key words and phrases. equilateral triangles, geometric covering, regular hexagon, area loss, finite enclosure, exact computer-assisted certificate.

where indices lie in $\mathbb{Z}/6\mathbb{Z}$. Write

$$e_{i,i+1} = [V_i, V_{i+1}], \quad r_i = [O, V_i], \quad M_i = \frac{1}{2}V_i,$$

and put

$$S = \partial H \cup \bigcup_{i=0}^5 r_i.$$

Then

$$(1) \quad \begin{aligned} \mathcal{H}^1(\partial H) &= 6, \\ \mathcal{H}^1\left(\bigcup_{i=0}^5 r_i\right) &= 6, \\ \mathcal{H}^1(S) &= 12. \end{aligned}$$

Theorem 1.1 (Main theorem). *The hexagon H cannot be covered by seven open equilateral triangles of side length one.*

Corollary 1.2 (Soifer's Hexagon Problem). *For every $L > 1$, the regular hexagon $H_L = LH$ cannot be covered by seven closed unit equilateral triangles.*

The seven points $O, V_0, \dots, V_5$ are pairwise at distance at least one, whereas any two points in an open unit equilateral triangle are at distance strictly less than one. Hence the seven covering triangles admit a unique labeling

$$U_C, U_0, \dots, U_5$$

such that

$$O \in U_C, \quad V_i \in U_i.$$

We call U_C the *C triangle* and U_i the *V triangle* at V_i . Set

$$T_C = \overline{U_C}, \quad T_i = \overline{U_i}.$$

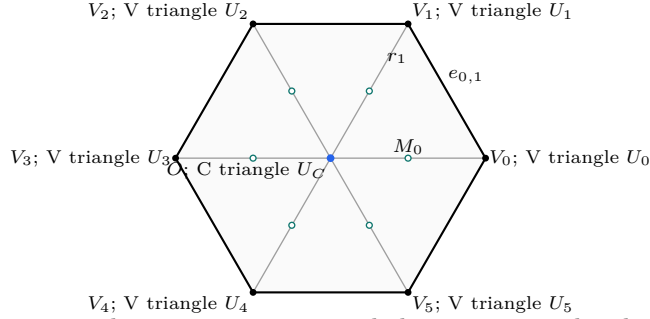


FIGURE 1. The center, vertices, radial segments, and radial mid-points of H .

1.2. Finite classifications. Set

$$N_E(T_C) := |\{i \in \mathbb{Z}/6\mathbb{Z} : \mathcal{H}^1(T_C \cap e_{i,i+1}) > 0\}|.$$

The only possible values are 0, 1, 2. We call the corresponding C triangle types CE0, CE1, CE2; thus

$$T_C \text{ has type CE0} \iff N_E(T_C) = 0,$$

$$T_C \text{ has type CE1} \iff N_E(T_C) = 1,$$

$$T_C \text{ has type CE2} \iff N_E(T_C) = 2.$$

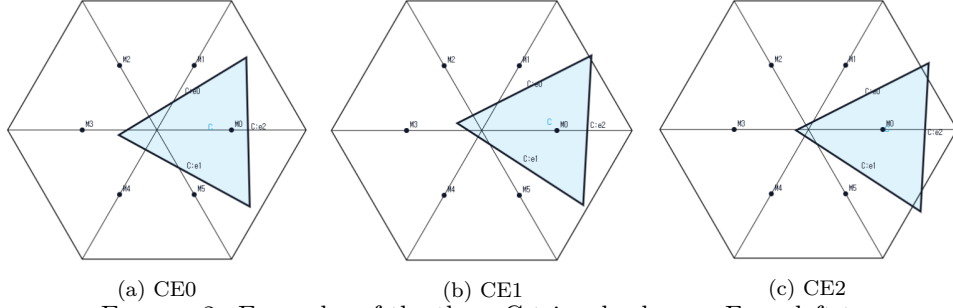


FIGURE 2. Examples of the three C triangle classes. From left to right, the C triangle has positive-length trace on zero, one, and two edges of ∂H ; point contacts do not count. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

For the V triangle T_i , let $o(T_i)$ be the number of its triangle vertices outside H and put

$$n(T_i) = |\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\}|,$$

with indices modulo 6. The raw pair (3,0) says that all three vertices of T_i lie outside H ; it does not say that T_i is disjoint from H , since $V_i \in \text{int}(T_i) \cap H$. Proposition 2.5 replaces every such triangle by a same-orientation translate with exactly the same trace in H , still containing V_i in its interior, and with at most two vertices outside H . We make this exact-trace normalization before defining any reach. The normalized exhaustive types used below are

$(o(T_i), n(T_i))$	established type
(1, 0)	Vd0
(2, 0)	Vd0
(1, 1)	Vd1
(1, 2)	Vd2
(2, 1)	T3-like.

The actual maximal reaches are

$$A_i := \max\{x \in [0, 1] : V_i + x(V_{i-1} - V_i) \in T_i\},$$

$$B_i := \max\{x \in [0, 1] : V_i + x(V_{i+1} - V_i) \in T_i\},$$

$$C_i := \max\{x \in [0, 1] : V_i + x(O - V_i) \in T_i\}.$$

The same numbers are the suprema of the corresponding reaches in the open triangle U_i . The V triangle T_i is *supercritical* when $A_i + B_i > 1$, and

$$(2) \quad N_+ := |\{i \in \mathbb{Z}/6\mathbb{Z} : A_i + B_i > 1\}|.$$

Lowercase reaches are used only after a selection satisfying

$$0 \leq a_i \leq A_i,$$

$$0 \leq b_i \leq B_i,$$

$$0 \leq c_i \leq C_i.$$

They never define N_+ .

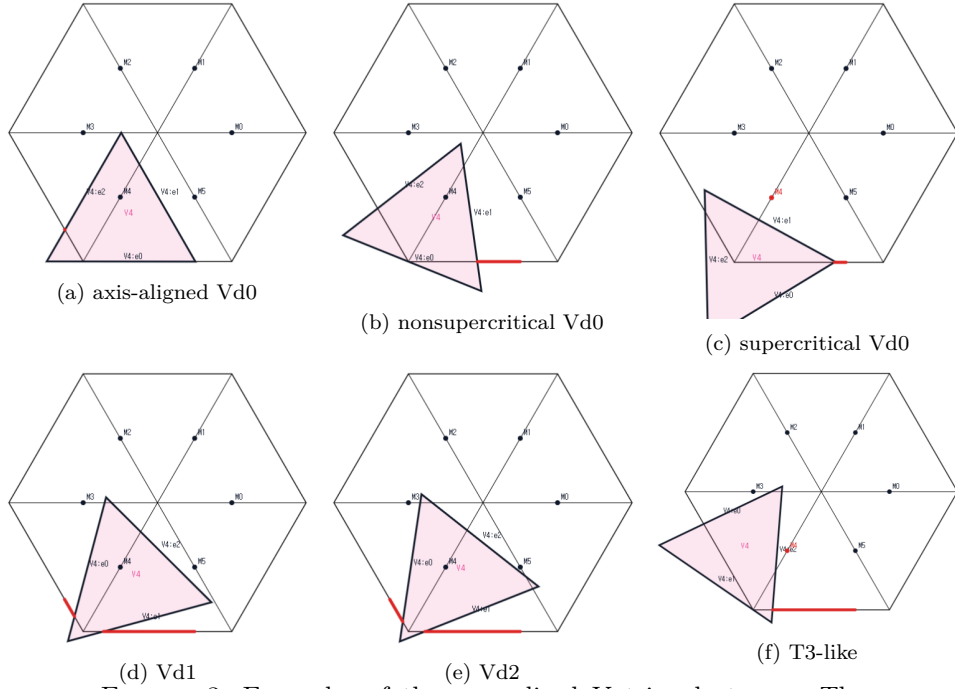


FIGURE 3. Examples of the normalized V triangle types. The top row shows an axis-aligned Vd0, a nonsupercritical Vd0 with $A_i + B_i \leq 1$, and a supercritical Vd0 with $A_i + B_i > 1$. The bottom row shows Vd1, Vd2, and T3-like. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

Parametrize $e_{i,i+1}$ from V_i to V_{i+1} by

$$X_i(x) := V_i + x(V_{i+1} - V_i), \quad 0 \leq x \leq 1.$$

The incident open traces are

$$U_i \cap e_{i,i+1} = X_i([0, B_i)),$$

$$U_{i+1} \cap e_{i,i+1} = X_i((1 - A_{i+1}, 1]).$$

Define the *boundary gap*

$$J_i := e_{i,i+1} \setminus ((U_i \cap e_{i,i+1}) \cup (U_{i+1} \cap e_{i,i+1})).$$

The displayed trace identities give

$$J_i \neq \emptyset \iff B_i + A_{i+1} \leq 1.$$

Equality gives a singleton gap, which remains uncovered. Put

$$N_{\text{gap}} := |\{i \in \mathbb{Z}/6\mathbb{Z} : J_i \neq \emptyset\}|.$$

For the case tables, put

$$d := |\{i \in \mathbb{Z}/6\mathbb{Z} : (o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}\}|,$$

$$t := |\{i \in \mathbb{Z}/6\mathbb{Z} : (o(T_i), n(T_i)) = (2, 1)\}|,$$

and

$$N_{\text{sp}} := |\{i \in \mathbb{Z}/6\mathbb{Z} : n(T_i) > 0\}| = d + t.$$

The common trace calculation proves that the number of V triangles that are either supercritical or meet an adjacent radial segment in positive length is

$$N_+ + N_{\text{sp}}.$$

In particular, every CE1/CE2 configuration with $N_+ + N_{\text{sp}} \geq 3$ is impossible by a trace-length bound on S .

1.3. Exhaustive routing. Zero actual gaps mean that the six open V triangles cover ∂H . The zero-gap route depends only on N_+ , not the C or V types. For nonzero gaps, N0 and the skeleton budget first reduce to one supercritical role and at most one positive-support role. The remaining distinction is whether the supercritical role or its midpoint supplier is based at the C midpoint. This replaces a separate case tree for each V-type count and each CE class.

N_{gap}	N_+	Additional condition	Closing mechanism
0	0	Arbitrary V types	Strict overlap
0	1	Arbitrary V types	Nine-point F
0	≥ 2	Arbitrary V types	Cyclic area loss
≥ 1	0	Arbitrary V types	N0, through B/C
≥ 1	≥ 1	$N_+ \geq 2$ or $N_+ + N_{\text{sp}} \geq 3$	Skeleton budget
≥ 1	1	$N_{\text{sp}} \leq 1$	Midpoint supplier: B/C, D, replacement, or Vd2 length

TABLE 1. Disjoint routes for the original open cover. Nonzero gaps force CE1 or CE2; both classes share the same placement reduction. Replacement preserves the skeleton and ends at N0, without preserving gap rank.

The exact structural normalization is theorem 2.15. The last row is resolved by theorem 5.17; the local CE1 and CE2 calculations remain separate inside the B/C path theorem. The independent complementary-gap construction A is retained in Appendix D, outside the minimum proof route.

1.4. Organization of the proof. Section 2 proves the structural classification, strict handoff selection, and midpoint facts. Appendix A proves the signed-center normal form used by the later arguments. Section 3 gives the complete trace-length argument. Section 4 proves the local and cyclic area-loss estimates. Section 5 gives the finite residual-hull principle and the nonzero-gap and zero-gap enclosure obstructions. Section 6 then applies the zero-gap alternatives before the nonzero-gap midpoint-supplier reduction of Table 1 and closes the proof. Appendices A–E supply the corresponding solved optimizations; Appendix F is the exact polynomial certificate for the two mixed zero-gap intersections.

2. COMMON GEOMETRY AND STRUCTURAL REDUCTIONS

The introduction fixes the exact C triangle and V triangle classifications, the actual maximal reaches A_i, B_i, C_i , the selected lower bounds a_i, b_i, c_i , and the boundary gaps J_i . This section proves that those definitions give the finite structural alternatives used by all three mechanisms. Coordinate evaluations needed in the proofs are collected in Appendix A.

2.1. Open and closed formulations.

Proposition 2.1 (Compactness and scaling equivalence). *The following assertions are equivalent.*

- (1) H is covered by seven open equilateral triangles of side 1.
- (2) For some $\varepsilon > 0$, H is covered by seven closed equilateral triangles of side $1 - \varepsilon$.
- (3) For some $L > 1$, $H_L := LH$ is covered by seven closed equilateral triangles of side 1.

Proof. The uniform inward displacement for (1) $\Rightarrow$ (2) is the compactness-margin calculation theorem A.1. Enlarging each closed triangle about its incenter gives (2) $\Rightarrow$ (1), and dilation by $(1 - \varepsilon)^{-1}$ identifies (2) with (3). $\square$

Lemma 2.2 (Distinct triangles). *The seven triangles can be labelled so that*

$$O \in U_C, \quad V_i \in U_i \quad (0 \leq i \leq 5),$$

and these seven triangles are distinct. Consequently

$$O \in \text{int}(T_C), \quad V_i \in \text{int}(T_i).$$

Proof. The seven points $O, V_0, \dots, V_5$ are pairwise at distance at least 1, whereas two points of an open unit equilateral triangle are at distance strictly less than its diameter 1. Selecting a covering triangle for each point therefore gives a bijection. $\square$

2.2. C triangle and V triangle classifications. Point contacts have zero trace length, so they do not create an additional positive boundary trace in the definition of $N_E(T_C)$.

Proposition 2.3 (Exhaustive center types). *The closed C triangle T_C has exactly one of the types CE0, CE1, CE2; equivalently,*

$$N_E(T_C) \leq 2.$$

Proof. If $N_E(T_C) \geq 3$, two of the traced edges are nonadjacent in the boundary six-cycle. Choose a relative-interior point from each positive-length trace. By theorem A.6, those points are more than unit distance apart, so a unit equilateral triangle cannot contain both. $\square$

Lemma 2.4 (Local wedge). *Every original V triangle has $1 \leq o(T_i) \leq 3$. If $\mathcal{H}^1(T_i \cap r_j) > 0$ for some $j \in \{i-1, i+1\}$, at least one vertex of T_i belongs to H ; if both adjacent radial traces have positive length, at least two vertices belong to H .*

Proof. This is the wedge-incidence calculation in theorem A.7. It is stated for the closed T_i because closure may add a tangency, although it adds no trace length. $\square$

Raw roles with $(o(T_i), n(T_i)) = (3, 0)$ admit an exact-trace normalization.

Proposition 2.5 (Exact-trace normalization of raw $(3, 0)$ roles). *Let U be an original open V triangle at V_i , put $T = \overline{U}$, and suppose $(o(T), n(T)) = (3, 0)$. There is a same-orientation translate $\hat{U}$, with closure $\hat{T}$, such that*

$$(3) \quad V_i \in \hat{U}, \quad (o(\hat{T}), n(\hat{T})) = (2, 0),$$

and

$$(4) \quad T \cap H = \hat{T} \cap H, \quad U \cap H = \hat{U} \cap H.$$

In particular, all three actual maximal reaches and the inequality $A_i + B_i > 1$ are unchanged.

Proof. The two-orientation affine calculation and the trace-preserving translation are given in theorem A.8. $\square$

We apply this translation simultaneously and reuse U_i, T_i for the normalized triangles.

Proposition 2.6 (Exhaustive normalized vertex types). *Every normalized V triangle has exactly one of the four types Vd0, Vd1, Vd2, T3-like in the exact dictionary of the introduction.*

Proof. Lemma 2.4 gives $o \leq 2$ when $n \geq 1$ and $o \leq 1$ when $n = 2$, while $o \geq 1$. Proposition 2.5 replaces the only additional raw pair, $(3, 0)$, by $(2, 0)$. $\square$

The next construction is a trace majorant, not a new open covering triangle.

Proposition 2.7 (T3-like closed-trace domination). *Let T be the closure of an original T3-like V triangle at V_i . There is a same-orientation translate $\hat{T}$ such that*

$$V_i \in \text{relint}(a \text{ side of } \hat{T}), \quad T \cap H \subsetneq \hat{T} \cap H,$$

and $(o(\hat{T}), n(\hat{T})) = (2, 1)$, while

$$\text{int}(T) \cap (H \setminus \{V_i\}) \subseteq \text{int}(\hat{T}).$$

Since V_i becomes a boundary point, $\hat{T}$ is only a closed-trace majorant.

Proof. See the two-orientation feasibility calculation in theorem A.2. $\square$

2.3. Initial reaches and boundary gaps. The exact reach definitions are those in the introduction. The following elementary fact records how their closed maxima and open suprema are used.

Lemma 2.8 (Initial reach segments). *On the three segments from V_i toward V_{i-1}, V_{i+1}, O , the closed parameter sets are respectively*

$$[0, A_i], \quad [0, B_i], \quad [0, C_i],$$

and the open parameter sets are

$$[0, A_i), \quad [0, B_i), \quad [0, C_i).$$

Thus the closed maxima equal the corresponding open-triangle suprema.

Proof. Each intersection with a segment issuing from the interior point V_i is convex. Closedness of T_i , openness of U_i , and $T_i = \overline{U_i}$ give the stated endpoints. $\square$

The gap J_i is the exact set difference defined in the introduction. The name includes the equality case $B_i + A_{i+1} = 1$: then J_i is a singleton missed by both incident open V triangles.

Lemma 2.9 (Exhaustiveness of the gap split). *Every J_i is empty or one closed interval, with singleton intervals retained. Under a hypothetical cover,*

$$N_{\text{gap}} = |\{i \in \mathbb{Z}/6\mathbb{Z} : J_i \neq \emptyset\}|,$$

and every nonempty J_i lies in a positive-length C triangle trace. Consequently a closed C triangle T_C of type CE0, CE1, CE2 permits respectively at most 0, 1, 2 gaps, and

$$N_{\text{gap}} = 0 \iff U_0, \dots, U_5 \text{ cover } \partial H.$$

Proof. The two incident open traces are initial intervals from opposite endpoints, so their complement is exactly the interval displayed in the introduction. A nonincident V triangle cannot contain a relative-interior point of the edge: that point is more than unit distance from its distinguished vertex. Hence a gap lies in U_C ; openness turns even a singleton V-gap into a positive-length C trace. Proposition 2.3 gives the count and the final equivalence. $\square$

Lemma 2.10 (T3-like V triangles are nonsupercritical). *Every original T3-like V triangle satisfies*

$$(5) \quad A_i + B_i \leq 1.$$

It is therefore not counted by N_+ .

Proof. This is the second assertion of theorem A.2. $\square$

2.4. Strict handoffs and midpoint forcing.

Proposition 2.11 (Strict boundary handoffs). *Assume that $U_0, \dots, U_5$ cover ∂H . There are choices*

$$\xi_i \in (1 - A_{i+1}, B_i), \quad a_i := 1 - \xi_{i-1}, \quad b_i := \xi_i,$$

such that

$$0 < a_i < A_i, \quad 0 < b_i < B_i, \quad a_i^2 + a_i b_i + b_i^2 < 1.$$

The selected anchors lie in U_i . If exactly one actual V triangle is supercritical, every strict selection has that same unique selected supercritical index. If at least

two actual V triangles are supercritical, some simultaneous strict selection retains at least two.

Proof. The overlap intervals and the cyclic selector are proved in theorem A.3. Geometrically, each ξ_i lies in the overlap of two adjacent open edge traces, which makes both associated reach demands strict. $\square$

Lemma 2.12 (Boundary deficit and overlap identities). *For the actual reaches put $s_i = 1 - A_i - B_i$ and $\omega_i = B_i + A_{i+1} - 1$. Then*

$$A_{i+1} - A_i = s_i + \omega_i, \quad B_i - B_{i+1} = s_{i+1} + \omega_i.$$

On a nonsupercritical gap-free path the A_i increase and the B_i decrease strictly. Weak handoffs give the weak version. Around the full cycle, $\sum_i \omega_i = -\sum_i s_i$.

Proof. Expand the definitions and telescope. Nonsupercriticality gives $s_i \geq 0$, and a gap-free edge has $\omega_i > 0$ because its incident traces are open. A supercritical role has $s_i < 0$; propagation through it is not licensed. $\square$

Lemma 2.13 (Self-midpoint obstruction). *Let u, v be unit vectors meeting at 120° . If a closed unit equilateral triangle contains*

$$0, \quad au, \quad bv, \quad \frac{u+v}{2},$$

then $a + b \leq 1$. Consequently

$$M_i \in T_i \implies A_i + B_i \leq 1.$$

If the boundary points and midpoint are contained with positive margin, the conclusion is strict.

Proof. This is the four-point caliper optimization theorem A.10. $\square$

Proposition 2.14 (CE1/CE2 exactly-one-midpoint property). *If $O \in \text{int}(T_C)$ and T_C is CE1 or CE2, then*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1.$$

Proof. Normalize one positive boundary trace. The signed-center optimization theorem A.11 gives the exact midpoint set $\{M_0\}$; undoing the symmetry proves the assertion. $\square$

The interior hypothesis is essential: $\text{conv}\{O, V_0, V_1\}$ has a boundary edge overlap and contains both M_0 and M_1 , but O is not interior to it.

2.5. Exhaustive structural routing.

Proposition 2.15 (Exhaustive structural reduction). *Every hypothetical cover admits the distinct C triangle and V triangles used in Table 1, and its actual maximal reaches determine exactly one row of that table.*

Proof. Lemma 2.2 and Propositions 2.3 and 2.6 give the exhaustive, mutually exclusive roles. The actual reaches determine N_+ , while the open traces determine N_{gap} .

For zero gaps, boundary completeness holds independently of center type; the three possibilities $N_+ = 0, 1, \geq 2$ are disjoint. For a nonzero gap, the C triangle is CE1 or CE2. First separate $N_+ = 0$, then the high-count states $N_+ \geq 2$ or $N_+ + N_{\text{sp}} \geq 3$. Every survivor has $N_+ = 1, N_{\text{sp}} \leq 1$. Thus the rows of table 1 are

disjoint and exhaustive. The later midpoint-supplier lemma resolves the last row by relative placement, rather than by a list of V-type patterns. $\square$

3. TRACE-LENGTH MECHANISM

For a closed triangle T and a rectifiable target E , define

$$L_E(T) := \mathcal{H}^1(T \cap E).$$

Lemma 3.1 (Connected open-cover budget). *Let X be compact and connected, with a finite Borel measure μ positive on every nonempty relatively open subset. For a finite open cover with at least two nonempty relative pieces,*

$$\mu(X) < \sum_j \mu(U_j \cap X) \leq \sum_j \mu(T_j \cap X).$$

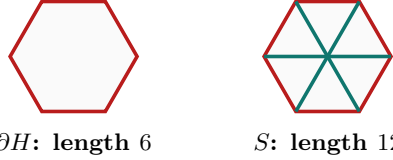
Consequently closed-piece upper bounds whose sum is at most $\mu(X)$ already contradict the open cover.

Proof. If the nonempty relative pieces were pairwise disjoint, one and the union of the others would disconnect X . Some pair therefore overlaps in a nonempty relatively open set, of positive measure. Integrating the covering multiplicity gives strict inequality. Closure only enlarges each piece. $\square$

For $X = \partial H$ or S , use length; for $X = H$, use area. Each measure has the stated full-support property, and the seven distinguished roles meet the corresponding targets in positive measure. In particular,

$$(6) \quad \mathcal{H}^1(E) < L_E(T_C) + \sum_{i=0}^5 L_E(T_i), \quad E \in \{\partial H, S\},$$

whenever these targets are covered. Empty center boundary traces are simply omitted from the collection. The local geometric caps remain separate from this common final budget argument.



the full-skeleton estimate is used only under its stated branch hypotheses

FIGURE 4. The perimeter and full-skeleton targets. The skeleton is used only under the hypotheses of the corresponding trace budget.

3.1. Skeleton interfaces.

Lemma 3.2 (Center skeleton cap). *If $N_E(T_C) \in \{1, 2\}$ and*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1,$$

then

$$L_S(T_C) < \frac{3}{2}.$$

Proof. This is the signed-center trace optimization theorem B.1. $\square$

Lemma 3.3 (Positive-support skeleton cap). *Let T_i be the closure of an original open V triangle. If*

$$\exists j \in \{i-1, i+1\} \quad \mathcal{H}^1(T_i \cap r_j) > 0,$$

then

$$L_S(T_i) < \frac{3}{2}.$$

Proof. Normalize to $i = 0$ and translate the hexagon by V_0 . The five components of S on which a unit triangle containing V_0 can have positive trace become the corresponding local components of the translated skeleton. The positive adjacent-radial trace becomes a positive boundary trace there. Proposition 2.14 and Lemma 3.2 give the bound; every nonlocal component is more than unit distance from V_0 , apart from zero-measure endpoints. $\square$

Lemma 3.4 (No-support skeleton cap). *If*

$$\begin{aligned} A_i + B_i &\leq 1, \\ \mathcal{H}^1(T_i \cap r_{i-1}) &= 0, \quad \mathcal{H}^1(T_i \cap r_{i+1}) = 0, \end{aligned}$$

then

$$L_S(T_i) \leq 2.$$

Proof. See theorem B.2; geometrically, the two incident boundary traces and the own radial trace are the only components that can contribute positive length. $\square$

Lemma 3.5 (Supercritical skeleton cap). *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Proof. The exact admissible-cell and polynomial-sign calculation is theorem B.3. $\square$

Lemma 3.6 (Positive-support rescuer). *Suppose $N_E(T_C) \in \{1, 2\}$ and, after symmetry,*

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

If $N_+ \geq 2$, then for a supercritical index $s \neq 0$ there is $j \in \{s-1, s+1\}$ such that

$$M_s \in U_j, \quad \mathcal{H}^1(T_j \cap r_s) > 0.$$

Proof. Choose a supercritical $s \neq 0$. Lemma 2.13 gives $M_s \notin T_s$, and the displayed midpoint identity gives $M_s \notin T_C$. The open cover therefore puts M_s in another U_j . Diameter locality restricts j to $s-1$ or $s+1$, and openness makes its trace on r_s have positive length. $\square$

Theorem 3.7 (Direct CE1/CE2 skeleton count). *If*

$$\begin{aligned} N_E(T_C) &\in \{1, 2\}, \\ |T_C \cap \{M_0, \dots, M_5\}| &= 1, \end{aligned}$$

and

$$(7) \quad N_+ + N_{\text{sp}} \geq 3,$$

then the seven triangles do not cover S .

Proof. The three geometric caps above make the total trace strictly smaller than $\mathcal{H}^1(S) = 12$; the exact budget is theorem B.4. This contradicts (6). $\square$

3.2. Boundary trace register. The exact evaluation is in Appendix B; the following table is the complete interface used by the routing argument.

Theorem 3.8 (Boundary trace table). *For closures of original open triangles, the following bounds hold.*

	<i>exact hypothesis</i>	<i>established class</i>	<i>exact bound</i>
	$N_E(T_C) = 0$	CE0	$L_{\partial H}(T_C) = 0$
	$N_E(T_C) = 1$	CE1	$L_{\partial H}(T_C) \leq \frac{\sqrt{3}}{2} - \frac{3}{4}$
	$N_E(T_C) = 2$	CE2	$L_{\partial H}(T_C) < \frac{1}{2}$
(8)	$(o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\}$ $A_i + B_i \leq 1$	Vd0	$L_{\partial H}(T_i) \leq 1$
	$(o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\}$ $A_i + B_i > 1$	Vd0	$L_{\partial H}(T_i) \leq \frac{2}{\sqrt{3}}$
	$(o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}$	Vd1 or Vd2	$L_{\partial H}(T_i) < \frac{1}{2}$
	$(o(T_i), n(T_i)) = (2, 1)$	T3-like	$L_{\partial H}(T_i) < 1.$

For every V triangle,

$$(9) \quad L_{\partial H}(T_i) = A_i + B_i.$$

Proof. The locality identity and every row are proved in theorem B.7. $\square$

Corollary 3.9 (Branches closed by the common budgets). *The following exact alternatives are impossible.*

- (1) $N_E(T_C) = 0$, $N_+ = 1$, and $d \geq 1$;
- (2) $N_E(T_C) \in \{1, 2\}$, $N_+ = 0$, and $d \geq 1$;
- (3) $N_E(T_C) = 1$, $N_+ = 1$, and $d \geq 1$;
- (4) $N_E(T_C) = 2$, $N_+ = 1$, and $d \geq 2$;
- (5) $N_E(T_C) = 2$, $N_+ = 1$, and there is an index i such that

$$(o(T_i), n(T_i)) = (1, 2)$$

and

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \emptyset;$$

- (6) $N_E(T_C) \in \{1, 2\}$ and $N_+ + N_{\text{sp}} \geq 3$;
- (7) $N_E(T_C) \in \{1, 2\}$ and $N_+ \geq 2$.

Proof. The perimeter sums for items (1)–(5) are theorem B.11. Item (6) is Theorem 3.7. For item (7), normalize the unique C triangle midpoint and apply Lemma 3.6; its rescuer is distinct from the two supercritical triangles, so item (6) applies. $\square$

Proposition 3.10 (Vd2 adjacent-midpoint branch). *Suppose the C triangle is CE1 or CE2 and $N_+ = 1$. If a Vd2 role contains an adjacent midpoint, the perimeter is not covered.*

Proof. By theorem B.6, that role contributes strictly less than $1/3$. The common boundary caps give

$$L_{\partial H}(T_C) + \sum_i L_{\partial H}(T_i) < \frac{1}{2} + \frac{2}{\sqrt{3}} + \frac{1}{3} + 4 < 6.$$

Apply theorem 3.1. The argument needs no CE2-only formula. $\square$

3.3. Routing conclusion. The zero-gap $N_+ = 1$, $d \geq 1$ clause is retained as an independent alternative; the routing table now sends every zero-gap $N_+ = 1$ state to theorem 5.24.

Proposition 3.11 (Branches closed by the trace-length mechanism). *The following routing entries and hybrid subcase are impossible.*

- (1) $N_{\text{gap}} = 0$ and $N_+ = 0$;
- (2) $N_{\text{gap}} = 0$, $N_+ = 1$, and $d \geq 1$;
- (3) $N_E(T_C) \in \{1, 2\}$ and $N_+ + N_{\text{sp}} \geq 3$;
- (4) $N_E(T_C) \in \{1, 2\}$, $N_+ = 0$, and $d \geq 1$;
- (5) $N_E(T_C) = 1$, $N_+ = 1$, and $(d, t) = (1, 0)$;
- (6) the subcase of $N_E(T_C) = 2$, $N_+ = 1$, and $(d, t) = (1, 0)$ in which the unique non-Vd0 triangle T_i satisfies

$$(o(T_i), n(T_i)) = (1, 2)$$

and

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \emptyset.$$

Proof. For item (1), Lemma 2.9 makes the incident open V traces overlap on every edge, so

$$B_i + A_{i+1} > 1 \quad (0 \leq i \leq 5).$$

Summing contradicts $N_+ = 0$. Items (2)–(5) are clauses (1), (6), (2), and (3), respectively, of Corollary 3.9; their numerical budgets are evaluated in Appendix B. Item (6) is Proposition 3.10. $\square$

4. AREA-LOSS ESTIMATE

The area-loss method records how much of each V triangle must lie outside H . The local estimate applies to every V triangle. A final cyclic argument adds these losses without imposing a center type.

Put $A_\Delta = \sqrt{3}/4$ and normalize all areas by A_Δ . At V_0 define

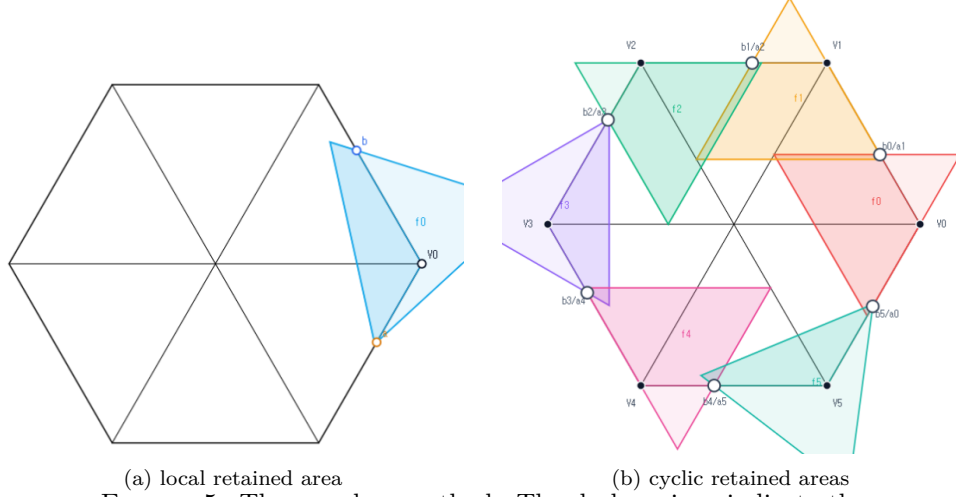
$$\mathcal{T}(a, b) := \left\{ T : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ \{V_0, V_0 + a(V_5 - V_0), V_0 + b(V_1 - V_0)\} \subseteq T \end{array} \right\},$$

$$\mathcal{L}(T) := \frac{\text{area}(T \setminus H)}{A_\Delta}, \quad \mathcal{L}(a, b) := \inf_{T \in \mathcal{T}(a, b)} \mathcal{L}(T),$$

where $a, b \in [0, 1]$ and $\mathcal{T}(a, b) \neq \emptyset$. Hexagon symmetry gives the corresponding family at every V_i . In particular, if

$$0 \leq a_i \leq A_i, \quad 0 \leq b_i \leq B_i,$$

then the rotated copy of T_i belongs to $\mathcal{T}(a_i, b_i)$.



(a) local retained area (b) cyclic retained areas
 FIGURE 5. The area-loss method. The dark regions indicate the portions retained in H ; the exact quantities are $\mathcal{L}(T)$ and $\mathcal{L}(a, b)$.

4.1. Local area loss.

Theorem 4.1 (Local square loss). *For every feasible (a, b) ,*

$$(10) \quad \mathcal{L}(a, b) \geq \min\{a, b\}^2.$$

If $a + b > 1$, then

$$(11) \quad \mathcal{L}(a, b) \geq \max\{a, b\}^2.$$

The same bounds hold for every $T \in \mathcal{T}(a, b)$.

Proof. The two exhaustive orientation families give these minima. Their polygon areas and the factorizations proving both inequalities are computed in theorem C.2. $\square$

4.2. Cyclic accumulation. Use the strict handoffs from theorem 2.11:

$$\xi_i \in (1 - A_{i+1}, B_i), \quad a_i := 1 - \xi_{i-1}, \quad b_i := \xi_i.$$

They satisfy

$$0 < a_i < A_i, \quad 0 < b_i < B_i, \quad a_i^2 + a_i b_i + b_i^2 < 1.$$

Lemma 4.2 (Cyclic area-loss certificate). *The six actual V triangles satisfy*

$$|\{i \in \mathbb{Z}/6\mathbb{Z} : a_i + b_i > 1\}| \geq 2 \implies \sum_{i=0}^5 \mathcal{L}(T_i) > 1.$$

Proof. Apply the local bounds around the six-cycle. The exact sum, including the strict contribution from the second ascent, are given in theorem C.3. $\square$

Proposition 4.3 (Branch closed by area). *The branch $N_{\text{gap}} = 0$, $N_+ \geq 2$ is impossible.*

Proof. The six V triangles cover ∂H . The at-least-two clause of theorem 2.11 supplies the selection for theorem 4.2. Thus the V triangles have total normalized inside area below five. The C triangle contributes at most one, whereas H has normalized area six. Apply theorem 3.1 with area. $\square$

5. FINITE-ENCLOSURE OBSTRUCTIONS

The finite-enclosure method finds compact sets that the V triangles miss. Under a hypothetical cover those sets lie in the open C triangle, while their least equilateral enclosure has side at least one.

Put

$$h := \frac{\sqrt{3}}{2}.$$

For $X \in \mathbb{R}^2$ and $r \geq 0$, put

$$\mathbb{B}(X, r) := \{Y \in \mathbb{R}^2 : \|Y - X\| \leq r\}.$$

For every nonempty compact $K \subseteq \mathbb{R}^2$, define

$$\Lambda(K) := \inf \left\{ s > 0 : \begin{array}{l} \exists T [T \text{ is a closed equilateral triangle of side } s, \\ K \subseteq T] \end{array} \right\}.$$

Thus $\Lambda(K)$ is the least equilateral enclosure side for K .

Lemma 5.1 (Compact set in an open unit triangle). *If K is compact, U is an open unit equilateral triangle, and $K \subset U$, then*

$$\Lambda(K) < 1.$$

Proof. The distance from K to each side line of U is positive. Moving all three lines inward by a smaller common distance gives a closed equilateral triangle of side below one that still contains K . $\square$

5.1. Four-anchor and neighboring-arm capacities. Normalize at V_0 and define the four-anchor hull

$$K(a, b, c) := \text{conv}\{V_0, X_5(1-a), X_0(b), (1-c)V_0\},$$

$$\mathcal{A} := \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \leq 1\}.$$

For

$$0 \leq a, b \leq 1, \quad a^2 + ab + b^2 \leq 1,$$

put

$$c_{\max}(a, b) := \max\{c \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

For $0 \leq a, c \leq 1$, also put

$$M_c(a) := \max\{b \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

The exact admissible set, attainment, and the evaluation of these extrema are proved in theorem A.9.

The neighboring-arm capacities are the variational quantities

$$C_+(a, b) := \max \left\{ c \in [0, 1] : \begin{array}{l} \exists T [T \text{ is a closed unit equilateral triangle,} \\ \{V_0, X_5(1-a), X_0(b), (1-c)V_1\} \subseteq T] \end{array} \right\},$$

$$C_-(a, b) := C_+(b, a), \quad 0 \leq a, b \leq 1, \quad a + b \leq 1.$$

Only a bound, not the full piecewise neighboring capacity, is needed below. The direct four-caliper proof is theorem D.12.

Definition 5.2 (Fixed total radial endpoint). Parametrize r_i by $Z_i(c) = (1 - c)V_i$, measured from V_i toward O . Let $u_{j \rightarrow i}$ be the largest parameter in the actual positive trace of T_j on r_i for $j = i - 1, i + 1$, with value zero when no such positive trace exists. Put

$$\gamma_i := \max\{C_i, u_{i-1 \rightarrow i}, u_{i+1 \rightarrow i}\}, \quad d_i := 1 - \gamma_i, \quad \hat{P}_i := d_i V_i.$$

Thus $\hat{P}_i$ is the innermost endpoint reached by any local V triangle on r_i , rather than just the endpoint of the own V triangle. This distinction is essential when a nonsupercritical V triangle supports an adjacent radial segment.

Lemma 5.3 (Fixed radial forcing and demand recovery). *One has $0 < d_i < 1$ and $\hat{P}_i \notin \bigcup_j U_j$. Under skeleton coverage, $\hat{P}_i \in U_C$. More generally, for every $\gamma_i \leq \Gamma \leq 1$, the point $(1 - \Gamma)V_i$ is missed by every open V triangle and is center-forced under skeleton coverage. If an arbitrary open unit candidate U' contains O and $\hat{P}_i$, and its exit on r_i is d'_i measured from O , then $\gamma_i > 1 - d'_i$. If both neighboring endpoints are at most $1 - d'_i$, it follows that*

$$C_i > 1 - d'_i.$$

It is sufficient that each neighboring V triangle be nonsupercritical with both actual boundary reaches greater than d'_i .

Proof. A local open triangle containing the total endpoint would extend its trace farther toward O . Nonlocal triangles are excluded by diameter. The last assertion follows from common-pair domination, which bounds a neighboring endpoint by $1 - \min\{A_j, B_j\}$. The full endpoint and strictness argument is in theorem D.1. $\square$

Let selected lower bounds satisfy

$$0 \leq a_j \leq A_j, \quad 0 \leq b_j \leq B_j \quad (0 \leq j \leq 5),$$

$$a_j^2 + a_j b_j + b_j^2 \leq 1 \quad (0 \leq j \leq 5).$$

A neighboring term is used only when the actual V triangle has the corresponding positive radial trace:

$$\mathcal{H}^1(T_{i-1} \cap r_i) > 0 \implies a_{i-1} + b_{i-1} \leq 1,$$

$$\mathcal{H}^1(T_{i+1} \cap r_i) > 0 \implies a_{i+1} + b_{i+1} \leq 1.$$

Define

$$\Gamma_i := \max\left(\{c_{\max}(a_i, b_i)\} \cup \{C_+(a_{i-1}, b_{i-1}) : \mathcal{H}^1(T_{i-1} \cap r_i) > 0\} \cup \{C_-(a_{i+1}, b_{i+1}) : \mathcal{H}^1(T_{i+1} \cap r_i) > 0\}\right), \quad D_i := (1 - \Gamma_i)V_i.$$

The points D_i are the radial witnesses used below.

Lemma 5.4 (Type-aware radial forcing). *For every i ,*

$$D_i \in r_i \subseteq S, \quad D_i \notin \bigcup_{j=0}^5 U_j.$$

Consequently,

$$S \subseteq U_C \cup \bigcup_{j=0}^5 U_j \implies D_i \in U_C.$$

Proof. The variational capacities give $\gamma_i \leq \Gamma_i \leq 1$, including only actual positive adjacent traces. Apply theorem 5.3. The point O when $\Gamma_i = 1$ is handled by that lemma as well; no positive-distance claim is made there. $\square$

Lemma 5.5 (Common-pair domination). *If*

$$p, q \geq 0, \quad p + q \leq 1,$$

then

$$C_+(p, q), C_-(p, q) \leq 1 - \min\{p, q\} \leq c_{\max}(p, q).$$

Proof. This is the direct diagonal-capacity comparison in theorem D.14. $\square$

Corollary 5.6 (Uniform common-pair radial forcing). *Suppose $p, q \geq 0$, $p + q \leq 1$, and*

$$p \leq a_i \leq A_i, \quad q \leq b_i \leq B_i \quad (0 \leq i \leq 5), \quad 0 < c_* := c_{\max}(p, q) < 1.$$

Then $(1 - c_)V_i \notin \bigcup_{j=0}^5 U_j$ for every i , without a restriction on the normalized V types. Under a hypothetical cover of H ,*

$$(1 - c_*)V_i \in U_C \quad (0 \leq i \leq 5), \quad \mathbb{B}(O, h(1 - c_*)) \subset U_C.$$

Proof. Convexity permits reselecting every boundary pair as (p, q) . Apply theorem 5.4 to these common lower bounds, including only the actual positive neighboring traces. By theorem 5.5, every included C_+ or C_- term is at most c_* , while the own-ray term equals c_* . Hence each maximum is exactly c_* and the forced points are the stated symmetric ones. Their convex hull contains the displayed disk. In particular, a Vd1, Vd2, or T3-like crossing does not change any witness coordinate. $\square$

Lemma 5.7 (Open trace endpoint). *Let U be an open equilateral triangle and $T = \overline{U}$. If $T \cap [V, O]$ is the closed interval from V to P , then $P \notin U$. If a noncentral point $Q \in [V, O]$ lies in U , then $\mathcal{H}^1(U \cap [V, O]) > 0$.*

Proof. The first point is an endpoint of the closed convex intersection. For the second assertion, intersect a small ball about Q contained in U with the radial arm. $\square$

5.2. Fixed witnesses and the case register. An obstruction for the original C triangle is not, by itself, an enclosure inequality. We therefore separate two assertions: the original cover forces one fixed finite set into U_C , and every open unit equilateral candidate fails to contain that same set. Candidate center parameters are recomputed in the second assertion; the witness points are never recomputed.

The actual gap on $e_{i,i+1}$ is

$$J_i = X_i([B_i, 1 - A_{i+1}]).$$

Both endpoints are missed by the incident open V triangles and by all nonincident V triangles. Equality $B_i = 1 - A_{i+1}$ is a singleton gap. Convexity fills a gap from its endpoints, so a candidate containing the endpoints of all actual gaps completes the boundary with the original six V triangles. This observation does not assert that it completes the whole skeleton.

Lemma 5.8 (The origin is implicit in the path witnesses). *For $G = X_0(t)$, $0 < t < 1$, and $d_2, d_4 > 0$,*

$$G + \frac{1-t}{d_2}(d_2V_2) + \frac{1}{d_4}(d_4V_4) = 0.$$

Consequently $O \in \text{conv}\{G, d_2V_2, d_4V_4\}$.

Proof. Use $V_2 = V_1 - V_0$ and $V_4 = -V_1$, and normalize the three positive coefficients to sum to one. $\square$

There are four active witness recipes and three principal mechanisms: a non-supercritical boundary path, a supported rescuer, and the zero-gap asymmetric construction. The complementary-gap disk is retained in Appendix D as an independent alternative A, not as a dependency of the all-nonsupercritical theorem. In the nonzero-gap rows normalize the C triangle's distinguished radial midpoint to M_0 . When $N_+ = 1$, let σ be the unique supercritical index. All point counts below are upper bounds; coincident gap endpoints can reduce them.

Table 2: Fixed finite-enclosure recipes. Ordinary nonsupercritical V triangles are not subdivided by type.

Conditions	Fixed witness set	Disk and count
B: two gaps; $T_1, \dots, T_5$ nonsupercritical	$\{X_5(B_5), X_0(1 - A_1), \hat{P}_2, \hat{P}_4\}$	No disk; at most 4
C: one gap; $T_1, \dots, T_5$ nonsupercritical	$\{M_0, X_0(B_0), X_0(1 - A_1), \hat{P}_2, \hat{P}_3, \hat{P}_4\}$	No disk; at most 6
D: center-based T3-like or Vd1 rescuer; one or two gaps	$\{O, \varepsilon V_1, Y(A_0), Y(1 - B_5)\}$	No disk; at most 4
F: zero gaps, $N_+ = 1$	Six common radial points and Q_-, Q_0, Q_+	Disk; at most 9

Replacement is a skeleton-preserving operation, not a fifth witness recipe. For $N_+ = 1$ and $\sigma = 0$, use C with one gap and B with two gaps. This includes both adjacent and nonadjacent Vd placements: their location no longer changes the active point construction. When $\sigma \neq 0$, midpoint rescue and the retained length filters give the D or R placements specified below. A Vd2 rescuer retains its perimeter exit; the four-point D theorem does not assume endpoint inequalities that have only been proved for T3-like and Vd1 triangles.

5.3. B and C: a nonsupercritical-path enclosure theorem. In B the gaps are on $e_{5,0}, e_{0,1}$; in C the single gap is on $e_{0,1}$. In both clauses the path consists of $T_1, \dots, T_5$ and satisfies $A_i + B_i \leq 1$. The role T_0 may itself be supercritical or nonsupercritical. Every witness is computed from the original V triangles.

Lemma 5.9 (Two-gap common-pair adapter for a candidate). *Let an open unit candidate U' contain $X_5(B_5), X_0(1 - A_1), \hat{P}_2, \hat{P}_4$. Then $O \in U'$ and the candidate is CE2 with common midpoint M_0 . For its own signed variables put $p = W - \alpha$, $q = R - \delta$. In the two-gap path configuration,*

$$A_i > q, \quad B_i > p \quad (1 \leq i \leq 5).$$

Thus every own or permitted neighboring contribution on r_2, r_4 is at most $c_{\max}(p, q)$.

Proof. Use theorem 5.8. The two boundary anchors force positive traces on both adjacent edges, hence CE2. Their strict containment gives $B_5 > p$, $A_1 > q$. The four gap-free edges propagate these inequalities along the nonsupercritical path. Common-pair domination also bounds both possible neighboring contributions. $\square$

Theorem 5.10 (Four-point two-gap enclosure). *Under the B path hypotheses, the fixed set*

$$K_B := \{X_5(B_5), X_0(1 - A_1), \hat{P}_2, \hat{P}_4\}$$

satisfies $\Lambda(K_B) \geq 1$. Skeleton coverage forces $K_B \subset U_C$. There is no disk in this recipe.

Proof. Forcing uses actual gap endpoints and total radial endpoints. For any candidate containing K_B , the preceding adapter and the signed CE2 inequality $c_{\max}(p, q) < 1 - \min\{\alpha, \delta\}$ place $\hat{P}_2$ beyond exit δ or $\hat{P}_4$ beyond exit α . The full arbitrary-candidate argument is theorem D.2. Neither point is recomputed from the candidate center. $\square$

For C define

$$K_C := \{M_0, X_0(B_0), X_0(1 - A_1), \hat{P}_2, \hat{P}_3, \hat{P}_4\}.$$

The midpoint is a structural anchor belonging to the original open C triangle. The other points are missed by the V triangles. The origin is implicit by theorem 5.8; the construction uses at most six points even though a candidate must also contain O .

Lemma 5.11 (The fixed transverse witness is center-forced). *In the C placement, with the original C triangle's distinguished midpoint M_0 , skeleton coverage implies $K_C \subset U_C$.*

Proof. Use the two actual gap endpoints, total radial forcing on r_2, r_3, r_4 , and the distinguished midpoint. No Vd0 locality is assumed. $\square$

Proposition 5.12 (CE2 transverse threshold terminal). *No open unit CE2 candidate contains K_C .*

Proof. The fixed midpoint and implicit origin normalize every candidate. Its boundary witnesses give a common path pair (X, Q) , where $X = R - \delta$ and $Q = (\eta + \alpha + \delta)/(2R)$. The two thresholds in theorem D.3 exclude the total endpoint on r_2 or r_4 using common-pair domination. $\square$

Proposition 5.13 (CE1 transverse return with neighboring support). *No open unit CE1 candidate contains K_C .*

Proof. In the candidate's signed domain, both boundary reaches of every relevant path role exceed $\min\{X, Q\} > \alpha/R > \alpha$. Neighbor estimates and fixed-endpoint containment first recover the own demands at T_4, T_3 . The first selected return step gives $B_3 > L_1$, and the positive identity

$$R(L_1 - \delta) = (1 - 2\alpha)\eta + (W - 2\alpha)(\alpha + \delta) + 4R\alpha^2$$

excludes both neighboring contributions on r_2 . This recovers the third own demand before the full return is invoked. The exact argument is theorem D.3; the conditional local return remains theorem D.23. $\square$

Theorem 5.14 (Six-point type-independent transverse enclosure). *Under the C path hypotheses, $\Lambda(K_C) \geq 1$. The theorem imposes no V -type restriction on $T_1, \dots, T_5$ and does not require T_0 itself to be supercritical. No disk is used.*

Proof. Every candidate contains O, M_0 and a boundary point openly, hence is CE1 or CE2. The preceding propositions exclude both classes. $\square$

Theorem 5.15 (Center-aligned path obstruction). *Suppose a skeleton cover has at least one actual gap and its C triangle's unique midpoint is M_k . If every T_i with $i \neq k$ is nonsupercritical, the configuration is impossible.*

Proof. Normalize $k = 0$. The signed normal form puts all actual gaps on $e_{5,0}$ and $e_{0,1}$. With two gaps apply theorem 5.10; with one gap reflect if necessary and apply theorem 5.14. The midpoint belongs openly to U_C . Neither path theorem requires T_0 to be supercritical. $\square$

Theorem 5.16 (Every skeleton cover has a supercritical V triangle). *A skeleton cover by the seven original open roles satisfies $N_+ \geq 1$.*

Proof. If $N_+ = 0$ and there are no gaps, strict overlaps give $6 < \sum_i (A_i + B_i) \leq 6$. If a gap is present, all six roles are nonsupercritical, so theorem 5.15 applies. Thus N_0 uses B/C, not alternative A. The local transverse return does not use N_0 , so the later replacement argument introduces no circular dependency. $\square$

Lemma 5.17 (A unique midpoint supplier). *In a nonzero-gap skeleton cover, the preceding N_0 theorem and the skeleton budget leave only $N_+ = 1$, $N_{sp} \leq 1$. Write M_k for the C midpoint and σ for the supercritical index. If $\sigma \neq k$, there is a unique positive-support role T_τ , where*

$$\tau \in \{\sigma - 1, \sigma + 1\}, \quad M_\sigma \in U_\tau.$$

If T_τ is T3-like, then $\tau = k$. If $\tau \neq k$, the edge shared by T_τ, T_σ is center-free, in both CE1 and CE2.

Proof. N_0 excludes $N_+ = 0$. Two supercritical roles would require a distinct positive-support rescuer by theorem 3.6, and theorem 3.7 would exclude them. The same count bound gives $N_{sp} \leq 1$.

If $\sigma \neq k$, neither the C triangle nor T_σ contains M_σ . Diameter excludes nonlocal roles; a Vd0 role has no positive adjacent trace and cannot contain that point openly. Hence its unique supplier is an adjacent positive-support role. A T3-like supplier also misses its own midpoint. Unless $\tau = k$, neither the center nor the other roles could cover this second midpoint: all other roles are Vd0. Finally, when $k \notin \{\tau, \sigma\}$, the shared edge is not incident to V_k , whereas every open center boundary trace is incident to V_k by the signed normal form. $\square$

5.4. D: a four-point supported-rescuer enclosure. Let $Y(t) = (1 - t)V_0 + tV_5 = X_5(1 - t)$. The geometric lemma below is stated independently of any V type or covering configuration.

Theorem 5.18 (Four-point rescuer geometry). *Suppose $a \geq 0$, $\varepsilon > 0$, $\beta \geq 0$, and*

$$s = a + \varepsilon \leq 1, \quad a \leq \varepsilon, \quad \beta \leq \frac{\varepsilon}{s}.$$

Then

$$\Lambda\{O, \varepsilon V_1, Y(a), Y(1 - \beta)\} \geq 1.$$

Proof. The convex combination

$$\frac{\varepsilon}{s}Y(a) + \frac{a}{s}(\varepsilon V_1) = \frac{\varepsilon}{s}V_0$$

forces M_0 into any candidate containing the four points. Its left boundary trace and exit on r_1 then give the incompatible inequalities $1 - \beta < R + \alpha < a/s \leq 1 - \beta$. The reflected one-edge normal form and the endpoint cases are checked in theorem D.4. $\square$

Lemma 5.19 (Scalar rescuer ratio test). *For $0 \leq x, c \leq 1/2$ and $M = (c + \sqrt{c^2 - 8c + 4})/2$,*

$$x \leq 1 - M \iff x^2 + (c - 2)x + c \geq 0.$$

Proof. The smaller root is $1 - M$, while the other root is at least one. The quadratic is nonnegative on $[0, 1/2]$ exactly before its smaller root. $\square$

For the two actual adapters normalize T_0 as the rescuer of M_1 , T_1 as uniquely supercritical, and $T_2, \dots, T_5$ as nonsupercritical ordinary roles on the four center-free edges. Write

$$T_0 \cap r_1 = \{(1 - t)V_1 : c \leq t \leq u\}, \quad 0 < c \leq \frac{1}{2} \leq u < 1, \quad \varepsilon = 1 - u.$$

Let $M = M_c^{\sup} = (c + \sqrt{c^2 - 8c + 4})/2$, with the value $1/2$ at $c = 1/2$ understood as the continuous endpoint convention. The local requirements are

$$\varepsilon V_1 \in U_C, \quad C_1 \geq c, \quad A_0 + \varepsilon \leq 1, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M.$$

The separate inequality $A_0 \leq 1 - M$ follows from the last two and is not an additional hypothesis.

The strict-supercritical envelope gives $B_1 < M$; the four ordinary handoffs give $B_5 \leq B_1$. Thus $B_5 < 1 - A_0$, and the two left-gap endpoints are $Y(A_0)$ and $Y(1 - B_5)$. The fixed set is

$$K_D = \{O, \varepsilon V_1, Y(A_0), Y(1 - B_5)\}.$$

It lies in U_C and satisfies the four-point theorem with geometric parameters $a = A_0$, $\beta = B_5$. No sum greater than four is needed. There is no disk, and no distinction between one and two actual gaps in this enclosure step.

T3-like local input. In its translated chart put

$$1 \leq D_T \leq 2/\sqrt{3}, \quad R_T = (D_T + \sqrt{4 - 3D_T^2})/2.$$

The actual supported endpoints are

$$c = \frac{D_T(1 + A_0) - 1}{R_T}, \quad u = 1 - \frac{R_T}{D_T} + A_0, \quad \varepsilon = \frac{R_T}{D_T} - A_0.$$

The T3-like chart calculation verifies the local requirements. In particular, with $x = A_0 D_T / R_T$ and $c = x + \theta$, the ratio test becomes the single quadratic $2x^2 + (\theta - 1)x + \theta \geq 0$.

Vd1 local input. With $t \geq 1$ and $d_t = \sqrt{t^2 + t + 1}$, the actual corner data give

$$c = \frac{t(1 - B_0)}{t + 1}, \quad u = \frac{d_t - A_0 - tB_0 - 1}{t}, \quad \varepsilon = \frac{t + 1 - d_t + A_0 + tB_0}{t}.$$

The Vd1 chart calculation verifies the same requirements for CE1 and CE2; it uses no CE2-only center inequality. These two chart calculations remain in Appendix D; their global ending is identical.

Proposition 5.20 (T3-like supported-rescuer terminal). *Suppose $N_+ = 1$, there is exactly one T3-like role and no Vd1/Vd2 role, and there is at least one actual gap. Then the skeleton is not covered.*

Proof. If the supercritical role is center-aligned, use theorem 5.15. Otherwise theorem 5.17 puts the T3-like supplier at the center midpoint and, after reflection, the supercritical role at T_1 . The actual-or-bounded frontier forces εV_1 into U_C , and radial coverage gives $C_1 \geq c$. The local calculation theorem D.29 gives the remaining ratio and size bounds. Apply the common adapter theorem D.28. It handles either gap rank without propagating a common pair through the supercritical role. $\square$

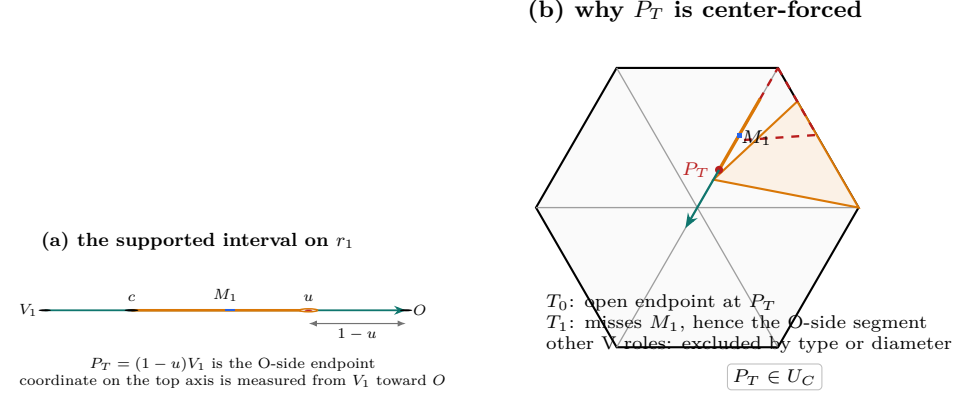


FIGURE 6. Construction of the T3-like rescuer endpoint. The interval $[c, u]$ is parametrized from V_1 toward O , so its O-side endpoint is $P_T = (1 - u)V_1$. The T3-like open role omits its endpoint, the adjacent supercritical role contains V_1 but misses M_1 and therefore misses the O-side segment by convexity, and all remaining V roles are excluded. Thus P_T is forced into the C triangle.

5.5. The remaining placement router.

Proposition 5.21 (One-Vd placement assembly). *Suppose the C triangle is CE1 or CE2, $N_+ = 1$, exactly one V triangle is Vd1 or Vd2, and there is an actual gap. The skeleton is not covered.*

Proof. An additional positive-support role closes by the skeleton budget. Otherwise theorem 5.17 gives the unique supplier or the center-aligned placement. The latter closes by theorem 5.15. A Vd2 supplier closes by theorem 3.10. A center-based Vd1 supplier has its endpoint forced by the common frontier and $C_1 \geq c$ by radial

coverage; theorems D.28 and D.32 then give D. An away Vd1 supplier has a center-free shared edge, so theorem D.10 applies and contradicts N0. The latter preserves only the skeleton, not all of H or the input gap rank. $\square$

Proposition 5.22 (All nonzero-gap finite-enclosure rows). *Every nonzero-gap row of table 1 assigned to finite enclosure is impossible.*

Proof. Use N0 for $N_+ = 0$, the common skeleton budget for the high-count states, and theorem 5.17 for the remaining placements. The center-aligned case is B/C; a T3-like supplier uses theorem 5.20; the Vd1/Vd2 cases use theorem 5.21, with its explicit Vd2 length exit. $\square$

5.6. The zero-gap nine-point obstruction. Assume

$$N_{\text{gap}} = 0, \quad N_+ = 1.$$

No normalized V-type restriction is imposed. The same nine points work when Vd1, Vd2, or T3-like roles meet adjacent radial segments. Rotate the unique supercritical role to index 4. The exact-one handoff calculation theorem E.2 gives

$$(12) \quad \xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

Define

$$(13) \quad a := 1 - \xi_3, \quad b := \xi_4, \quad p := 1 - b, \quad q := 1 - a.$$

Then

$$(14) \quad \begin{aligned} 0 < a, b < 1, \quad a &= a_4 < A_4, \quad b = b_4 < B_4, \\ a + b &> 1, \quad a^2 + ab + b^2 < 1, \\ a_i &\geq p, \quad b_i \geq q \quad (0 \leq i \leq 5). \end{aligned}$$

Define the two boundary anchors

$$P_A := V_4 + a(V_3 - V_4), \quad P_B := V_4 + b(V_5 - V_4),$$

and the feasible union

$$\mathcal{R}_{AB}(a, b) := \bigcup \left\{ T \cap H : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ V_4, P_A, P_B \in T \end{array} \right\}.$$

Put

$$c_* := c_{\max}(p, q), \quad r_* := h(1 - c_*), \quad P_i^{\text{rad}} := (1 - c_*)V_i \quad (0 \leq i \leq 5).$$

The bounds in theorem E.3 give $0 < c_* < 1$. Applying theorem 5.6 to the common pair gives

$$(15) \quad \mathbb{B}(O, r_*) \subseteq \text{conv}\{P_0^{\text{rad}}, \dots, P_5^{\text{rad}}\} \subset U_C.$$

The two circles adjacent to the supercritical frontier are

$$\mathcal{C}_- := \partial\mathbb{B}(X_2(b), 1), \quad \mathcal{C}_+ := \partial\mathbb{B}(X_5(1 - a), 1).$$

Define ℓ_- and ℓ_+ as the corresponding maximal nondegenerate line segments of the interior frontier:

$$\{\ell_-\} := \left\{ [X, Y] : \begin{array}{l} X \neq Y, \quad [X, Y] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H), \\ [X, Y] \cap \mathcal{C}_- \neq \emptyset, \\ \nexists X', Y' \quad [X, Y] \subsetneq [X', Y'] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H) \end{array} \right\},$$

$$\{\ell_+\} := \left\{ [X, Y] : \begin{array}{l} X \neq Y, \quad [X, Y] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H), \\ [X, Y] \cap \mathcal{C}_+ \neq \emptyset, \\ \nexists X', Y' [X, Y] \subsetneq [X', Y'] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H) \end{array} \right\}.$$

The three frontier points are the singleton intersections

$$\{Q_0\} := \ell_- \cap \ell_+, \quad \{Q_-\} := \ell_- \cap \mathcal{C}_-, \quad \{Q_+\} := \ell_+ \cap \mathcal{C}_+.$$

Explicit construction of the zero-gap points and disks. One may choose every strict handoff explicitly as

$$\xi_i = \frac{1 - A_{i+1} + B_i}{2}.$$

The exact-one handoff lemma preserves the supercritical index under this choice. At index 4 use the outgoing/incoming chart

$$\Psi_4(u, v) = V_4 + u(V_5 - V_4) + v(V_3 - V_4).$$

Put $\rho = a^2 + ab + b^2$, $\Delta = \sqrt{4\rho - 3}$ and define

$$\begin{aligned} \alpha_F &= \frac{h(b + 2a - b\Delta)}{2\rho}, & \beta_F &= \frac{h(b - a + (a + b)\Delta)}{2\rho}, \\ \gamma_F &= \frac{h(-b + a + (a + b)\Delta)}{2\rho}, & \delta_F &= \frac{h(2b + a - a\Delta)}{2\rho}. \end{aligned}$$

These are frontier coefficients, not the signed C-triangle parameters. The selected first-root parameters are

$$\begin{aligned} \lambda_- &= 2(\alpha_F + b\beta_F) - \frac{2}{3}\sqrt{9(\alpha_F + b\beta_F)^2 - 9b^2 + 9b - 6}, \\ \mu_+ &= 2(\delta_F + a\gamma_F) - \frac{2}{3}\sqrt{9(\delta_F + a\gamma_F)^2 - 9a^2 + 9a - 6}. \end{aligned}$$

Then the same three frontier witnesses are

$$\begin{aligned} Q_- &= \Psi_4(b - \beta_F\lambda_-, \alpha_F\lambda_-), & Q_+ &= \Psi_4(\delta_F\mu_+, a - \gamma_F\mu_+), \\ Q_0 &= \Psi_4\left(\frac{2b + a - a\Delta}{\Delta + 3}, \frac{b + 2a - b\Delta}{\Delta + 3}\right). \end{aligned}$$

The exact fixed-line calculation ensures that the radicals are real and selects the stated roots. No sampled point or new root selection is used. The finite set and its compact comparison set are

$$K_F = \{(1 - c_*)V_i : 0 \leq i \leq 5\} \cup \{Q_-, Q_0, Q_+\},$$

$$\mathcal{D}_F = \{x : \|x\| \leq h(1 - c_{\max}(1 - b, 1 - a))\}, \quad \mathcal{D}_F \cup \{Q_-, Q_0, Q_+\} \subseteq \text{conv}(K_F).$$

When $c_* \leq 2/3$, the six radial points already suffice because $\Lambda(\mathcal{D}_F) = 3(1 - c_*) \geq 1$. For $c_* > 2/3$, the unchanged asymmetric enclosure certificate is needed.

Its smaller comparison disk is also explicit. With $p = 1 - b$, $q = 1 - a$, $s = p + q$, $m = \min\{p, q\}$, and $\chi = s^4 - s^2 + pq$, put

$$\bar{c} = \begin{cases} s - \frac{\chi}{4s^3 - 2s + m}, & \chi \leq 0, \\ s - \frac{\chi}{1 - m}, & \chi > 0. \end{cases}$$

Then

$$\overline{\mathcal{D}}_F = \{x : \|x\| \leq h(1 - \bar{c})\} \subseteq \mathcal{D}_F,$$

using the exact bound $c_* \leq \bar{c} < 1$. This smaller disk is a comparison inside the tangent-residual certificate, not an additional global witness recipe. The two unit

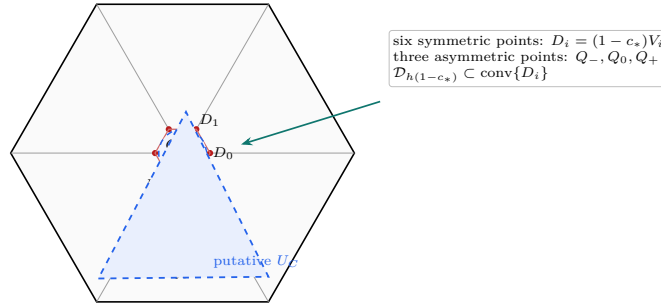
disks defining the frontier circles are $\mathbb{B}(X_2(b), 1)$ and $\mathbb{B}(X_5(1-a), 1)$; they are auxiliary distance constructions, not disks asserted to lie in the C triangle.

Their exact construction and exclusions in theorems E.1, E.4 and E.5 use contained vertices and actual handoffs, not Vd0 locality, and give

$$(16) \quad Q_-, Q_0, Q_+ \in U_C.$$

Define the public witness set

$$(17) \quad K_{\text{wit}}(a, b) := \mathbb{B}(O, r_*) \cup \{Q_-, Q_0, Q_+\} \subset U_C.$$



$$K_{\text{wit}} = \mathcal{D}_{h(1-c_*)} \cup \{Q_-, Q_0, Q_+\} \subset U_C \text{ under a hypothetical cover}$$

FIGURE 7. The type-independent zero-gap nine-point witness. The six symmetric radial points force a centered disk into the convex C triangle, while the three asymmetric frontier points supply the missing directional support. Marker shapes separate the two constructions: circles are radial witnesses and triangles are strict-supercritical AB-frontier witnesses. The displayed placement is illustrative; the point definitions are the exact ones in the preceding lemmas.

Theorem 5.23 (Witness enclosure). *Every pair (a, b) satisfying equation (14) satisfies*

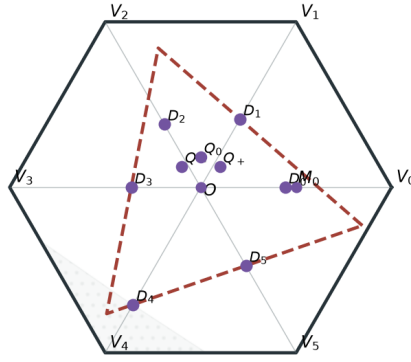
$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

Proof. The disk closes the range $c_* \leq 2/3$. For $c_* > 2/3$, the Newton inner points reduce the enclosure to four supporting contacts: the lines through two consecutive points and the two exposed disk tangencies. The line bounds are analytic; the paired tangent residuals use the exact certificate and simultaneous radius transfer. The complete calculation is theorem E.10. $\square$

Theorem 5.24 (Zero-gap obstruction). *There is no hypothetical cover with*

$$N_{\text{gap}} = 0, \quad N_+ = 1.$$

Proof. The construction gives the compact containment $K_{\text{wit}}(a, b) \subset U_C$, while theorem 5.23 gives $\Lambda(K_{\text{wit}}(a, b)) \geq 1$. This contradicts theorem 5.1. $\square$

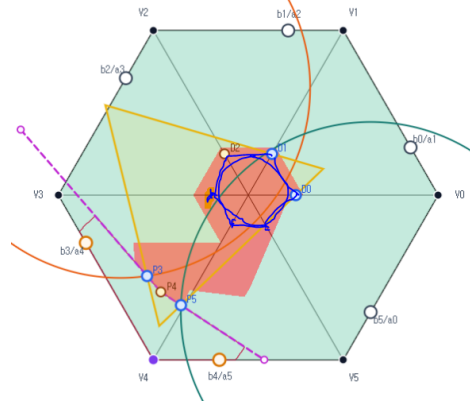


Trace-exact witness.

$$N_{\text{gap}} = 0$$

$$N_+ = 1$$

$$(d, t) = (0, 0)$$



Core configuration.

$$N_{\text{gap}} = 0$$

$$N_+ = 1$$

$$(d, t) = (0, 0)$$

FIGURE 8. Case F: sampled zero-gap obstruction data.

Proposition 5.25 (Cases excluded by the nine-point obstruction). *The entire $N_{\text{gap}} = 0$, $N_+ = 1$ row of table 1 is impossible, independently of the center class and normalized V-type pattern.*

Proof. This is theorem 5.24; the equality $N_{\text{gap}} = 0$ supplies full boundary coverage by theorem 2.9. $\square$

6. COMPLETION OF THE PROOF

Proof of Theorem 1.1. Suppose seven open unit equilateral triangles cover H . Their center and vertex roles are distinct. By theorem 5.16, $N_+ \geq 1$.

If there are no actual gaps, the V triangles cover the perimeter. Exactly one supercritical role is excluded by the type-independent nine-point theorem theorem 5.24; at least two are excluded by theorem 4.3.

If there is a gap, theorem 5.17 reduces the cover to $N_+ = 1, N_{\text{sp}} \leq 1$. Let M_k be the C midpoint and σ the supercritical index. If $\sigma = k$, apply theorem 5.15. Otherwise M_σ has a unique adjacent positive-support supplier. A T3-like supplier must be based at V_k and is excluded by theorem 5.20. A Vd2 supplier is excluded by theorem 3.10. A center-based Vd1 supplier uses D; an away Vd1 supplier admits the two-vertex replacement and contradicts N0. These last alternatives are assembled in theorem 5.21. Thus every possibility contradicts the cover. $\square$

The zero-gap certificate uses coverage of all of H . Replacement needs only skeleton coverage and is never used to invoke that certificate. Corollary 1.2 follows from Proposition 2.1.

APPENDIX A. STRUCTURAL, SHARED-LOCAL, AND SIGNED-CENTER OPTIMIZATION

This appendix solves the coordinate and compactness problems used by the geometric interfaces in Section 2. It also proves the shared equilateral-enclosure calculus used by the later optimization appendices. Each calculation is attached to the body statement it supplies.

A.1. Open–closed shrinking feasibility.

Lemma A.1 (Uniform shrinking margin). *If seven open unit equilateral triangles cover H , then, for some $\varepsilon > 0$, seven closed equilateral triangles of side $1 - \varepsilon$ also cover H .*

Proof. For $p \in H$ put

$$m_\nu(p) := \text{dist}(p, \mathbb{R}^2 \setminus U_\nu) \quad (\nu \in \{C, 0, \dots, 5\}), \quad m(p) := \max_{\nu \in \{C, 0, \dots, 5\}} m_\nu(p).$$

The continuous function m is positive on compact H , hence

$$\eta := \min_{p \in H} m(p) > 0.$$

Choose

$$0 < \rho < \min \left\{ \eta, \frac{\sqrt{3}}{6} \right\}.$$

Moving every side of U_ν inward by ρ gives a closed equilateral triangle K_ν of side $1 - 2\sqrt{3}\rho$. For each $p \in H$, some $m_\nu(p) \geq \eta > \rho$, so $p \in K_\nu$. Thus $\varepsilon = 2\sqrt{3}\rho$ is feasible. This supplies the only quantitative step in Proposition 2.1. $\square$

A.2. T3-like trace-majorant feasibility. The calculation uses the local wedge and the two orientation charts developed in the exact-trace module below. It is placed here beside the body warning that the output is a closed majorant, not a replacement open triangle.

Lemma A.2 (T3-like translation calculation). *The closed-trace majorant in Proposition 2.7 exists. In the calculation chart, every original T3-like triangle also satisfies $A_i + B_i \leq 1$.*

Proof. Use the wedge chart of theorem A.7 and the orientation forms (25)–(26) in Subsection A.6. In Type I, direct inversion shows that when exactly two vertices lie outside W , the sole wedge vertex has both coordinates strictly below 1: the relevant bounds are $(1 - t)/z$ and t/z , with strictness from $\alpha > 0$ or $\beta > 0$. The support argument in theorem A.7 then excludes positive-length contact with either adjacent radial line. Thus a T3-like triangle has Type II.

Reflect in $x = y$ if necessary so that $T_0 \cap r_1$ has positive length. The endpoint cases $t = 0, 1$ give no such interval, hence $0 < t < 1$. The exact axis reaches are

$$A_0 = \beta, \quad B_0 = z - \alpha - \beta,$$

and the support condition is

$$Q_x > 1, \quad Q_y \leq 1,$$

where Q is (28). In particular

$$A_0 + B_0 = z - \alpha < z \leq 1.$$

Define $\widehat{T}$ by replacing α with 0 and retaining t, β . This changes only the translation. At the origin, $\widehat{U} = 0$ and $0 < \widehat{V} = \beta < z$, so V_0 lies in the relative interior of a side. For $(x, y) \in W$,

$$\widehat{U} = (1-t)x + ty \geq 0, \quad \widehat{V} = V, \quad \widehat{U} + \widehat{V} = U + V - \alpha.$$

Thus $T \cap W \subset \widehat{T} \cap W$. The old x -axis reach is B_0 and the new reach is $B_0 + \alpha$, so the inclusion is strict in H .

The two outside vertices remain outside, while V_0 and the new wedge vertex lie in H . Moreover $\widehat{Q}_x > Q_x > 1$, and $Q_x > 1$ gives

$$tD_t(1 - \widehat{Q}_y) > D_t(1 - z) + (1-t)\alpha > 0.$$

Hence $\widehat{Q}_y < 1$ and $(o(\widehat{T}), n(\widehat{T})) = (2, 1)$. At every nonzero point of W , $\widehat{U} > 0$; the other two displayed slack relations show that old interior points other than V_0 remain interior. Reflection proves the other supported-arm case. $\square$

A.3. Strict-handoff feasibility.

Lemma A.3 (Cyclic strict-handoff selection). *Under the hypotheses of Proposition 2.11, all six overlap intervals are nonempty, and the selectors can preserve the stated one- or two-supercritical-index alternatives.*

Proof. Since V_i is interior to a diameter-one triangle, $0 < A_i < 1$ and $0 < B_i < 1$. No nonincident V triangle contains a relative-interior point of $e_{i,i+1}$. Thus the relatively open incident traces cover the edge and overlap, so

$$(1 - A_{i+1}, B_i) \neq \emptyset.$$

Choosing ξ_i in this interval places both selected anchors in U_i . Their squared distance is $a_i^2 + a_i b_i + b_i^2$, which is strictly below 1 because both are interior points of the same open unit triangle.

Put $\ell_i := 1 - A_{i+1}$ and $u_i := B_i$. Then

$$(18) \quad A_i + B_i > 1 \iff \ell_{i-1} < u_i, \quad a_i + b_i > 1 \iff \xi_{i-1} < \xi_i.$$

If i is actually nonsupercritical, then

$$\xi_i < u_i \leq \ell_{i-1} < \xi_{i-1},$$

so no selection makes it supercritical. Also

$$(19) \quad \sum_{i=0}^5 (a_i + b_i) = \sum_{i=0}^5 (1 - \xi_{i-1} + \xi_i) = 6.$$

If p is the sole actual supercritical index, the other five selected sums are below 1, and (19) forces the p th sum above 1.

For two nonadjacent supercritical indices p, q , choose

$$\ell_{r-1} < z_r < u_r \quad (r \in \{p, q\}),$$

and then

$$\xi_{r-1} \in (\ell_{r-1}, \min\{u_{r-1}, z_r\}), \quad \xi_r \in (\max\{\ell_r, z_r\}, u_r).$$

The pairs are disjoint and create ascents at p, q . Any three indices on a six-cycle contain two nonadjacent indices. If the only two supercritical indices are adjacent, say $p, p+1$, the remaining inequalities give

$$\ell_{p-1} < \ell_{p+1}, \quad \max\{\ell_{p-1}, \ell_p\} < \min\{u_p, u_{p+1}\}.$$

Choose ξ_p between the latter bounds and choose $\xi_{p-1} < \xi_p < \xi_{p+1}$ in their overlap intervals. By (18), both selected indices remain supercritical. $\square$

A.4. Shared equilateral-enclosure optimization. Let R denote counterclockwise rotation through $2\pi/3$. For a nonempty compact $K \subset \mathbb{R}^2$, put

$$h_K(n) := \max_{x \in K} \langle x, n \rangle.$$

Proposition A.4 (Equilateral enclosure gauge). *The least side length $\Lambda(K)$ of a closed equilateral triangle containing K is*

$$(20) \quad \Lambda(K) = \frac{2}{\sqrt{3}} \min_{\|n\|=1} \sum_{j=0}^2 h_K(R^j n).$$

It is monotone under inclusion, translation invariant, and positively homogeneous. If K is a polygonal hull with at least two distinct points, a minimizing normal can be chosen perpendicular to a hull edge.

Proof. For fixed outward normals n, Rn, R^2n , the least containing triangle has the corresponding three support numbers, and its side is $2/\sqrt{3}$ times their sum. Translation invariance follows from

$$n + Rn + R^2n = 0,$$

while monotonicity and homogeneity are immediate. On an open angular cell with fixed support points, the support sum is $A \cos \theta + B \sin \theta$ and its second derivative is its negative. For a nonsingleton the support sum is positive, so an interior critical point is a maximum; a minimum occurs at a support tie, hence at a hull-edge normal. $\square$

A.4.1. Finite caliper criterion. Let J denote counterclockwise rotation through $\pi/2$. For a nonempty finite set F define

$$(21) \quad W_F(N) := \sum_{k=0}^2 \max_{z \in F} \langle z, R^k N \rangle.$$

Theorem A.5 (Finite caliper certificate). *If F has at least two distinct points, then F is contained in a closed unit equilateral triangle if and only if there are distinct $p, q \in F$ and $\varepsilon \in \{-1, 1\}$ such that*

$$(22) \quad W_F(\varepsilon J(q - p)) \leq \frac{\sqrt{3}}{2} \|p - q\|.$$

A singleton is contained in equilateral triangles of arbitrarily small positive side length. For $F_x = \{O, A, B, x\}$, every clause is a finite polynomial clause of degree at most four in the coordinates of A, B, x ; at most $12 \cdot 4^3$ clauses are needed.

Proof. Apply Proposition A.4 to $\text{conv} F$. A minimizing normal is perpendicular to a hull edge $[p, q]$ and, after cyclic rotation, equals $\varepsilon J(q - p)/\|p - q\|$. Homogeneity gives (22); a nondegenerate segment follows by checking its two normals.

For the polynomial assertion fix (p, q, ε) , put $N = \varepsilon J(q - p)$ and $N_k = R^k N$, and select support labels $s_k \in F_x$. Retain the strict condition

$$\|p - q\|^2 > 0.$$

On the cell

$$(23) \quad \langle s_k - y, N_k \rangle \geq 0 \quad (y \in F_x, \ k = 0, 1, 2),$$

the test becomes

$$(24) \quad \sum_{k=0}^2 \langle s_k, N_k \rangle \leq \frac{\sqrt{3}}{2} \|p - q\|.$$

The left side is polynomial of degree at most two. Both sides are nonnegative on an active cell, so one squaring gives degree at most four. There are at most six unordered pairs, two normal signs for each pair, and 4^3 support triples for each pair-normal candidate. $\square$

A.5. Center and V triangle incidence feasibility.

Lemma A.6 (Separation of nonadjacent boundary edges). *Relative-interior points on two nonadjacent edges of H are more than unit distance apart.*

Proof. Up to symmetry take

$$x = (1 - t)V_0 + tV_1, \quad 0 < t < 1.$$

For $y = (1 - s)V_2 + sV_3$, $0 < s < 1$, direct expansion gives

$$\|x - y\|^2 - 1 = (1 - t)(2 - t) + s(s + t) > 0.$$

For the other orbit, $y = (1 - s)V_3 + sV_4$, put $u = t + s$ and obtain

$$\|x - y\|^2 - 1 = (u - 1)^2 + 2 > 0.$$

These are the two nonadjacent-edge configurations. Since the diameter of a unit equilateral triangle is 1, the result supplies Proposition 2.3. $\square$

Lemma A.7 (Local wedge calculation). *The incidence conclusions of Lemma 2.4 hold.*

Proof. Normalize to $i = 0$ and use the affine chart

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0).$$

Then $V_0 = (0, 0)$,

$$\|(x, y)\|^2 = x^2 + y^2 - xy,$$

and, in the closed unit ball about V_0 , membership in H is membership in

$$W := \{(x, y) : x \geq 0, \ y \geq 0\}.$$

Indeed,

$$x^2 + y^2 - xy = \left(y - \frac{x}{2}\right)^2 + \frac{3x^2}{4} = (x - y)^2 + xy.$$

The adjacent radial segments are

$$r_1 = \{(1, y) : 0 \leq y \leq 1\}, \quad r_5 = \{(x, 1) : 0 \leq x \leq 1\}.$$

Suppose $T_0 \cap r_1$ has positive length and let m be the largest x -coordinate of a vertex of T_0 . If $m > 1$, a maximizing vertex (m, y) lies in the unit ball and has $0 < y < 1$, so it lies in H . If $m = 1$, positive-length contact with the support line $x = 1$ forces a side there, and both endpoints satisfy $1 + y^2 - y \leq 1$ and lie in H . The argument for r_5 is symmetric. If both arms are met but only one vertex lies in H , that vertex would have $x > 1$ and $y > 1$, contradicting $(x - y)^2 + xy > 1$.

Finally, if all three vertices lay in convex H , then $T_0 \subset H$, contrary to the extreme point V_0 lying in $\text{int}(T_0)$. Hence $o(T_0) \geq 1$, completing the calculation. $\square$

A.6. Exact-trace normalization. Set

$$D_t := 1 - t + t^2, \quad z := \sqrt{D_t}, \quad 0 \leq t \leq 1.$$

After relabelling the vertices, every orientation belongs to one of the two families

	first side vector	second side vector
I	$z^{-1}(1, t)$	$z^{-1}(1 - t, 1)$
II	$z^{-1}(t, -(1 - t))$	$z^{-1}(1, t)$.

In Type I write

$$(25) \quad U = \alpha + y - tx, \quad V = \beta + x - (1 - t)y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\},$$

and in Type II write

$$(26) \quad U = \alpha + (1 - t)x + ty, \quad V = \beta + tx - y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\}.$$

In both charts

$$(27) \quad V_0 \in \text{int}(T) \iff \alpha > 0, \quad \beta > 0, \quad \alpha + \beta < z.$$

The Type-II vertex in the wedge is

$$(28) \quad Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z - \alpha) + (1 - t)\beta}{D_t} \right).$$

Lemma A.8 (Raw $(3, 0)$ trace-normalization calculation). *The translation asserted in Proposition 2.5 exists and preserves both the open and closed traces in H .*

Proof. Normalize to $i = 0$. A Type-II triangle cannot have $o = 3$, because the point Q in (28) lies in the wedge; together with the diameter-one condition, Lemma 2.4 puts it in H . Hence T has Type-I form. Its vertices are

$$(29) \quad \begin{aligned} Q_0 &= \left(-\frac{(1 - t)\alpha + \beta}{D_t}, -\frac{\alpha + t\beta}{D_t} \right), \\ Q_1 &= \left(\frac{z - (1 - t)\alpha - \beta}{D_t}, \frac{tz - \alpha - t\beta}{D_t} \right), \\ Q_2 &= \left(\frac{(1 - t)z - (1 - t)\alpha - \beta}{D_t}, \frac{z - \alpha - t\beta}{D_t} \right). \end{aligned}$$

Thus $o = 3$ is equivalent to

$$(30) \quad \alpha + t\beta > tz, \quad (1 - t)\alpha + \beta > (1 - t)z,$$

which forces $0 < t < 1$. Put $s = \alpha + \beta$ and $L = z - s > 0$. Then

$$(31) \quad \alpha > \frac{tL}{1 - t}, \quad \beta > \frac{(1 - t)L}{t}.$$

For $(x, y) \in W$ satisfying

$$(32) \quad (1 - t)x + ty \leq L,$$

the minimum of U is attained at $(L/(1 - t), 0)$ and that of V at $(0, L/t)$. Hence

$$(33) \quad T \cap W = \{(x, y) \in W : (1 - t)x + ty \leq L\}.$$

Keep t, s fixed and set

$$(34) \quad \hat{\alpha} := \frac{tL}{1-t}, \quad \hat{\beta} := s - \hat{\alpha}.$$

Then $0 < \hat{\alpha} < \alpha$, $\hat{\beta} > \beta > 0$, and $\hat{\alpha} + \hat{\beta} = s < z$. Changing these affine constants is a translation and preserves $V_0 \in \text{int}(\hat{T})$. Its third inequality remains (32); $\hat{U} = 0$ only at the endpoint $(L/(1-t), 0)$ of that common boundary and $\hat{V} > 0$. Therefore

$$\hat{T} \cap W = T \cap W,$$

and their interiors have the same trace on W . The local wedge calculation identifies these with the closed and open traces in H .

Equation (34) places Q_1 on the boundary of H ; Q_0 remains outside and the second strict inequality in (31) keeps Q_2 outside. Hence $(o(\hat{T}), n(\hat{T})) = (2, 0)$. Since the three reach segments lie in H , their open and closed traces, and therefore A_i, B_i, C_i and $A_i + B_i > 1$, are unchanged. $\square$

A.7. The local admissible set and radial envelope. Let u, v be unit vectors meeting at 120° and define

$$K(a, b, c) := \text{conv}\{0, au, bv, c(u+v)\}.$$

This is the translate of the body normalization at V_0 , with $u = V_5 - V_0$ and $v = V_1 - V_0$. The exact local admissible set is

$$(35) \quad \mathcal{A} := \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \leq 1\}.$$

This set records the three reach lower bounds that a closed unit V triangle can realize.

Proposition A.9 (Exact local admissible set). *Put*

$$\begin{aligned} s &:= a + b, & d &:= a^2 + ab + b^2, & m &:= \min\{a, b\}, & M &:= \max\{a, b\}, \\ & & q &:= s^4 - s^2 + ab, \end{aligned}$$

and

$$\begin{aligned} F_L &:= c^4 - c^2 + mc - m^2, \\ F_T &:= (s^2 - 1)c^2 + Mc - M^2, \\ F_S &:= (m^2 - 1)c^2 + (2mM^2 + M)c + M^4 - M^2. \end{aligned}$$

Then

$$(36) \quad \mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S,$$

where all variables lie in $[0, 1]$ and

$$\begin{aligned} \mathcal{A}_L &:= \{d \leq 1, s \leq 1, q \leq 0, F_L \leq 0\}, \\ \mathcal{A}_T &:= \{d \leq 1, s \leq 1, q \geq 0, F_T \leq 0, c \leq 2M\}, \\ \mathcal{A}_S &:= \{d \leq 1, s > 1, F_S \leq 0, c \leq 1/2\}. \end{aligned}$$

The set is coordinatewise down-closed. Define its maximal radial demand by

$$c_{\max}(a, b) := \max\{c \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

Then

$$(37) \quad c_{\max}(a, b) = \begin{cases} 1, & a = b = 0, \\ C_L(m), & s \leq 1, \quad q \leq 0, \quad (a, b) \neq (0, 0), \\ \frac{2M}{1 + \sqrt{4s^2 - 3}}, & s \leq 1, \quad q > 0, \\ \frac{M(2mM + 1) - M\sqrt{4d - 3}}{2(1 - m^2)}, & s > 1, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

Proof. First assume $s > 0$ and $cs \leq ab$. When $a, b > 0$,

$$c(u + v) = \frac{c}{a}(au) + \frac{c}{b}(bv) + \left(1 - \frac{c}{a} - \frac{c}{b}\right)0 \in \text{conv}\{0, au, bv\};$$

the zero-coordinate cases follow by continuity. The distance from au to bv is $\sqrt{d}$. With $T = -bu - av$, the triangle $\text{conv}\{au, bv, T\}$ is equilateral of side $\sqrt{d}$ and

$$0 = \frac{b^2}{d}(au) + \frac{a^2}{d}(bv) + \frac{ab}{d}T.$$

Thus the least enclosure side is $\sqrt{d}$. When $a = b = 0$, it is c .

Now assume $cs > ab$. The hull has cyclic vertices $0, bv, c(u + v), au$. Proposition A.4 reduces its least enclosure side to its four edge normals. For the edges through 0 one obtains

$$L_{0A} = b + \max\{a, c\}, \quad L_{B0} = a + \max\{b, c\}.$$

For the edge from au to $c(u + v)$ put

$$R_a := \sqrt{a^2 - ac + c^2}, \quad n_0 := \frac{1}{2R_a}(\sqrt{3}a, 2c - a).$$

The three support values in the rotated directions are

$$\frac{\sqrt{3}ac}{2R_a}, \quad \frac{\sqrt{3}a(a - c)_+}{2R_a}, \quad \frac{\sqrt{3}c \max\{b, (c - a)_+\}}{2R_a}.$$

Consequently

$$L_{AC} = \begin{cases} (a^2 + bc)/R_a, & a \geq c, \\ c(a + b)/R_a, & a < c \leq a + b, \\ c^2/R_a, & c > a + b, \end{cases}$$

and L_{CB} is obtained by interchanging a, b . Hence

$$\Lambda(K(a, b, c)) = \min\{L_{0A}, L_{B0}, L_{AC}, L_{CB}\}.$$

Assume by symmetry that $a = m \leq b = M$. Reading the active support points in these four formulas gives

range		active supports in cyclic order			unit frontier
s	q	first	second	third	equation
$s < 1$	$q < 0$	$\{0\}$	$\{c(u + v)\}$	$\{au, c(u + v)\}$	$F_L = 0$
$s < 1$	$q > 0$	$\{0\}$	$\{bv, c(u + v)\}$	$\{au\}$	$F_T = 0$
$s = 1$		$\{0, bv\}$	$\{bv, c(u + v)\}$	$\{au\}$	$c = M$
$s > 1$		$\{bv\}$	$\{bv, c(u + v)\}$	$\{au\}$	$F_S = 0$

The inactive-support inequalities reduce to $q \leq 0$, $q \geq 0$, and $s > 1$. The positive unit-frontier equations square to $F_L = 0$, $F_T = 0$, $F_S = 0$.

It remains to select the geometric components introduced by squaring. In Cell T the roots are

$$c_- := \frac{2M}{1 + \sqrt{4s^2 - 3}}, \quad c_+ := \frac{2M}{1 - \sqrt{4s^2 - 3}}.$$

The polynomial inequality also has a formal high component, while the geometric fiber is the component joined to $c = 0$. Since

$$c_- \leq 2M \leq c_+,$$

the exact selector is $c \leq 2M$.

In Cell S ,

$$F_S(a, b, 0) = M^2(M^2 - 1) < 0, \quad F_S(a, b, M) = M^2(s^2 - 1) > 0.$$

Moreover $4F_S(a, b, 1/2)$ increases with m , and at $m = 1 - M$ equals $M^2(2M - 1)^2$. Because $s > 1$, one has $F_S(a, b, 1/2) > 0$. The quadratic opens downward, so $c \leq 1/2$ selects exactly the smaller-root component containing 0.

In Cell L , the fiber from $c = 0$ reaches one root in $[\sqrt{3}/2, 1]$; uniqueness there follows from $4c^3 - 2c + m > 0$. At $q = 0$ the L and T formulas meet at $c = s$. This proves (36), including all equality and degenerate cases by continuity. Convexity of any containing triangle proves coordinatewise down-closedness. Solving the three selected frontier equations gives (37). $\square$

For later variational use, the raw forward envelope retains its established notation

$$M_c(a) := \max\{b \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

No closed-form evaluation of M_c is needed in the geometric body.

A.8. Midpoint caliper optimization.

Lemma A.10 (Four-point self-midpoint caliper). *If $a + b > 1$, then*

$$\Lambda\left(\text{conv}\left\{0, au, \frac{u+v}{2}, bv\right\}\right) > 1.$$

This supplies Lemma 2.13.

Proof. Put

$$K := \text{conv}\left\{0, au, \frac{u+v}{2}, bv\right\}.$$

By Theorem A.5, it suffices to check a normal to each of the four hull edges. For the edge from 0 to au , the sum of the three supports is

$$S_{0A} = \max\left\{\frac{\sqrt{3}}{4}, \frac{\sqrt{3}a}{2}\right\} + \frac{\sqrt{3}b}{2} > \frac{\sqrt{3}}{2}.$$

If $a \geq 1/2$, this is $(\sqrt{3}/2)(a + b)$; if $a < 1/2$, then $b > 1/2$. The edge through bv is symmetric.

For the edge from au to $(u + v)/2$, put $D_a := \sqrt{4a^2 - 2a + 1}$. Projection gives

$$S_{AM} = \begin{cases} \frac{\sqrt{3}(a+b)}{2D_a}, & a \leq 1/2, \\ \frac{\sqrt{3}(2a^2+b)}{2D_a}, & a > 1/2. \end{cases}$$

The first value exceeds $\sqrt{3}/2$ because $D_a \leq 1$. In the second case, $b > 1 - a$ and

$$(2a^2 - a + 1)^2 - D_a^2 = a^2(2a - 1)^2 \geq 0,$$

so it again exceeds $\sqrt{3}/2$. The remaining edge is symmetric. Formula (20) now gives $\Lambda(K) > 1$.

If $a + b = 1$ while all four points are contained with positive margin, both boundary reach demands can be increased slightly, contradicting the strict case just proved. $\square$

A.9. Signed-center optimization. The CE1 and CE2 cases share one signed cell. The parameters below are used only in the appendices; the geometric body uses the exact C triangle type and midpoint interfaces.

Proposition A.11 (Signed CE1/CE2 center normal form). *Let T_C be a closed unit equilateral triangle with $O \in \text{int}(T_C)$ and a positive-length boundary trace. Normalize such a trace to $e_{0,1}$ and use affine coordinates*

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0).$$

There are parameters

$$\begin{aligned} 0 < R < 1, \quad W &:= 1 - R, \quad E := \sqrt{1 - RW}, \\ \eta &:= 1 - E, \quad P := E(1 - E), \end{aligned}$$

and nonnegative slacks α, δ , with $k := \eta + \alpha + \delta$, such that

$$\begin{aligned} F_0 &= R + \alpha - a + Wb, \\ (38) \quad F_1 &= Rb + Wa - k, \quad T_C = \{F_0, F_1, F_2 \geq 0\}. \\ F_2 &= W + \delta - b + Ra, \end{aligned}$$

Define

$$(39) \quad \Delta_R := P - \alpha - W\delta, \quad \Delta_L := P - R\alpha - \delta.$$

Then

$$(40) \quad T_C \cap e_{0,1} = \left[\frac{k}{R}, W + \delta \right], \quad \mathcal{H}^1(T_C \cap e_{0,1}) = \frac{\Delta_R}{R},$$

and the possible companion trace is

$$(41) \quad T_C \cap e_{5,0} = \left[\frac{k}{W}, R + \alpha \right] \quad \text{when } \Delta_L > 0, \quad \mathcal{H}^1(T_C \cap e_{5,0}) = \frac{[\Delta_L]_+}{W}.$$

The normalized trace requires $\Delta_R > 0$, and

$$(42) \quad T_C \text{ is CE1} \iff \Delta_L \leq 0, \quad T_C \text{ is CE2} \iff \Delta_L > 0.$$

The six radial exits from O are

$$\begin{aligned} d_0^C &= E - \alpha - \delta, \quad d_1^C = \frac{\delta}{R}, \quad d_2^C = \delta, \\ (43) \quad d_3^C &= \min \left\{ \frac{\alpha}{R}, \frac{\delta}{W} \right\}, \quad d_4^C = \alpha, \quad d_5^C = \frac{\alpha}{W}. \end{aligned}$$

Moreover,

$$(44) \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

For the closure of an original open C triangle, $\alpha, \delta > 0$.

Proof. Write $T_C \cap e_{0,1} = [s, t]$ with $0 \leq s < t \leq 1$. The two sides through the endpoints have a positive slope ratio. Reflect across the perpendicular bisector if necessary, call the ratio $R \leq 1$, and put $W = 1 - R$. The three side equations are

$$F_1 = Rb + Wa - Rs, \quad F_2 = -b + Ra + t, \quad F_0 = Wb - a + E + Rs - t,$$

where $E = \sqrt{1 - RW}$ is the unit-side relation. If $R = 1$, then $F_0(O) = s - t < 0$, so $0 < R < 1$.

Set $\alpha := F_0(O)$ and $\delta := F_2(O)$. Then

$$t = W + \delta, \quad Rs = \eta + \alpha + \delta = k,$$

which gives (38). The gradients are the three equilateral side-normal directions and $F_0 + F_1 + F_2 = E$.

On $e_{0,1}$, $a = 0$ and the active inequalities are $Rb \geq k$ and $b \leq W + \delta$. Thus its length is

$$W + \delta - \frac{k}{R} = \frac{P - \alpha - W\delta}{R}.$$

On $e_{5,0}$, $b = 0$ and the active inequalities are $Wa \geq k$ and $a \leq R + \alpha$, giving signed length

$$R + \alpha - \frac{k}{W} = \frac{P - R\alpha - \delta}{W}.$$

This proves the two trace formulas.

The positive right trace implies

$$\delta < \frac{P}{W} = \frac{ER}{1 + E} < \frac{R}{2}.$$

On $e_{1,2}$, parametrized by $b = 1 + q, a = q$, one has $F_2 = \delta - R - Wq < 0$. The center classification leaves $e_{5,0}$ as the only possible companion edge, and its signed length proves (42); $\Delta_L = 0$ is only point contact.

Substituting the six ray parametrizations into (38) gives (43). Finally,

$$W(\alpha + \delta) \leq \alpha + W\delta < P$$

implies

$$E - \alpha - \delta > E - \frac{P}{W} = \frac{E(E - R)}{W} > \frac{1}{2},$$

where the last inequality is equivalent to $(E - R)^2 > 0$. Also

$$\alpha < P < \min \left\{ \frac{R}{2}, \frac{W}{2} \right\}, \quad \delta < \frac{R}{2}.$$

Therefore $d_0^C > 1/2$ and $d_i^C < 1/2$ for $i = 1, \dots, 5$, proving the midpoint formula. Interior containment of O in the original open triangle makes all three side slacks positive, so $\alpha, \delta > 0$. $\square$

Lemma A.12 (CE2 total slack). *In the CE2 sign range, with $T := \alpha + \delta$,*

$$(45) \quad T < \frac{\eta}{E}.$$

Proof. The CE2 inequalities are $\alpha + W\delta < P$ and $R\alpha + \delta < P$. Multiply them by W and R and add. Since $W + R^2 = W^2 + R = E^2$, this gives $E^2T < P = E\eta$. $\square$

Corollary A.13 (Common center-boundary formula). *The full center-boundary contribution is*

$$(46) \quad L_{\partial H}(T_C) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}.$$

In CE1,

$$L_{\partial H}(T_C) \leq P \leq \frac{\sqrt{3}}{2} - \frac{3}{4},$$

and in CE2,

$$L_{\partial H}(T_C) = \frac{E}{1+E} - \frac{E^2(\alpha + \delta)}{RW} < \frac{1}{2}.$$

Proof. The first formula sums (40) and (41). In CE1, $R\alpha + \delta \geq P$; multiplication by W , together with $\alpha \geq RW\alpha$, gives $\alpha + W\delta \geq WP$. Thus $\Delta_R \leq RP$ and the contribution is at most P . Since $E \geq \sqrt{3}/2$ and $E(1-E)$ decreases there, the stated cap follows.

In CE2, direct addition gives

$$L_{\partial H}(T_C) = \frac{P - E^2(\alpha + \delta)}{RW} = \frac{E}{1+E} - \frac{E^2(\alpha + \delta)}{RW} < \frac{1}{2}.$$

□

Lemma A.14 (Adjacent-edge diameter transfer). *For $0 \leq q \leq 1$, the zero-radial forward envelope is*

$$M_0(q) = \frac{-q + \sqrt{4 - 3q^2}}{2}.$$

Here M_0 without an argument remains the radial midpoint. If a unit equilateral triangle reaches parameters q, b on adjacent boundary edges, then $b \leq M_0(q)$. The function is strictly decreasing and, for $q > 0$,

$$M_0(q) < 1 - \frac{q}{2}, \quad 1 - M_0(q) > \frac{q}{2}.$$

Proof. The squared distance between the boundary points is $q^2 + qb + b^2 \leq 1$. Solving the quadratic in b gives the formula. Differentiation gives

$$M'_0(q) = \frac{-1 - 3q/\sqrt{4 - 3q^2}}{2} < 0.$$

For $q > 0$, $\sqrt{4 - 3q^2} < 2$, which gives both strict comparisons. □

APPENDIX B. TRACE-LENGTH OPTIMIZATION

This appendix evaluates the trace quantities whose exact interfaces appear in Section 3. The center and V triangle coordinates are used only here; the body retains the geometric hypotheses and final bounds.

B.1. Conditional skeleton caps.

Lemma B.1 (Center-skeleton calculation). *Under the hypotheses of Lemma 3.2, one has $L_S(T_C) < 3/2$.*

Proof. Normalize the unique midpoint as M_0 and a positive boundary trace to $e_{0,1}$. Use Proposition A.11; thus

$$W = 1 - R, \quad E = \sqrt{1 - RW}, \quad \eta = 1 - E, \quad P = E(1 - E),$$

and the positive trace condition is

$$(47) \quad \alpha + W\delta < P.$$

The two possible boundary contributions are

$$\ell_{01} = W + \delta - \frac{\eta + \alpha + \delta}{R}, \quad \ell_{50} = \frac{[P - (R\alpha + \delta)]_+}{W}.$$

The six radial contributions are bounded by

i	0	1	2	3	4	5
$\mathcal{H}^1(T_C \cap r_i)$	$E - \alpha - \delta$	δ/R	δ	$\min\{\alpha/R, \delta/W\}$	α	α/W

Hence

$$L_S(T_C) \leq K_R + \mathcal{E}(\alpha, \delta), \quad K_R := W + E - \frac{\eta}{R},$$

where

$$\mathcal{E}(\alpha, \delta) := -\frac{\alpha}{R} + \frac{\alpha}{W} + \delta + \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\} + \frac{[P - (R\alpha + \delta)]_+}{W}.$$

Splitting according to the minimum and positive part gives

$$(48) \quad \mathcal{E}(\alpha, \delta) \leq \frac{P}{W}.$$

If $\alpha/R \leq \delta/W$, the expression without the positive part is $\alpha/W + \delta$; when that part is active, the excess over P/W is $\alpha - R\delta/W \leq 0$. The reflected order gives $\delta - W\alpha/R \leq 0$. If the positive part vanishes, (47) gives the same bound strictly.

Put $A := 1 + R - R^2$. Then

$$\frac{3}{2} - K_R - \frac{P}{W} = \frac{1 + A - 2EA}{2RW}.$$

Both sides of $1 + A > 2EA$ are positive, and

$$(1 + A)^2 - 4A^2E^2 = R^2W^2(5 + 4R - 4R^2) > 0.$$

Therefore $K_R + P/W < 3/2$, as required. $\square$

Lemma B.2 (No-support skeleton calculation). *The hypotheses of Lemma 3.4 imply $L_S(T_i) \leq 2$.*

Proof. The boundary contribution is $A_i + B_i \leq 1$ by (9). If $P = sV_j \in r_j \setminus \{O\}$ and $j \notin \{i-1, i, i+1\}$, then

$$\|V_i - P\|^2 = 1 + s^2 - 2s \cos \frac{(j-i)\pi}{3} > 1.$$

The adjacent radial traces have zero length by hypothesis, so only r_i can add positive trace, of length at most one. $\square$

For the remaining cap, reflect the local directions if necessary so that $0 \leq A_i \leq B_i \leq 1$. The actual reach triple of a supercritical V triangle lies in the selected cell

$$(49) \quad \begin{aligned} A_i^2 + A_i B_i + B_i^2 &\leq 1, \\ C_i &\leq \frac{1}{2}, \\ (A_i^2 - 1)C_i^2 + (2A_i B_i^2 + B_i)C_i + (B_i^4 - B_i^2) &\leq 0, \end{aligned}$$

where C_i belongs to the connected component containing 0 in the corresponding one-dimensional fiber. This is Cell S of Proposition A.9. The selector is essential: the quadratic is negative at $C_i = 0$, positive at $C_i = 1/2$, and opens downward.

Lemma B.3 (Supercritical skeleton calculation). *Every V triangle with $A_i + B_i > 1$ has $L_S(T_i) < 3/2$.*

Proof. The corner normal form, Proposition B.5, makes Vd1 and Vd2 triangles nonsupercritical. Lemma 2.10 gives the same conclusion for T3-like triangles. Hence a supercritical triangle is Vd0 and has no positive adjacent-radial support. Reflect so that $A_i \leq B_i$. Then

$$L_S(T_i) = A_i + B_i + C_i.$$

Set

$$s := A_i + B_i, \quad \delta := B_i - A_i, \quad \gamma_0 := \frac{3}{2} - s.$$

The diameter constraint gives

$$1 < s \leq \frac{2}{\sqrt{3}}, \quad 0 \leq \delta \leq \sqrt{4 - 3s^2}.$$

Let $P(\gamma)$ be the quadratic in (49). Substitution and multiplication by 16 yield

$$\begin{aligned} 16P(\gamma_0) = Q(s, \delta) &= \delta^4 + 8\delta^3s - 6\delta^3 + 14\delta^2s^2 - 18\delta^2s + 5\delta^2 \\ &\quad - 8\delta s^3 + 30\delta s^2 - 34\delta s + 12\delta \\ &\quad + s^4 - 6s^3 - 19s^2 + 60s - 36. \end{aligned}$$

Its derivative factors as

$$\frac{\partial Q}{\partial \delta} = 2(2\delta + 4s - 3)(\delta^2 + \delta(4s - 3) + (s - 1)(2 - s)) \geq 0.$$

Therefore

$$Q(s, \delta) \geq Q(s, 0) = (s - 1)(s^3 - 5s^2 - 24s + 36) > 0.$$

The cubic decreases on $[1, 2/\sqrt{3}]$ and at the right endpoint equals $88/3 - 136\sqrt{3}/9 > 0$. Hence $P(\gamma_0) > 0$. The selected smaller-root component in (49) forces $C_i < \gamma_0$, so $A_i + B_i + C_i < 3/2$. $\square$

Lemma B.4 (Common skeleton-budget calculation). *Under the hypotheses of Theorem 3.7, the total skeleton trace of the seven triangles is strictly less than 12.*

Proof. The center cap and the three V triangle caps give

$$\begin{aligned} L_S(T_C) + \sum_{i=0}^5 L_S(T_i) &< \frac{3}{2} + \frac{3}{2}(N_+ + N_{\text{sp}}) + 2(6 - N_+ - N_{\text{sp}}) \\ &= \frac{27 - N_+ - N_{\text{sp}}}{2} \leq 12. \end{aligned}$$

The two classes counted by N_+ and N_{sp} are disjoint: every Vd1, Vd2, or T3-like triangle has positive adjacent-radial support and is nonsupercritical, while every supercritical triangle is Vd0. $\square$

B.2. Vd1 and Vd2 corner optimization. Use temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0),$$

so $V_0 = (0, 0)$, $V_5 = (1, 0)$, $V_1 = (0, 1)$, $O = (1, 1)$, and $\|(x, y)\|^2 = x^2 + y^2 - xy$.

Proposition B.5 (Vd1/Vd2 corner normal form). *Let T be the closure of an original Vd1 or Vd2 triangle at V_0 , and let A_0, B_0 be its exact reaches toward V_5, V_1 . There is a unique $t > 0$ such that, with $d = \sqrt{t^2 + t + 1}$,*

$$(50) \quad T = \left\{ (x, y) : \begin{array}{l} x - (t+1)y \leq A_0, \\ ty - (t+1)x \leq tB_0, \\ tx + y \leq d - A_0 - tB_0 \end{array} \right\}.$$

The raw traces, parametrized from the indicated outer vertex, are

$$(51) \quad \begin{array}{l} r_0 \\ r_1 \\ r_5 \end{array} \left| \begin{array}{l} 0 \leq s \leq \frac{d - A_0 - tB_0}{t+1} \\ \frac{t(1 - B_0)}{t+1} \leq s \leq \frac{d - A_0 - tB_0 - 1}{t} \\ \frac{1 - A_0}{t+1} \leq s \leq d - A_0 - tB_0 - t. \end{array} \right.$$

Intersecting with $[0, 1]$ gives the actual traces, and

$$(52) \quad A_0 + B_0 < \frac{1}{2}.$$

Proof. The triangle has one vertex W outside H and two vertices U, Z in H . Because V_0 is an interior convex combination, both coordinates of W are negative. The axis endpoints $(A_0, 0), (0, B_0)$ lie on the two sides through W . Put $p = U - W$ and $q = Z - W$. Their order selects

$$q = (p_x - p_y, p_x), \quad p_x > p_y > 0.$$

Writing $t = (p_x - p_y)/p_y$ gives uniquely

$$p = \frac{(t+1, 1)}{d}, \quad q = \frac{(t, t+1)}{d},$$

which yields (50); substitution of $(s, s), (s, 1), (1, s)$ yields (51).

If r_1 has positive trace, clearing its endpoint inequality gives

$$(t+1)A_0 + tB_0 < (t+1)(d-1) - t^2.$$

Since the left side is at least $t(A_0 + B_0)$,

$$A_0 + B_0 < \frac{(t+1)(d-1) - t^2}{t} < \frac{1}{2}.$$

The last inequality follows from

$$\left(t^2 + \frac{3t}{2} + 1\right)^2 - (t+1)^2(t^2 + t + 1) = \frac{t^2}{4} > 0.$$

Reflection treats support on r_5 . □

Lemma B.6 (Vd2 adjacent-midpoint cap). *If an original open V triangle has Vd2 closure T_i and*

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \emptyset,$$

then its exact boundary reaches satisfy

$$A_i + B_i < \frac{1}{3}.$$

Proof. In Proposition B.5, reflection sends

$$(t, A_0, B_0, M_1, M_5) \mapsto (1/t, B_0, A_0, M_5, M_1).$$

Thus assume $t \geq 1$ and use $A_0 + B_0 \leq A_0 + tB_0$.

If $M_5 = (1, 1/2)$ lies in T , the third half-plane gives

$$A_0 + B_0 \leq A_0 + tB_0 \leq d - t - \frac{1}{2} \leq \sqrt{3} - \frac{3}{2} < \frac{1}{3}.$$

If $M_1 = (1/2, 1)$ lies in T , the second and third half-planes give

$$B_0 \geq \frac{t-1}{2t}, \quad A_0 + B_0 \leq S(t) := d - t - \frac{1}{2t}.$$

Positive-length contact with r_5 gives

$$A_0 + B_0 < R(t) := \frac{2(t+1)d - 3t^2 - t - 2}{2t}.$$

Direct differentiation shows S increasing and R decreasing on $[1, \infty)$. For S' , the sign reduces to

$$(2t+1)^2 t^4 - (t^2 + t + 1)(2t^2 - 1)^2 = (t+1)(t^3 + 3t^2 - 1) > 0,$$

while $R' < 0$ follows from $d > t + 1/2$. Splitting at $t = 4/3$ gives

$$\begin{aligned} 1 \leq t \leq \frac{4}{3} : \quad A_0 + B_0 &\leq \frac{8\sqrt{37} - 41}{24} < \frac{1}{3}, \\ t \geq \frac{4}{3} : \quad A_0 + B_0 &< \frac{7\sqrt{37} - 39}{12} < \frac{1}{3}. \end{aligned}$$

The final comparisons are $8\sqrt{37} < 49$ and $7\sqrt{37} < 43$. $\square$

B.3. Boundary contributions.

Lemma B.7 (Boundary-trace table calculation). *All identities and bounds in Theorem 3.8 hold.*

Proof. The two incident traces of a V triangle are initial intervals of lengths A_i, B_i . A relative-interior point of a nonincident edge is more than unit distance from V_i , while $V_i \in \text{int}(T_i)$. Hence no other edge has positive-length trace and

$$L_{\partial H}(T_i) = A_i + B_i.$$

The $N_E(T_C) = 0$ row is the definition. The other C triangle rows are Corollary A.13. If a Vd0 triangle is nonsupercritical, its boundary trace is at most 1. If it is supercritical, its two incident endpoints have squared distance

$$A_i^2 + A_i B_i + B_i^2 \leq 1,$$

and therefore

$$\frac{3}{4}(A_i + B_i)^2 \leq 1.$$

This gives $A_i + B_i \leq 2/\sqrt{3}$. The Vd1/Vd2 row is (52). For an original T3-like triangle, the Type-II calculation in theorem A.2 gives $A_i + B_i = z - \alpha < z < 1$ because $0 < t < 1$ and $\alpha > 0$. $\square$

B.4. Common perimeter budgets.

Lemma B.8 (Master perimeter deficit). *Put*

$$\theta := \frac{2}{\sqrt{3}} - 1.$$

Suppose a branch has center contribution $L_C := L_{\partial H}(T_C)$, exactly n supercritical Vd0 triangles, and $m \geq 1$ distinct nonsupercritical V triangles $T_{i_1}, \dots, T_{i_m}$ such that

$$L_{\partial H}(T_{i_j}) < q_j < 1 \quad (1 \leq j \leq m).$$

Suppose also that each of the remaining $6 - n - m$ V triangles contributes at most 1. If

$$(53) \quad L_C + n\theta \leq \sum_{j=1}^m (1 - q_j),$$

then the seven triangles do not cover ∂H .

Proof. The supercritical Vd0 cap is $2/\sqrt{3}$, so the total contribution is strictly less than

$$L_C + n\frac{2}{\sqrt{3}} + \sum_{j=1}^m q_j + (6 - n - m) = 6 + L_C + n\theta - \sum_{j=1}^m (1 - q_j) \leq 6.$$

The strictness comes from $m \geq 1$ and the distinguished strict caps, in contradiction with perimeter subadditivity. $\square$

Lemma B.9 (Small center slacks in the one-Vd branch). *Assume there is exactly one supercritical Vd0 triangle, exactly one Vd1 or Vd2 triangle, and four nonsupercritical Vd0 triangles. Every candidate surviving the perimeter budget is CE2 and satisfies*

$$(54) \quad \alpha + \delta < \frac{1}{24}, \quad \alpha + \delta < \frac{\min\{R, W\}}{6}.$$

Proof. The exceptional V triangle has strict cap $1/2$. Perimeter coverage and Lemma B.8 force

$$L_C + \theta > \frac{1}{2}.$$

Put $\kappa := 1/2 - \theta = 3/2 - 2/\sqrt{3}$. The CE1 cap gives

$$L_C + \theta \leq \left(\frac{\sqrt{3}}{2} - \frac{3}{4} \right) + \left(\frac{2}{\sqrt{3}} - 1 \right) < \frac{1}{2},$$

so every survivor is CE2.

Set $T := \alpha + \delta$. The CE2 formula gives

$$T < M(E) := \frac{RW}{E^2} \left(\frac{E}{1+E} - \kappa \right) = \frac{1-E}{E^2} (E - \kappa(1+E)).$$

Since

$$M'(E) = -\frac{E-2\kappa}{E^3} < 0 \quad \left(\frac{\sqrt{3}}{2} \leq E < 1 \right),$$

where $2\kappa < \sqrt{3}/2$, one has

$$T < M(\sqrt{3}/2) = \frac{8\sqrt{3}}{9} - \frac{3}{2} < \frac{1}{24}.$$

The last comparison follows from $64\sqrt{3} < 111$.

Also $E/(1+E) < 1/2$ gives

$$T < \frac{RW}{E^2}\theta.$$

Because $E^2 = W + R^2 = R + W^2$, the factor RW/E^2 is at most both R and W ; and $\theta < 1/6$. This proves (54). $\square$

Lemma B.10 (CE2 endpoints dominate total slack). *Assume $\Delta_R > 0$ and $\Delta_L > 0$. Put*

$$T := \alpha + \delta, \quad x := \frac{\eta + T}{W}, \quad y := \frac{\eta + T}{R}.$$

Then

$$(55) \quad T < \frac{1}{2} \min\{x, y\}.$$

Proof. Equation (45) gives $T < \eta/E$. Since

$$E^2 - (2R - 1)^2 = 3RW > 0,$$

one has $|2R - 1|T < \eta$. Therefore

$$x - 2T = \frac{\eta - (1 - 2R)T}{W} > 0, \quad y - 2T = \frac{\eta - (2R - 1)T}{R} > 0.$$

$\square$

Lemma B.11 (Perimeter-budget substitutions). *Items (1)–(5) of Corollary 3.9 are impossible.*

Proof. For item (1), use $L_C = 0$, one supercritical cap $2/\sqrt{3}$, and one strict Vd1/Vd2 cap $1/2$ in Lemma B.8; the required comparison is

$$\frac{2}{\sqrt{3}} - 1 < \frac{1}{2}.$$

For item (2), $L_C < 1/2$, the distinguished V triangle has strict cap $1/2$, and the remaining five contribute at most 1.

For item (3), the CE1 cap, the exceptional cap, the supercritical cap, and four unit caps give

$$\left(\frac{\sqrt{3}}{2} - \frac{3}{4}\right) + \frac{1}{2} + \frac{2}{\sqrt{3}} + 4 < 6.$$

For item (4), the CE2 sum is strictly less than

$$\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{2}{\sqrt{3}} + 3 < 6.$$

For item (5), Lemma B.6 gives the strict $1/3$ cap, and the total is strictly less than

$$\frac{1}{2} + \frac{1}{3} + \frac{2}{\sqrt{3}} + 4 < 6.$$

Each inequality contradicts the length-6 perimeter requirement. $\square$

APPENDIX C. AREA-LOSS OPTIMIZATION

This appendix evaluates the three area optimizations used in section 4. The body fixes the feasible triangles, the loss functional, and the strict cyclic handoffs; here we introduce coordinates only to solve those problems.

C.1. Wedge chart.

Lemma C.1 (Local wedge reduction). *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \quad W = \{(x, y) : x \geq 0, y \geq 0\}.$$

If a closed unit equilateral triangle T contains V_0 , then

$$T \cap H = T \cap W.$$

Normalized Euclidean area is twice the corresponding (x, y) -coordinate area.

Proof. The two basis vectors have length one and angle 120° , so

$$\|(x, y)\|^2 = x^2 + y^2 - xy, \quad |\det(V_1 - V_0, V_5 - V_0)| = \frac{\sqrt{3}}{2}.$$

The hexagon is

$$H = \{0 \leq x, y \leq 2, |x - y| \leq 1\} \subset W.$$

Conversely, if $(x, y) \in T \cap W$, then the diameter-one condition gives $x^2 + y^2 - xy \leq 1$. Hence

$$|x - y|^2 \leq x^2 + y^2 - xy \leq 1, \quad \frac{3}{4}x^2 \leq x^2 + y^2 - xy,$$

and the same estimate holds for y . Thus $|x - y| \leq 1$ and $0 \leq x, y \leq 2/\sqrt{3} < 2$, proving $(x, y) \in H$. The determinant identity proves the area assertion. $\square$

C.2. Local square-loss optimization.

Lemma C.2 (Solved local square-loss optimization). *For the feasible family $\mathcal{T}(a, b)$ of section 4, the minimum loss satisfies equations (10) and (11).*

Proof. In the coordinates of Lemma C.1, physical rotation through 60° is $R(x, y) = (x - y, x)$. Put

$$0 \leq t \leq 1, \quad D = 1 - t + t^2, \quad z = \sqrt{D}.$$

After choosing one of the six directed sides, every orientation has one of the following two forms:

$$(56) \quad \begin{array}{c|cc} & d & Rd \\ \hline \text{I} & z^{-1}(1, t) & z^{-1}(1 - t, 1) \\ \text{II} & z^{-1}(t, -(1 - t)) & z^{-1}(1, t). \end{array}$$

Indeed these forms cover, respectively, the physical direction sectors $[120^\circ, 180^\circ]$ and $[60^\circ, 120^\circ]$; their endpoints cover the common boundary orientations.

For Type I introduce

$$U = \alpha + y - tx, \quad V = \beta + x - (1 - t)y.$$

Then T is the simplex $U, V \geq 0, U + V \leq z$. Containment of $(0, 0), (0, a), (b, 0)$ gives

$$(57) \quad \alpha \geq tb, \quad \beta \geq (1 - t)a, \quad \alpha + \beta + (1 - t)b \leq z, \quad \alpha + \beta + ta \leq z.$$

The two inequalities defining W become

$$(1-t)U + V \geq (1-t)\alpha + \beta, \quad U + tV \geq \alpha + t\beta.$$

For fixed t , lowering α or β therefore enlarges $T \cap W$. The outside loss is minimized at $\alpha = tb$, $\beta = (1-t)a$. At that placement the portion outside W has vertices, in the (U, V) -plane,

$$(0, 0), \quad (a + tb, 0), \quad (tb, (1-t)a), \quad (0, b + (1-t)a).$$

Since the Jacobian of $(x, y) \mapsto (U, V)$ has absolute value D ,

$$(58) \quad \mathcal{L}(T) \geq \mathcal{L}_I(a, b; t) = \frac{(1-t)a^2 + tb^2 + 2t(1-t)ab}{D}.$$

If $a \leq b$, then

$$D(\mathcal{L}_I - a^2) = t\{b^2 - a^2 + (1-t)(a^2 + 2ab)\} \geq 0;$$

the reflected factorization proves $\mathcal{L}_I \geq b^2$ when $b \leq a$. This proves (10) in Type I.

Suppose now that $a + b > 1$ and $a \leq b$. Set

$$r = \frac{a}{b}, \quad t_0 = \frac{1-r^2}{1+2r}.$$

Since $b > 1/(1+r)$, feasibility in (58) and (57) excludes $0 \leq t < t_0$: for $0 < t < t_0$,

$$\left(\frac{1+r(1-t)}{1+r}\right)^2 - D = \frac{t(1+2r)(t_0-t)}{(1+r)^2} > 0,$$

so $b + (1-t)a > z$, while $t = 0$ would give $a + b \leq 1$. Hence $t \geq t_0$, and

$$D(\mathcal{L}_I - b^2) = (1-t)b^2\{r^2 - 1 + t(2r+1)\} \geq 0.$$

If $b \leq a$, put $r = b/a$ and $t_1 = (r^2 + 2r)/(1+2r)$. The identical reflected calculation excludes $t > t_1$ and gives

$$D(\mathcal{L}_I - a^2) = ta^2\{r^2 + 2r - t(1+2r)\} \geq 0.$$

Thus (11) holds in Type I.

For Type II put

$$U = \alpha + (1-t)x + ty, \quad V = \beta + tx - y.$$

Again T is the simplex $U, V \geq 0$, $U + V \leq z$. Its exact reaches on the positive y - and x -axes are

$$(59) \quad A = \beta, \quad B = z - \alpha - \beta, \quad A + B \leq z \leq 1.$$

In particular Type II is impossible when $a + b > 1$. Its wedge intersection is the quadrilateral with vertices

$$(0, 0), \quad (B, 0), \quad \left(\frac{B + (1-t)A}{D}, \frac{A + tB}{D}\right), \quad (0, A).$$

Hence

$$(60) \quad \mathcal{L}(T) = \mathcal{L}_{II}(A, B; t) = 1 - \frac{(1-t)A^2 + tB^2 + 2AB}{D}.$$

Let $S = A + B$. If $A \leq B$, write $A = rS$, $B = (1-r)S$ with $0 \leq r \leq 1/2$. Since $S^2 \leq D$, and with

$$C = (1-t)r^2 + t(1-r)^2 + 2r(1-r),$$

we have $\mathcal{L}_{\Pi} \geq 1 - C$. The exact factorization

$$1 - C - r^2 D = (1 - t)(1 - 2r + tr^2) \geq 0$$

gives $\mathcal{L}_{\Pi} \geq r^2 D \geq A^2$. Reflection gives $\mathcal{L}_{\Pi} \geq B^2$ when $B \leq A$. Since $A \geq a$ and $B \geq b$, this proves (10). The two orientation types are exhaustive, so the theorem follows. $\square$

C.3. Cyclic loss optimization. The variables ξ_i, a_i, b_i are the strict handoffs already fixed in section 4; in particular, no endpoint selection is relaxed here.

Lemma C.3 (Solved cyclic area-loss optimization). *The implication in theorem 4.2 hold.*

Proof. Put $m = \min_i \xi_i$ and $M = \max_i \xi_i$. The reflection $y_i = 1 - \xi_{-i-1}$ swaps each V triangle's two coordinates and preserves the number of ascents and the total loss. Hence assume $m \leq 1 - M$. Every selected V triangle coordinate is then at least m , so (10) gives loss at least m^2 for every V triangle.

Choose an ascent out of a minimum plateau, $\xi_{p-1} = m < \xi_p$. Its V triangle contains the coordinate $1 - m$ and loses at least $(1 - m)^2$ by (11). A second ascent has one coordinate strictly larger than $1/2$ and hence loses strictly more than $1/4$. The other four V triangles lose at least m^2 each, so the total loss is strictly larger than

$$4m^2 + (1 - m)^2 + \frac{1}{4} = 5 \left(m - \frac{1}{5} \right)^2 + \frac{21}{20} > 1.$$

$\square$

APPENDIX D. NONZERO-GAP FINITE-ENCLOSURE OPTIMIZATION

This appendix solves the point, radial, disk, and interval calculations used by the nonzero-gap rows of table 2. The geometric witnesses and their case predicates remain in the body.

D.1. Fixed endpoints and arbitrary candidates. The new enclosure theorems use actual trace endpoints, fixed before an arbitrary candidate is chosen. This subsection supplies the new arguments; the subsequent local capacity and selected-return calculations are retained. In particular, the two-gap comparison pair belongs to the candidate, but the four-point set being tested belongs to the original configuration.

Lemma D.1 (Total radial endpoint and maximum-to-own-demand). *The assertions of theorem 5.3 hold for arbitrary V types.*

Proof. No closed V triangle contains O : its distinguished vertex is interior and at distance one from O , so moving that vertex slightly away from O inside the open triangle would contradict diameter one. Each own trace has positive length. Hence $0 < \gamma_i < 1$ and $0 < d_i < 1$.

If a local open role contained $Z_i(\gamma_i)$, openness would give a point $Z_i(c)$ with $c > \gamma_i$ in that role. A role with no positive neighboring trace would acquire one; otherwise its maximal endpoint would increase. Both contradict the definition of γ_i . The nonlocal roles are excluded by

$$\|d_i V_i - V_{i \pm 2}\|^2 = 1 + d_i + d_i^2 > 1, \quad \|d_i V_i - V_{i+3}\| = 1 + d_i > 1.$$

Thus every total endpoint is missed by all open V roles. For $\gamma_i \leq \Gamma < 1$, the same proof with $d = 1 - \Gamma$ excludes $(1 - \Gamma)V_i$: openness would give a local endpoint strictly above $\Gamma \geq \gamma_i$. For $\Gamma = 1$, the point is O , which was already excluded even from the closed V triangles.

Now let an arbitrary open unit candidate contain O and $\hat{P}_i$. If its radial exit is d'_i , open containment gives $d_i < d'_i$ and hence $\gamma_i > 1 - d'_i$. If both neighboring endpoints are at most $1 - d'_i$, the own term must realize this strict demand. For a nonsupercritical neighbor with both actual boundary reaches above d'_i , the bound $C_{\pm}(A_j, B_j) \leq 1 - \min\{A_j, B_j\} < 1 - d'_i$ excludes its contribution. No skeleton-cover hypothesis for this new candidate is used. $\square$

Lemma D.2 (Fixed four-point two-gap enclosure). *The fixed set K_B in theorem 5.10 has $\Lambda(K_B) \geq 1$.*

Proof. Suppose an open unit candidate U' contains K_B . The origin-hull identity puts O in it, and its two boundary anchors give positive traces on both $e_{5,0}$ and $e_{0,1}$. Thus the candidate is CE2 in the common-vertex normalization. Recompute $R, W, E, \eta, P, \alpha, \delta$ from this candidate and put

$$p = W - \alpha, \quad q = R - \delta, \quad c_* = c_{\max}(p, q).$$

Strict containment of the two anchors yields $B_5 > p, A_1 > q$. The four intervening gap-free edges and nonsupercriticality give

$$A_i > q, \quad B_i > p \quad (1 \leq i \leq 5).$$

Every local contributor on r_2 or r_4 belongs to this path, and common-pair comparison gives $\gamma_2, \gamma_4 \leq c_*$. The exact signed CE2 inequality proved below in theorem D.19 is

$$c_* < 1 - \min\{\alpha, \delta\}.$$

If $\delta \leq \alpha$, then $\|\hat{P}_2\| > \delta$, although the candidate exit on r_2 is δ . Otherwise $\|\hat{P}_4\| > \alpha$ contradicts its exit on r_4 . The same fixed points exclude every candidate. $\square$

Lemma D.3 (Fixed six-point transverse enclosure). *The set K_C in theorem 5.14 has $\Lambda(K_C) \geq 1$ for arbitrary V types on the nonsupercritical path.*

Proof. Suppose an open unit candidate U' contains K_C . Its origin is implicit in the witness hull, and M_0 fixes the unique-midpoint normalization. Write its signed parameters as $R, W, E, \eta, P, \alpha, \delta$ and put

$$X = R - \delta, \quad Q = \frac{\eta + \alpha + \delta}{2R}.$$

The two gap endpoints give $B_0 > 2Q$ and $A_1 > X$. Endpoint diameter yields

$$A_0 \leq M_0(B_0) < M_0(2Q) < 1 - Q.$$

The five gap-free edges and nonsupercriticality imply

$$(61) \quad A_1 < A_2 < A_3 < A_4 < A_5 < A_0, \quad A_i > X, \quad B_i > Q \quad (1 \leq i \leq 5).$$

The pair (X, Q) is positive and has sum below one because it is strictly dominated by a nonsupercritical actual pair.

CE2 candidates. Use the selected high-radial root

$$e(d) = \frac{1-d}{2} \left(1 - \sqrt{4(1-d)^2 - 3} \right).$$

The strict CE2 center inequalities give

$$(62) \quad X > e(\alpha), \quad \min\{e(\alpha), e(\delta)\} < Q.$$

For completeness, $WX > \eta + \alpha$, and testing the low-root equation at this bound reduces the first assertion to

$$\pi(t) = (\eta + t)(1 - 2t) - 2Wt > 0 \quad (0 \leq t \leq P).$$

The polynomial is concave, $\pi(0) = \eta > 0$, and $\pi(P) = \eta(\eta + 2ER^2) > 0$. For the second assertion put $T = \alpha + \delta$. The weighted sum of the two strict center inequalities gives $T < \eta/E$. With $d = \min\{\alpha, \delta\}$,

$$\min\{e(\alpha), e(\delta)\} < \frac{2d}{1 - 2d} \leq \frac{T}{1 - T} < Q.$$

The final comparison follows from concavity of $\chi(T) = (\eta + T)(1 - T) - 2RT$, positive at zero and at η/E , where it has value $\eta(E - R)^2/E^2$.

If $e(\alpha) < Q$, both coordinates of (X, Q) exceed $e(\alpha)$. The high-radial threshold therefore gives $c_{\max}(X, Q) < 1 - \alpha$. Otherwise $e(\delta) < Q \leq e(\alpha) < X$ and it gives $c_{\max}(X, Q) < 1 - \delta$. Common-pair domination and equation (61) bound all local contributors on r_4 in the first case and r_2 in the second. Thus the candidate misses the corresponding fixed total endpoint.

CE1 candidates. Now $R\alpha + \delta \geq P$ and $\alpha + W\delta < P$. Set $m = \alpha/R$; the transverse exits are α, m, δ on r_4, r_3, r_2 . Combining the two signed inequalities gives $E^2\alpha < RP$, so $m < \eta/E$. Also

$$X > \frac{R}{1 + E} > \frac{\eta}{E} > m, \quad 2R(Q - m) = \eta + \delta - \alpha > 0.$$

Here $E > W$ proves the middle inequality, and $\alpha < P = E\eta < \eta$ proves the last sign. Consequently

$$(63) \quad \min\{X, Q\} > m > \alpha.$$

The neighboring capacities on r_4, r_3 are therefore below the candidate's required demands. Fixed-endpoint containment and theorem D.1 recover

$$(64) \quad C_4 > 1 - \alpha, \quad C_3 > 1 - m.$$

Invoke only the first T_4 step of the return proof, through equation (79). This step uses equation (61), the upper squeeze, and $C_4 > 1 - \alpha$, but not a radial hypothesis at T_2 . A nonselected branch already contradicts the boundary inequalities at T_1 . On the surviving selected branch,

$$B_3 > L_1, \quad L_1 = (2 - 4\alpha)Q - (1 - 4\alpha)\alpha.$$

The identity

$$(65) \quad R(L_1 - \delta) = (1 - 2\alpha)\eta + (W - 2\alpha)(\alpha + \delta) + 4R\alpha^2 > 0$$

follows by expansion, with every sign justified by $0 < \alpha < P < \min\{1/2, W/2\}$. Hence $B_3 > \delta$. Backward path monotonicity gives $B_1 > B_3 > \delta$, and $A_1, A_3 > X > \delta$. Both neighbors of r_2 have capacities below $1 - \delta$. Candidate containment now recovers

$$(66) \quad C_2 > 1 - \delta.$$

All three own-radial hypotheses of theorem D.23 have been established before the full return is used. That return gives $A_0 > 1 - Q$, contradicting the upper squeeze.

Its selected-component conditions remain in force; in particular, $\delta < 1/10$ is not asserted on the bare CE1 signed domain. Both candidate classes are excluded. $\square$

Lemma D.4 (Fixed four-point supported-rescuer proof). *The geometric statement of theorem 5.18 holds.*

Proof. If $a = 0$ or $\beta = 0$, the set contains O and a hexagon vertex at distance one, so it cannot lie in an open unit equilateral triangle. Otherwise suppose a candidate contains the four points. The convex combination in the theorem's body proof puts $(\varepsilon/s)V_0$ in it, and $\varepsilon/s \geq 1/2$ puts M_0 in it by convexity with O . The candidate has a positive boundary trace and unique midpoint M_0 . Use the left-active reflected signed normal form when needed; a positive right-edge trace is not an extra hypothesis. The left trace has endpoints $k/W, R + \alpha$, and the exit on r_1 is δ/R , with $k = \eta + \alpha + \delta$ and $\eta, \alpha, \delta > 0$. Containment of $Y(a)$ and εV_1 gives

$$Wa > k, \quad \delta > R\varepsilon,$$

so $\alpha < a - Rs$ and $R < a/s$. Since $s \leq 1$,

$$R + \alpha < a + R(1 - s) \leq a/s.$$

Containment of $Y(1 - \beta)$ would require $1 - \beta < R + \alpha$, whereas $\beta \leq \varepsilon/s$ gives $1 - \beta \geq a/s$. This is impossible. $\square$

Using the geometric lemma on an actual supported trace. Set the lemma's parameters to $a = A_0$, $\beta = B_5$ and $\varepsilon = 1 - u$. The local T3-like or Vd1 certificate gives

$$A_0 + \varepsilon \leq 1, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M_c^{\text{sup}}.$$

The strict-supercritical envelope and path monotonicity give $B_5 \leq B_1 < M_c^{\text{sup}}$. Since $M_c^{\text{sup}} \geq 1/2$, these imply $A_0 \leq \varepsilon$ and $B_5 < \varepsilon/(A_0 + \varepsilon)$. They also imply $A_0 \leq 1 - M_c^{\text{sup}}$ and hence $B_5 < 1 - A_0$. Thus the two boundary points are actual left-gap endpoints, and the other noncentral point is the already forced supported endpoint. This constructs $K_D \subset U_C$ for both gap ranks; the four-point lemma is its universal candidate obstruction.

Scope of the retained calculations below. The local admissible cells, CE1 selected-return calculations, and the two rescuer chart certificates below remain inputs to the fixed-witness proof. The older one-Vd radial-separation estimates and the boundary-tail budget are retained as alternative quantitative results, not as additional active finite-enclosure cases. In particular, an actual-center separation is not itself renamed as an unrestricted enclosure inequality.

D.2. Exact gap endpoints.

Lemma D.5 (Actual open-trace endpoints). *For every actual gap J_i ,*

$$J_i = X_i([B_i, 1 - A_{i+1}]).$$

The formula includes the singleton case $B_i = 1 - A_{i+1}$.

Proof. The definitions of A_{i+1} and B_i , together with convexity, give the closed traces $X_i([0, B_i])$ and $X_i([1 - A_{i+1}, 1])$. Because U_i, U_{i+1} are open, their edge traces are respectively $X_i([0, B_i))$ and $X_i((1 - A_{i+1}, 1])$. Taking the set difference in the definition of J_i gives the displayed closed interval, including both endpoints. $\square$

D.3. Common-pair reach adapters.

Lemma D.6 (Solved one-gap common-pair adapter). *The reaches defined from row A in theorem D.15 satisfy its displayed coordinatewise alternatives.*

Proof. On the five gap-free edges choose

$$x_i \in (1 - A_{i+1}, B_i) \quad (1 \leq i \leq 5).$$

Since $N_+ = 0$, one has $B_i \leq 1 - A_i$ for every i . The exact gap endpoints and the five handoffs therefore give

$$\ell \leq x_5 \leq x_4 \leq x_3 \leq x_2 \leq x_1 \leq r.$$

With $p = 1 - r = A_1$ and $q = \ell = B_0$, this yields

$$p \leq A_i, \quad q \leq B_i \quad (i \in \mathbb{Z}/6\mathbb{Z}).$$

This is stronger than the alternative asserted in the body and retains the singleton case $\ell = r$. $\square$

Lemma D.7 (Two-gap common-pair adapter). *In the B configuration, $T_1, \dots, T_5$ are nonsupercritical by hypothesis. For any candidate containing the fixed K_B , its signed CE2 pair $p = W - \alpha$, $q = R - \delta$ satisfies*

$$A_i > q, \quad B_i > p \quad (1 \leq i \leq 5).$$

Every local contribution on r_2 or r_4 is at most $c_{\max}(p, q)$.

Proof. The origin-hull identity and the two boundary anchors normalize the candidate as CE2. Their strict containment gives $A_1 > q$ and $B_5 > p$. The four intervening gap-free edges, combined with nonsupercriticality, give $A_i \geq A_1$ and $B_i \geq B_5$ on this path. Coordinatewise antitonicity and common-pair domination bound the own and both possible neighboring capacities. No T3-like count identity is used and no nonsupercriticality is assumed at T_0 . $\square$

D.4. Vd1 two-chart replacement. The replacement below uses two different vertex charts. It preserves the five skeleton pieces that can change, then invokes the all-nonsupercritical obstruction without assuming preservation of the input gap rank.

Lemma D.8 (Half-square admissibility). *If $(x, y, z) \in \mathcal{A}$ satisfies*

$$x \geq \frac{1}{2}, \quad 0 \leq z \leq \frac{1}{2},$$

then

$$y \leq 1 - z.$$

Proof. Suppose $y > 1 - z$. Then $x + y > 1$, so the selected supercritical cell of theorem A.9 applies. In the ordered half $x \leq y$ its necessary polynomial is

$$F(x, y, z) = (x^2 - 1)z^2 + (2xy^2 + y)z + y^4 - y^2 \leq 0.$$

It is nondecreasing in x , its derivative in y is positive for $y \geq 1 - z$, and

$$F\left(\frac{1}{2}, 1 - z, z\right) = \frac{z^2(2z - 5)(2z - 1)}{4} \geq 0.$$

This is a contradiction. In the reflected half $y < x$, apply the same argument to $F(y, x, z)$. At the joining corner,

$$F(1 - z, 1 - z, z) = z(1 - 2z) \geq 0,$$

so no selected feasible point is lost at equality. $\square$

Lemma D.9 (Two-vertex replacement from scalar margins). *Consider two adjacent V roles. Only the five pieces*

$$e_{5,0}, \quad e_{0,1}, \quad e_{1,2}, \quad r_0, \quad r_1$$

may be affected by their replacement, and the shared edge is center-free. Let $a, c, B, r \in [0, 1]$ be the boundary reach on $e_{5,0}$, the own-radial reach on r_0 , the boundary reach on $e_{1,2}$, and the radial demand on r_1 needed to overlap the unchanged center interval. If

$$a < \frac{1}{2}, \quad a + c < 1, \quad a + B < 1, \quad r < \max\{1 - a, 1 - B\},$$

there are two open unit equilateral replacements preserving these five skeleton pieces, with actual boundary sums below one and no positive adjacent support. No center-class formula is assumed.

Proof. Choose first $p_2 \in (a, 1 - B)$ with $\max\{p_2, 1 - p_2\} > r$. This is possible because the supremum on that interval is $\max\{1 - a, 1 - B\}$. Next choose $a < p_1 < \min\{p_2, 1/2, 1 - c\}$ and

$$0 < \varepsilon < \min\{p_1 - a, p_2 - p_1, 1 - B - p_2, 1 - p_1 - c, \max(p_2, 1 - p_2) - r\}.$$

All five margins are positive. Use the separate vertex charts

$$X_0(x, y) = V_0 + x(V_5 - V_0) + y(V_1 - V_0), \quad X_1(x, y) = V_1 + x(V_0 - V_1) + y(V_2 - V_1).$$

Both have metric $x^2 + y^2 - xy$. Define

$$\Delta_p^- = \text{conv}\{(0, 1 - p), (1, 1 - p), (0, -p)\}, \quad D_{p,\varepsilon}^- = \text{int}(\Delta_p^-) + (-\varepsilon, 0) \quad (p \leq 1/2),$$

$$\Delta_p^+ = \text{conv}\{(p, 0), (p, 1), (p - 1, 0)\}, \quad D_{p,\varepsilon}^+ = \text{int}(\Delta_p^+) + (0, -\varepsilon) \quad (p > 1/2).$$

Their edges have unit length. The minus half-planes are $x > -\varepsilon$, $y < 1 - p$, $y > x + \varepsilon - p$; the plus half-planes are $y > -\varepsilon$, $x < p$, $x > y + \varepsilon + p - 1$. Substitution on the axes and diagonal gives their actual reaches:

	A	B	C
$D_{p,\varepsilon}^-$	$p - \varepsilon$	$1 - p$	$1 - p$
$D_{p,\varepsilon}^+$	p	$1 - p - \varepsilon$	p

Both contain the chart origin. On the adjacent support lines $(z, 1)$ and $(1, z)$ the half-planes are incompatible for $0 \leq z \leq 1$, so neither has positive adjacent support. Set $U'_0 = X_0(D_{p_1,\varepsilon}^-)$ and use $U'_1 = X_1(D_{p_2,\varepsilon}^-)$ for $p_2 \leq 1/2$, or $U'_1 = X_1(D_{p_2,\varepsilon}^+)$ otherwise. The two charts and this split cannot be identified. Both actual boundary sums are $1 - \varepsilon$.

The first replacement reaches $p_1 - \varepsilon > a$ on $e_{5,0}$ and $1 - p_1 > c$ on r_0 . On the shared edge the reaches sum to at least

$$(1 - p_1) + (p_2 - \varepsilon) > 1,$$

so the open traces overlap. The second reaches more than B on $e_{1,2}$ and $\max(p_2, 1 - p_2) > r$ on r_1 . Thus all five pieces remain covered; all other pieces are unchanged. $\square$

Proposition D.10 (Vd1 input to the two-vertex replacement). *Assume a skeleton cover has a unique Vd1 role T_0 supplying M_1 , a unique supercritical role T_1 , and all other V roles are nonsupercritical Vd0. Suppose $e_{0,1}$ is center-free and no other V role has positive trace on r_0 or r_1 . The scalar replacement applies in CE1 and CE2 and yields a skeleton cover with $N'_+ = 0$.*

Proof. Let A_0, B_0, C_0 be the actual incoming, outgoing, and own-radial reaches of the Vd1 role. Its corner normal form supplies $\vartheta \geq 1$ and

$$\omega := \sqrt{\vartheta^2 + \vartheta + 1}$$

such that

$$C_0 = \frac{\omega - A_0 - \vartheta B_0}{\vartheta + 1}, \quad \lambda = \frac{\vartheta(1 - B_0)}{\vartheta + 1},$$

$$u_{\text{adj}} = \frac{\omega - A_0 - \vartheta B_0 - 1}{\vartheta}.$$

The strict midpoint and one-sided-support conditions give

$$A_0 + C_0 < 1, \quad A_0 < \lambda \leq \frac{1}{2},$$

$$B_0 < \frac{1}{2}, \quad u_{\text{adj}} < 1 - A_0.$$

For the first inequality, evaluation on the closed midpoint face reduces strictness to

$$(2\vartheta^2 + 4\vartheta + 1)^2 - 4(\vartheta + 1)^2(\vartheta^2 + \vartheta + 1) = 4\vartheta^3 + 4\vartheta^2 - 4\vartheta - 3 > 0.$$

The remaining inequalities follow directly from the displayed endpoint formulas and the strict midpoint inequalities.

The center-free shared edge and the supported interval give

$$A_1 \geq 1 - B_0 > \frac{1}{2}, \quad C_1 \geq \lambda$$

for the actual reaches of T_1 . The selected supercritical component has $C_1 \leq 1/2$, so theorem D.8 yields

$$B_1 \leq 1 - C_1 \leq 1 - \lambda.$$

Since $A_0 < \lambda$,

$$A_0 + B_1 < 1.$$

Define the C-triangle reach on r_1 , measured from O , by

$$d_1^C := \max\{x \in [0, 1] : xV_1 \in T_C\}, \quad c_1^{\text{req}} := 1 - d_1^C.$$

Only U_C, U_1 , and the adjacent trace of U_0 can contribute a positive interval to r_1 . Open skeleton coverage gives

$$c_1^{\text{req}} < \max\{C_1, u_{\text{adj}}\} \leq \max\{1 - B_1, 1 - A_0\}.$$

Apply theorem D.9 with

$$(a, c, B, r) = (A_0, C_0, B_1, c_1^{\text{req}}).$$

All four scalar conditions were proved above. The four untouched roles remain nonsupercritical Vd0, while the two new actual boundary sums are $1 - \varepsilon < 1$. The resulting skeleton cover contradicts N0. The original C triangle is fixed; its class and the input gap rank were not used in the construction. This operation does not claim to preserve coverage of the interior of H . $\square$

D.5. Direct neighboring-ray domination. The capacities keep their variational definitions. The following proof establishes the bound needed by all active witnesses without evaluating the cubic plateau and radical branches of the full function.

Lemma D.11 (Support-cell rotation). *A least equilateral enclosure of a nondegenerate polygon can be chosen with one side containing a polygon edge. For a centered disk together with finitely many points, it suffices to test point–point and point–disk ties, or a disk-only orientation.*

Proof. By theorem A.4, the side is the three-term support sum divided by h . On a polygon cell with fixed support vertices the sum is $g(\theta) = a \cos \theta + b \sin \theta > 0$, so $g'' = -g < 0$ and no interior minimum occurs. In the disk case a cell with q disk supports has sum $q\rho + \langle v, n(\theta) \rangle$. If a point is active its projection exceeds $\rho \geq 0$, and the inner product is positive, again giving a negative second derivative. A disk-only cell has the constant global minimum 3ρ . Each application must still identify its exposed ties. $\square$

Lemma D.12 (Sharp diagonal neighboring capacity). *For $0 \leq m \leq 1/2$, one has $C_+(m, m) = C_-(m, m) = 1 - m$.*

Proof. Use oblique coordinates with metric $x^2 + y^2 - xy$. The four points are $P = (0, 0)$, $A = (m, 0)$, $B = (0, m)$, and $Z = (c, 1)$. The case $m = 0$ is immediate. If $0 < m \leq 1/2$ and $c > 1 - m$, their hull has cyclic order P, A, Z, B . The preceding lemma reduces the enclosure to its four outward edge normals. At PA and BP , the side lengths are $c + m$ and $1 + m$, both greater than one.

Embed (x, y) as $(x - y/2, hy)$. At the other two edges take unnormalized normals $n_A = (h, 1/2 - c + m)$ and $n_B = (h(m - 1), m/2 + c - 1/2)$. Put

$$\begin{aligned} D_A &= (c - m)^2 - (c - m) + 1, & N_A &= m + c^2 - cm - c + 1, \\ D_B &= c^2 + cm - c + m^2 - 2m + 1, & N_B &= cm + c^2 - c - m + 1. \end{aligned}$$

Selecting the contained points (A, Z, P) in the three normal directions at n_A , and (B, P, Z) at n_B , gives support lower bounds $L_{AZ} \geq N_A/\sqrt{D_A}$ and $L_{ZB} \geq N_B/\sqrt{D_B}$. No active-support selector is assumed. With $q = c + m - 1 > 0$,

$$\begin{aligned} N_A^2 - D_A &= q^2((q + 1 - 3m)^2 + 4m^2 - 2m + 1) \\ &\quad + q(1 - 2m)(6m^2 - 2m + 1) + m^2(1 - 2m)^2 > 0, \\ N_B^2 - D_B &= q^4 + (2 - 2m)q^3 + (m^2 - 4m + 2)q^2 \\ &\quad + (1 - m)(1 - 2m)q > 0. \end{aligned}$$

Here $4m^2 - 2m + 1 \geq 3/4$, $6m^2 - 2m + 1 > 0$, and $m^2 - 4m + 2 \geq 1/4$. The numerators are positive: $N_A \geq (1 + m)(3 - m)/4$ and $N_B = 1 - m + (1 - m)q + q^2$. Thus every caliper requires side greater than one. Equality is realized by the unit triangle $\text{conv}\{(-m, 0), (1 - m, 0), (1 - m, 1)\}$. Reflection gives C_- . $\square$

Lemma D.13 (Type-aware radial calculation). *The points $D_i = (1 - \Gamma_i)V_i$ of theorem 5.4 are missed by every open V role.*

Proof. The definitions of the variational capacities give $\gamma_i \leq \Gamma_i$. Apply the actual-or-bounded frontier in theorem D.1, including its $\Gamma_i = 1$ endpoint. The actual support restrictions determine which terms may occur. $\square$

Lemma D.14 (Solved common-pair comparison). *The inequalities in theorem 5.5 hold.*

Proof. Put $m = \min(p, q)$. By convexity, any triangle containing the p, q boundary anchors also contains the shorter m, m anchors. Hence theorem D.12 gives $C_{\pm}(p, q) \leq 1 - m$. If $q = m$, the unit triangle used for equality contains $(p, 0)$, $(0, q)$, and $(1 - q, 1 - q)$ because $p \leq 1 - q$. The last point is on the own ray, so $c_{\max}(p, q) \geq 1 - q$. Reflect for $p = m$. Increasing an anchor shrinks the feasible family, giving antitonicity. $\square$

D.6. Optional alternative A: the complementary-gap disk.

Lemma D.15 (One-gap common-pair adapter). *Suppose $N_+ = 0$ and the unique actual gap is $J_0 = X_0([B_0, 1 - A_1])$. Put $p = A_1$, $q = B_0$. Then $p, q > 0$, $p + q \leq 1$, and $A_i \geq p$, $B_i \geq q$ for all i , without a V-type restriction.*

Proof. The five gap-free edges give $B_i + A_{i+1} > 1$, while $A_i + B_i \leq 1$. Thus

$$A_1 < A_2 < A_3 < A_4 < A_5 < A_0, \quad B_0 < B_5 < B_4 < B_3 < B_2 < B_1.$$

The actual gap gives $p + q \leq 1$. These inequalities prove the assertion. $\square$

Put

$$\begin{aligned} c_A &:= c_{\max}(A_1, B_0), & D_i^A &:= (1 - c_A)V_i, \\ G_A &:= \begin{cases} X_0(B_0), & B_0 \leq A_1, \\ X_0(1 - A_1), & A_1 < B_0, \end{cases} & K_A &:= \{D_0^A, \dots, D_5^A, G_A\}. \end{aligned}$$

The disk used in this recipe is exactly

$$\mathcal{D}_A := \left\{ x \in \mathbb{R}^2 : \|x\| \leq \frac{\sqrt{3}}{2} (1 - c_{\max}(A_1, B_0)) \right\}.$$

It is the inscribed disk of the regular hexagon formed by the six radial points. It need not itself be missed pointwise by the V triangles: its containment in U_C follows from convexity of the six forced points.

Theorem D.16 (Complementary-gap finite enclosure). *For $p, q > 0$, $p + q \leq 1$, $c_* = c_{\max}(p, q)$, and $J(p, q) = X_0([q, 1 - p])$, let G be its farther endpoint from O . Then*

$$\Lambda(\mathbb{B}(O, h(1 - c_*)) \cup \{G\}) \geq 1.$$

Consequently the all-nonsupercritical one-gap configuration forces $K_A \subset U_C$ and satisfies $\Lambda(K_A) \geq 1$.

Proof. The disk-point calculation in theorems D.17 and D.18 uses only the farther endpoint of the actual gap. Common-pair forcing puts the six D_i^A in U_C , and their hull contains the stated disk. The gap endpoint is center-forced. Monotonicity of Λ under convex hulls therefore proves the finite-set assertion. $\square$

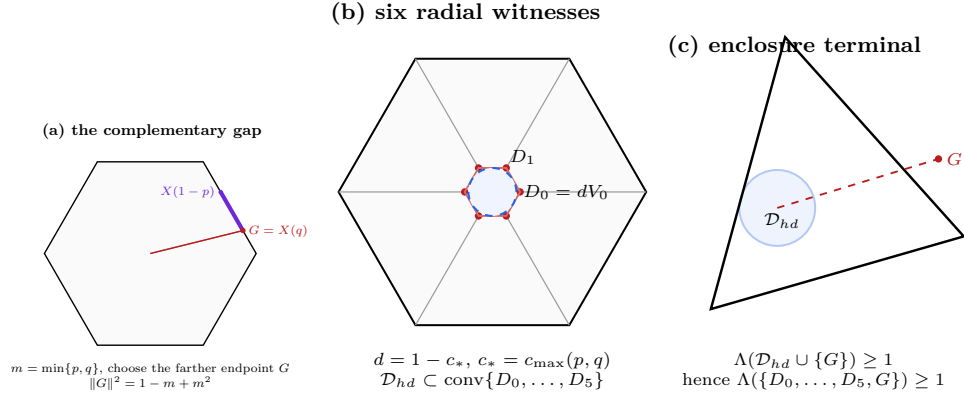
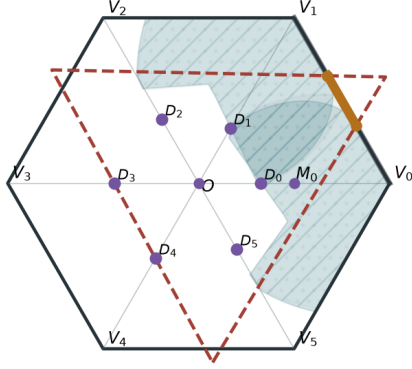
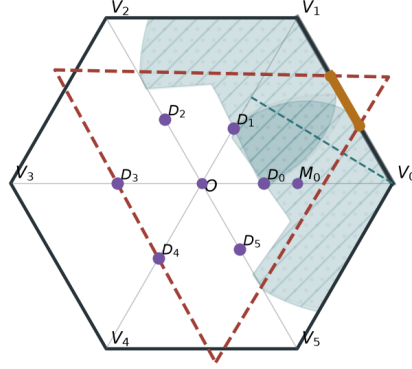


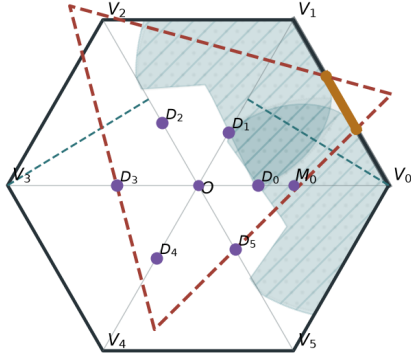
FIGURE 9. The complementary-gap enclosure mechanism. The farther endpoint G of $J(p, q)$ is center-forced. Common-pair domination produces the six center-forced points $D_i = (1 - c_*)V_i$, whose convex hull contains the disk $\mathcal{D}_{h(1-c_*)}$. The disk-plus-point formula then forces enclosing side length at least one. The disk is not an additional witness; it is contained in the convex hull of the six finite radial witnesses.



All Vd0.
 $N_{\text{gap}} = 1$
 $N_+ = 0$
 $(d, t) = (0, 0)$



One T3-like role.
 $N_{\text{gap}} = 1$
 $N_+ = 0$
 $(d, t) = (0, 1)$



Two T3-like roles.
 $N_{\text{gap}} = 1$
 $N_+ = 0$
 $(d, t) = (0, 2)$

FIGURE 10. Case A: sampled complementary-gap data.

D.7. Disk and complementary-gap optimization. Put $\mathcal{D}_\eta := \mathbb{B}(O, \eta)$.

Lemma D.17 (Disk-plus-point formula). *If $\|G\| = \rho$, then*

$$\Lambda(\mathcal{D}_\eta \cup \{G\}) = \begin{cases} 3\eta/h, & \rho \leq 2\eta, \\ \frac{3\eta + \sqrt{3(\rho^2 - \eta^2)}}{2h}, & \rho > 2\eta. \end{cases}$$

Proof. Apply theorem D.26. In the disk-only cell the support sum is 3η . Otherwise a point-disk tie gives the second active projection

$$\frac{-\eta + \sqrt{3(\rho^2 - \eta^2)}}{2},$$

which yields the second branch. $\square$

(b) $\rho > 2\eta$

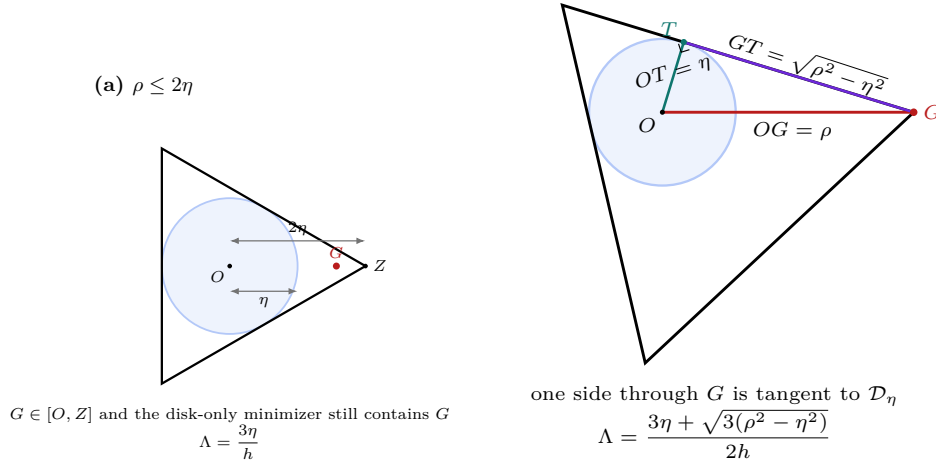


FIGURE 11. The two minimizing configurations in the disk-plus-point formula. For $\rho \leq 2\eta$, the centered equilateral triangle tangent to $\mathcal{D}_\eta$ already contains G . For $\rho > 2\eta$, the point G becomes a vertex of the minimizing triangle and one incident side is tangent to the disk. The tangent length $GT = \sqrt{\rho^2 - \eta^2}$ is the geometric source of the radical term.

Lemma D.18 (Solved complementary-gap optimization). *The set $\mathcal{D}(p, q) \cup J(p, q)$ of theorem D.16 has enclosure number at least one.*

Proof. Put $c = c_{\max}(p, q)$, $m = \min\{p, q\}$, and $d = 1 - c$. The farther endpoint G of $J(p, q)$ satisfies

$$\|G\|^2 = 1 - m + m^2.$$

By theorem D.17, the nontrivial branch reduces to

$$3c(1 - c) \geq m(1 - m).$$

Common-pair domination gives $c \geq 1 - m \geq 1/2$. For $c \leq h$, the left side is at least $3h(1 - h) > 1/4$. For $c > h$, the exact local support inequalities from theorem A.9 give

$$m \leq c \left(\frac{1}{2} - \sqrt{c^2 - \frac{3}{4}} \right).$$

At this upper bound,

$$\begin{aligned} & 3c(1 - c) - c \left(\frac{1}{2} - \sqrt{c^2 - \frac{3}{4}} \right) \left[1 - c \left(\frac{1}{2} - \sqrt{c^2 - \frac{3}{4}} \right) \right] \\ &= \frac{c(1 - c)}{2} \left(5 - 2c - 2c^2 + 2\sqrt{c^2 - \frac{3}{4}} \right) \geq 0. \end{aligned}$$

The whole interval $J(p, q)$ is contained in every convex enclosure of its farther endpoint and the centered disk. This proves the claim, including the singleton endpoint. $\square$

D.8. CE2 short-ray optimization.

Lemma D.19 (Solved CE2 short-ray optimization). *Under the hypotheses of theorem 5.10,*

$$c_{\max}(p, q) < 1 - e, \quad e := \min\{\alpha, \delta\},$$

and consequently $\{D_2, D_4\} \not\subseteq T_C$.

Proof. In the signed CE2 chart of Appendix A,

$$p = W - \alpha, \quad q = R - \delta, \quad M = \max\{R, W\}.$$

The two strict CE2 inequalities imply $P > (1 + M)e$. Since $E^2 = 1 - M + M^2$,

$$(1 + M)^2 - (3M - 1)^2 E^2 = 3M(1 - M)(3M^2 - 2M + 3) > 0.$$

Hence $\eta = P/E > (3M - 1)e$. The identity $RW = \eta + P$ gives

$$Wq > \eta + e > 3Me, \quad Rp > \eta + e > 3Me.$$

Thus

$$3e < p, q < 1 - 5e, \quad e < \frac{P}{1 + M} \leq \frac{2\sqrt{3} - 3}{6} < \frac{1}{12}.$$

Set $c = 1 - e$ and $f_c(t) = c^4 - c^2 + ct - t^2$. This function is concave, and

$$f_c(3e) = e(1 - 7e - 4e^2 + e^3) > 0,$$

$$f_c(1 - 5e) = e(2 - 15e - 4e^2 + e^3) > 0.$$

Therefore $f_c(p), f_c(q) > 0$. Since $c > p + q$ and $p + c, q + c > 1$, all four exact support-contact side lengths at radial demand c exceed one. Hence $c_{\max}(p, q) < 1 - e$.

If $e = \delta$, the forced point on r_2 lies strictly beyond the C exit; if $e = \alpha$, the forced point on r_4 does. Endpoint strictness therefore gives the asserted noncontainment. $\square$

D.9. High-radial and selected-chord calculations. The zero-radial diameter envelope is

$$(67) \quad M_0(x) = \frac{-x + \sqrt{4 - 3x^2}}{2}.$$

For

$$0 < e < 1 - \frac{\sqrt{3}}{2},$$

define the selected low root

$$(68) \quad \epsilon(e) = \frac{1-e}{2} \left(1 - \sqrt{4(1-e)^2 - 3} \right).$$

It satisfies

$$(69) \quad e < \epsilon(e), \quad \epsilon(e) < \frac{2e}{1-2e},$$

and, for $e \leq 1/8$,

$$(70) \quad \epsilon(e) \leq 2e + 5e^2.$$

These follow by substituting the proposed upper bounds in the selected quadratic and checking its sign.

Lemma D.20 (Direct high-radial threshold). *Let a nonsupercritical V triangle have incoming boundary reach A and outgoing boundary reach B . If*

$$C \geq 1 - e, \quad A > \epsilon(e),$$

then

$$\boxed{B \leq \epsilon(e)}.$$

Proof. The condition $C > 1/2$ forces $A + B \leq 1$. In the exact cells of Proposition A.9, the only selected component with $A > \epsilon(e)$ has $B \leq \epsilon(e)$; equivalently the two roots of the selected T -cell quadratic separate the feasible fiber. The selector $c \leq 2 \max\{A, B\}$ removes the formal high root. $\square$

For $0 \leq c \leq 1/2$, the strict-supercritical outgoing supremum is

$$M_c^{\sup} = \frac{c + \sqrt{c^2 - 8c + 4}}{2}.$$

It is not attained. Thus an actual strict-supercritical V triangle with own-radial reach at least c has outgoing boundary reach strictly below $M_c^{\sup}$.

Lemma D.21 (Selected-arc chord bounds). *Let $0 < e < 1/2$. On the selected Q_+ component for radial demand $1 - e$, let p be the incoming boundary reach and let q be the following incoming reach forced across a center-free edge. Then*

$$q - p = 1 - \sqrt{1 - x + x^2},$$

where

$$q = e + (1 - e)x, \quad p = e + (1 - e)x - 1 + \sqrt{1 - x + x^2}.$$

The selected arc is increasing and concave. In particular, on every interval used below,

$$q \geq p + \lambda(p - e)$$

for the chord slope λ joining the appropriate selected endpoints. The CE1 applications may take

$$\lambda > 1 - 4e \quad (e = \alpha), \quad \lambda > 1 - 5e \quad (e = \alpha/R).$$

Proof. The selected T -cell equation reduces, after putting $x = (q - e)/(1 - e)$, to

$$x(1 - x) = (q - p)(2 - q + p).$$

The selected component takes the smaller root, giving the displayed parameterization. Strict concavity follows by differentiation. A concave graph lies above its chords. For the first slope, equation (69) gives

$$\frac{e}{\epsilon(e) - e} > 1 - 4e.$$

For the second, use equation (70) when $e \leq 1/8$ and $\epsilon(e) < (1 - e)/2$ otherwise; the resulting slope is greater than $1 - 5e$ throughout $0 < e < 1 - \sqrt{3}/2$. $\square$

D.10. Exact direct output formulas. The preceding proofs refer to three local inequalities. We derive them here from Proposition A.9. This replaces the former presentation by iterated reach maps.

For a nonsupercritical triangle with incoming reach at least a and radial reach at least c , let

$$\mathcal{B}_c(a) = \max\{b : (a, b, c) \text{ is admissible and } a + b \leq 1\}.$$

This is an outgoing capacity for one triangle; it is never composed in the proof.

For $1/2 < c \leq \sqrt{3}/2$, put

$$h(c) = \frac{2c}{1 + \sqrt{16c^2 - 3}}.$$

For $\sqrt{3}/2 < c < 1$, put

$$\ell(c) = \frac{c}{2} \left(1 - \sqrt{4c^2 - 3}\right), \quad r(c) = c - \ell(c).$$

Also set

$$Q_-(a, c) = \frac{\sqrt{a^2 - ac + c^2}}{c} - a,$$

$$Q_+(a, c) = \frac{2(1 - a^2)c^2}{2ac^2 + c + \sqrt{(2ac^2 + c)^2 - 4(1 - c^2)(1 - a^2)c^2}}.$$

Proposition D.22 (Four direct high-radial outputs). *For $1/2 < c \leq \sqrt{3}/2$,*

$$\mathcal{B}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a < h(c), \\ h(c), & a = h(c), \\ Q_-(a, c), & h(c) < a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

For $\sqrt{3}/2 < c < 1$,

$$\mathcal{B}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a \leq \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ Q_-(a, c), & r(c) \leq a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

At $a = \ell(c)$, the selected value is $r(c)$; immediately to the right it is $\ell(c)$.

Proof. Since $c > 1/2$, any containing V triangle contains the radial midpoint. Therefore $a + b \leq 1$, and the supercritical cell of Proposition A.9 is absent. On the cap $b = 1 - a$, the T -cell inequality becomes

$$M(c - M) \leq 0,$$

where $M = \max\{a, b\}$. Thus the linear cap is feasible exactly when $a \leq 1 - c$ or $a \geq c$.

Off the linear cap, a maximal point lies on $F_L = 0$ or $F_T = 0$. The selected L -cell equation has the two ordered roots $\ell(c), r(c)$; the selector $q \leq 0$ retains the smaller coordinate. The ordered halves of $F_T = 0$ give Q_+ and Q_- . Their equality boundary $a = b$ is $h(c)$ in the low-radial range, while their $q = 0$ boundaries are $(\ell(c), r(c))$ and $(r(c), \ell(c))$ in the high range. These conditions produce the two displayed formulas.

The formal other root in the Q_+ quadratic violates the geometric selector $c \leq 2 \max\{a, b\}$ and is discarded. At the high-range junction, the selected T -cell point has value $r(c)$, whereas the adjacent selected L -cell point has value $\ell(c)$; this is the genuine downward jump stated above. $\square$

D.10.1. *Vd corner margins.* In the exact Vd1/Vd2 corner form, let the boundary reaches be a, b , let $t > 0$, and put $d = \sqrt{t^2 + t + 1}$. The own-radial reach is

$$c_0 = \frac{d - a - tb}{t + 1}.$$

If the role has positive support on the forward adjacent arm, its far endpoint there is

$$u_+ = \frac{d - a - tb - 1}{t}.$$

Since $d < t + 1$,

$$c_0 < 1 - \frac{a + tb}{t + 1} \leq 1 - \min\{a, b\},$$

and

$$u_+ < 1 - b - \frac{a}{t} < 1 - b.$$

Thus lower boundary demands $a \geq A$, $b \geq H$ imply

$$c_0 < 1 - \min\{A, H\}, \quad u_+ < 1 - H.$$

Reflection supplies the backward adjacent-arm statements. These two one-line inequalities are exactly the radial separations used in the one-Vd placement proof.

D.11. CE1 reverse-path calculation. The CE1 part of the seven-point enclosure theorem uses five boundary handoffs and the three transverse radial reaches at T_4, T_3, T_2 .

Proposition D.23 (CE1 three-transverse return certificate). *Use the signed CE1 center variables*

$$0 < R < 1, \quad w = 1 - R, \quad E = \sqrt{1 - Rw}, \\ \eta = 1 - E, \quad P = E(1 - E),$$

and write

$$A = \alpha, \quad D = \delta.$$

Assume

$$(71) \quad A > 0, \quad D > 0, \quad A + wD < P, \quad D + RA \geq P.$$

Put

$$(72) \quad X = R - D, \quad m = \frac{A}{R}, \quad h_0 = \frac{\eta + A + D}{2R}.$$

Let T_0 be a V triangle and let $T_1, \dots, T_5$ be nonsupercritical V triangles of arbitrary types. All reaches below are actual reaches. No supercriticality hypothesis on T_0 is used in this local return implication. Assume

$$A_1 \geq X,$$

$$B_i + A_{i+1} \geq 1 \quad (1 \leq i \leq 4), \quad B_5 + A_0 \geq 1,$$

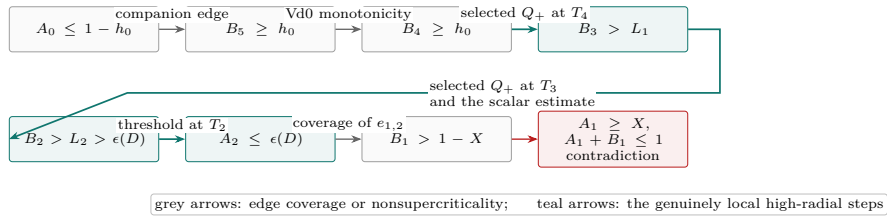
and

$$(73) \quad C_4 \geq 1 - A, \quad C_3 \geq 1 - m, \quad C_2 \geq 1 - D.$$

Then

$$(74) \quad \boxed{A_0 > 1 - h_0}.$$

CE1 reverse path: from a deficient terminal to the final contradiction



L_1 and L_2 are direct selected-arc lower bounds; no composed reach map is introduced.

FIGURE 12. Logical structure of the CE1 direct reverse-path certificate. The proof starts from the negation of the desired terminal bound at T_0 , moves through the actual boundary reaches, uses two selected-arc estimates and one high-radial threshold, and returns to T_1 with incompatible incoming and outgoing demands.

Proof. The sign difference in equation (71) gives

$$RD > wA,$$

so the center exit on r_3 is $m = A/R$. Moreover

$$(75) \quad 0 < A < P < \frac{1}{8}, \quad 0 < D < \frac{R}{2}, \quad 0 < m < \frac{w}{2} < \frac{1}{2}.$$

We first record

$$(76) \quad \epsilon(A) < X, \quad h_0 > A.$$

Indeed, $A + wD < P$ and $D = R - X$ give $A < wX - \eta$. Convexity of

$$wX - \frac{X}{2(1+X)}$$

on $R/2 < X < R$ gives

$$A < \frac{X}{2(1+X)}.$$

The selected-root bound equation (69) yields $\epsilon(A) < X$. Also

$$2R(h_0 - A) = \eta + D + (1 - 2R)A > 0;$$

when $R > 1/2$, use $A < P = \eta E$ for the last sign.

Suppose, contrary to equation (74), that

$$(77) \quad A_0 \leq 1 - h_0.$$

The assumed final handoff gives $B_5 \geq h_0$. Nonsupercriticality of T_5 and the handoff $B_4 + A_5 \geq 1$ imply

$$(78) \quad \boxed{B_4 \geq B_5 \geq h_0.}$$

The assumed transverse radial reach at T_4 is at least $1 - A$. Apply the exact local catalogue to the reflected boundary pair (B_4, A_4) . If the local point is not on the selected Q_+ component, the constant, Q_- , and linear components together with equation (76) give

$$1 - A_4 \geq 1 - \epsilon(A) > 1 - X.$$

The center-free edges then carry this lower bound backward to $B_1 > 1 - X$, contradicting $A_1 \geq X$ and $A_1 + B_1 \leq 1$.

Hence the T_4 state is selected Q_+ . The selected-chord estimate theorem D.21 and equation (78) give

$$(79) \quad B_3 > L_1, \quad L_1 = (2 - 4A)h_0 - (1 - 4A)A.$$

At this point the argument has used only the boundary handoffs, the upper-squeeze assumption, and the radial demand at T_4 . In particular, neither $C_2 \geq 1 - D$ nor $C_3 \geq 1 - m$ was used to obtain this first step. The fixed-witness proof uses this fact before recovering its third radial demand; the full return is invoked only afterwards.

The selected-state condition also forces

$$(80) \quad \boxed{X > \frac{1}{2}.}$$

To see this, selectedness gives

$$h_0 \leq \epsilon(A) < \frac{2A}{1-2A}.$$

If $X \leq 1/2$, the lower CE1 inequality gives

$$2Rh_0 = \eta + A + D \geq w(R + A).$$

Consequently

$$f_R(A) < 0, \quad f_R(z) = w(R + z)(1 - 2z) - 4Rz.$$

The quadratic is strictly concave and $f_R(0) > 0$. If $R \leq 1/2$, then

$$f_R(P) \geq \frac{1-E}{2}(6E^3 - 2E^2 - 5E + 2) > 0.$$

If $R > 1/2$, then $A < A_* = w/2 - \eta$ and

$$f_R(A_*) = \frac{1-E}{2}(E + 11R - 3ER - 5) > 0.$$

Concavity contradicts $f_R(A) < 0$ in both cases, proving equation (80). Thus

$$(81) \quad m < \frac{w}{2} < \frac{1}{4}.$$

If $B_3 \geq 1/2$, then $B_3 > 1 - X$ and the preceding immediate contradiction applies. Assume $B_3 < 1/2$. At T_3 the reflected incoming reach is $B_3 > h_0 > m$, while its assumed transverse radial reach is at least $1 - m$. If this state is not selected Q_+ , the exact local catalogue gives

$$B_2 \geq 1 - B_3 > \frac{1}{2} > 1 - X,$$

and the remaining center-free edge again contradicts T_1 .

Thus both T_4 and T_3 are selected Q_+ . A second use of theorem D.21 gives

$$B_2 > (2 - 5m)B_3 - (1 - 5m)m.$$

Since $2 - 5m > 0$, equation (79) yields

$$(82) \quad \boxed{B_2 > L_2, \quad L_2 = (2 - 5m)L_1 - (1 - 5m)m.}$$

We now prove the scalar estimate

$$(83) \quad \boxed{D < \frac{1}{10}, \quad L_2 > 2D + 3D^2 > \epsilon(D), \quad \epsilon(D) < X.}$$

The T_4 selected-state inequality and equation (70) give

$$(84) \quad D \leq D_h := A(4R - 1) + 10RA^2 - \eta.$$

If $D \geq 1/10$, then $A + wD < P$ gives

$$A < \bar{A} = \frac{w(5R - 1)}{10}.$$

The right side of equation (84) is increasing in A , while $\eta > Rw/2$, so

$$D < U(R) := \bar{A}(4R - 1) + 10R\bar{A}^2 - \frac{Rw}{2}.$$

Direct expansion gives

$$\frac{1}{10} - U(R) = -\frac{R}{10}P_4(R),$$

where

$$P_4(R) = 25R^4 - 60R^3 + 26R^2 + 22R - 14.$$

For $y = 2R - 1 \in (0, 1)$,

$$16P_4(R) = 25y^4 - 20y^3 - 106y^2 + 124y - 39 \leq -101y^2 + 124y - 39 < 0.$$

Hence $U(R) < 1/10$, a contradiction. This proves $D < 1/10$.

Put

$$\Psi(D) = L_2 - 2D - 3D^2, \quad J(D) = L_2 - \frac{23}{10}D.$$

Since $D < 1/10$,

$$\Psi(D) - J(D) = 3D \left(\frac{1}{10} - D \right) > 0.$$

For fixed R, A , differentiation of equation (82) gives

$$J'(D) = \frac{S(R, A)}{10R^2},$$

where

$$S(R, A) = 100A^2 - (40R + 50)A + R(20 - 23R).$$

The nonempty interval $P - RA \leq D \leq D_h$ implies

$$A > \frac{4Rw}{25R - 4} =: A_*.$$

On the relevant interval $\partial S / \partial A < -45$, and

$$S(R, A) < S(R, A_*) = -\frac{Rq_3(R)}{(25R - 4)^2},$$

where

$$q_3(R) = 8775R^3 - 14260R^2 + 7928R - 1120.$$

For $y = 2R - 1$,

$$8q_3(R) = 8775y^3 - 2195y^2 + 997y + 3007 > 0.$$

Thus J decreases with D , so $J(D) \geq J(D_h)$.

At $D = D_h$, one has $h_0 = 2A + 5A^2$. If $L_{2,h}$ is the corresponding value, then

$$L_{2,h} - A(1 + 4R) = \frac{A}{R^2}Q(R, A),$$

where

$$Q(R, A) = 100A^3R - 40A^2R^2 - 30A^2R + 12AR^2 - 15AR + 5A - 4R^3 + 5R^2 - R.$$

This polynomial is concave in A on $0 \leq A \leq Rw/2$. Its endpoint values are

$$Q(R, 0) = Rw(4R - 1) > 0$$

and

$$Q\left(R, \frac{Rw}{2}\right) = \frac{Rw}{2}(25R^5 - 30R^4 + 20R^3 - 3R^2 - 7R + 3) > 0.$$

The last positivity follows after putting $y = 2R - 1$:

$$32p(R) = 25y^5 + 65y^4 + 90y^3 + 106y^2 - 35y + 5 > 0.$$

Hence $L_{2,h} > A(1 + 4R)$.

Finally $A < P = \eta E$, $A < Rw/2$, and $1/E > 1 + Rw/2$ give

$$\frac{D_h}{A} < C_* := 4R - 2 + 5R^2w - \frac{Rw}{2}.$$

Moreover

$$1 + 4R - \frac{23}{10}C_* = \frac{230R^3 - 253R^2 - 81R + 112}{20} > 0.$$

Thus $J(D_h) > 0$, so

$$L_2 > 2D + 3D^2.$$

Substitution in the selected-root quadratic gives $\epsilon(D) < 2D + 3D^2$, and the argument proving $\epsilon(A) < X$ also gives $\epsilon(D) < X$. This proves equation (83).

The assumed transverse radial reach at T_2 is at least $1 - D$. Since $B_2 > \epsilon(D)$, theorem D.20, applied after reflection, gives

$$A_2 \leq \epsilon(D).$$

Coverage of $e_{1,2}$ therefore forces

$$B_1 \geq 1 - A_2 \geq 1 - \epsilon(D) > 1 - X,$$

contradicting $A_1 \geq X$ and $A_1 + B_1 \leq 1$. Hence equation (77) is impossible. $\square$

D.12. Transverse seven-point calculation.

Notation for retained own-endpoint calculations. In the older all-Vd0 specialization below put $\ell = B_0$, $r = 1 - A_1$, $P_i = (1 - C_i)V_i$, and

$$K_{\text{tr}} = \{O, M_0, X_0(\ell), X_0(r), P_2, P_3, P_4\}.$$

These auxiliary own endpoints are distinct from the total endpoints $\hat{P}_i$ of the active B/C theorems. Their Vd0-specific forcing is retained only for that specialization; the conditional local return below uses no V-type assumption once its own-radial demands are supplied.

Lemma D.24 (Endpoint upper squeeze). *If an open unit triangle U' contains K_{tr} , use its local signed-center variables and put*

$$X = R - \delta, \quad Q = \frac{\eta + \alpha + \delta}{2R}.$$

Then its incoming reach at V_0 satisfies

$$A_0(U') < 1 - Q.$$

Proof. The actual gap endpoints lie strictly inside the edge trace of U' . Writing its two corresponding closed reaches locally as $\hat{B}_0, \hat{A}_1$, one has

$$\hat{B}_0 = \ell > 2Q, \quad \hat{A}_1 = 1 - r > X.$$

The diameter inequality for the two boundary endpoints gives

$$A_0(U') \leq M_0(\hat{B}_0) < M_0(2Q) = -Q + \sqrt{1 - 3Q^2} < 1 - Q.$$

No radial point on r_0 occurs in this calculation. $\square$

Lemma D.25 (Solved CE2 transverse threshold). *For an open unit equilateral triangle U_C , put $T_C := \overline{U_C}$. Then*

$$N_E(T_C) = 2 \implies K_{\text{tr}} \not\subseteq U_C.$$

Proof. Nonsupercriticality and the five gap-free edges give

$$A_1 < A_2 < A_3 < A_4 < A_5 < A_0$$

for the hypothetical containing triangle. The endpoint squeeze therefore implies $B_2, B_4 > Q$. Containment of the actual radial endpoints P_2, P_4 gives

$$C_2 > 1 - \delta, \quad C_4 > 1 - \alpha.$$

The signed CE2 threshold calculations give

$$R - \delta > \epsilon(\alpha), \quad \min\{\epsilon(\alpha), \epsilon(\delta)\} < Q.$$

If $\epsilon(\alpha) < Q$, theorem D.20 at T_4 gives $B_4 < Q$; otherwise the same lemma at T_2 gives $B_2 < Q$. Both alternatives contradict the preceding strict lower bounds. $\square$

D.13. Disk, one-third, and supported-tail optimizations.

Proposition D.26 (Disk-finite-set calipers). *Let $P = \{p_1, \dots, p_m\}$, and let J denote rotation by $\pi/2$. Unless the disk is the active support in all three directions at a minimizing orientation, a minimizing normal belongs to the finite list*

$$\pm \frac{J(p_k - p_i)}{\|p_k - p_i\|} \quad (p_i \neq p_k)$$

or

$$\frac{\eta p_i \pm \sqrt{\|p_i\|^2 - \eta^2} J p_i}{\|p_i\|^2} \quad (\|p_i\| \geq \eta).$$

Thus one side of a least enclosing equilateral triangle either contains two points, or is tangent to the disk and contains one point.

Proof. Apply theorem D.11. A point-point tie has normal perpendicular to $p_k - p_i$; solving $\langle p_i, n \rangle = \eta$ with $\|n\| = 1$ gives the displayed point-disk normals. Rotation through 120 degrees identifies the three supporting directions. A disk-only cell contributes 3η and is the only additional alternative. $\square$

Lemma D.27 (Half-edge one-third radial envelope). *If*

$$M \geq \frac{1}{2}, \quad 0 < m \leq M, \quad M + m < 1,$$

then

$$(85) \quad \boxed{c_{\max}(M, m) < 1 - \frac{m}{3}}.$$

Proof. Put $s = M + m$ and $c_0 = 1 - m/3$. Only the L and T cells of theorem A.9 can occur.

For $m \leq 3/8$, substitution into the L polynomial gives

$$F_L(c_0) = \frac{m}{81}(m^3 - 12m^2 - 63m + 27) > 0;$$

the cubic decreases on $[0, 3/8]$ and equals $891/512$ at $3/8$. Thus the selected L root is below c_0 . At the L/T transition the two roots agree. On the T cell,

$$c_T(s) = \frac{2(s - m)}{1 + \sqrt{4s^2 - 3}},$$

and, writing $r = \sqrt{4s^2 - 3}$,

$$c'_T(s) = \frac{2(r + 4ms - 3)}{r(1 + r)^2} < 0$$

because $m \leq s/2$ and $r + 2s^2 - 3 < 0$ for $s < 1$. Hence the T root remains below c_0 .

For $3/8 \leq m < 1/2$, one has

$$s^4 - s^2 + mM \geq \left(\frac{7}{8}\right)^4 - \left(\frac{7}{8}\right)^2 + \frac{3}{16} = \frac{33}{4096} > 0,$$

so only the T cell occurs. It decreases with M ; at $M = 1/2$, $s \geq 7/8$ and its value is at most $4/5 < 5/6 < c_0$. $\square$

Lemma D.28 (Supported endpoint to four-point enclosure). *Suppose T_0 has actual boundary endpoint $a = A_0$ toward V_5 and supported interval $[c, u]$ on r_1 , measured from V_1 toward O . Put $\varepsilon = 1 - u > 0$ and $M = M_c^{\text{sup}}$. Assume*

$$\varepsilon V_1 \in U_C, \quad A_0 + \varepsilon \leq 1, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M.$$

$$(86) \quad A_0 \leq 1 - M, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M.$$

The first inequality in the last display is a consequence, not an extra hypothesis. If T_1 is uniquely supercritical with $C_1 \geq c$, and the four center-free edges from $e_{1,2}$ through $e_{4,5}$ join nonsupercritical T_2, T_3, T_4, T_5 , then the skeleton is not covered. The conclusion applies to both possible nonzero gap ranks.

Proof. The strict-supercritical envelope gives $B_1 < M$, and the four handoffs give $B_5 \leq B_4 \leq B_3 \leq B_2 \leq B_1$. Thus $B_5 < 1 - A_0$, so the left-edge actual gap has endpoints $Y(A_0), Y(1 - B_5)$, with $Y(t) = (1 - t)V_0 + tV_5$. Under coverage these endpoints and $O, \varepsilon V_1$ belong to U_C . Since $M \geq 1/2$, the hypotheses imply

$$A_0 \leq \varepsilon, \quad B_5 < M \leq \frac{\varepsilon}{A_0 + \varepsilon}.$$

Apply theorem 5.18 with geometric parameters $a = A_0$, $\beta = B_5$. The fixed set has enclosure number at least one, contrary to compact containment in U_C . No sum greater than four or propagation through the supercritical role is used. $\square$

D.14. T3-like supported-tail adapter.

Lemma D.29 (Solved T3-like supported-tail adapter). *The exact hypotheses of theorem 5.20 imply the rescuer-tail hypotheses in theorem D.28.*

Proof. Normalize the center midpoint to M_0 and reflect so that T_0 is T3-like, $M_1 \in T_0$, and T_1 is uniquely supercritical. Write the T3-like interval on r_1 , measured from V_1 toward O , as $[c, u]$, with $c \leq 1/2 \leq u$. Its O-side endpoint is

$$P_T = (1 - u)V_1.$$

The open endpoint and the V-type restrictions exclude every V role from P_T , so $P_T \in U_C$.

Let A_0 be the actual boundary reach of T_0 on $e_{5,0}$. The translated T3-like normal form has

$$R_0 = \frac{D + \sqrt{4 - 3D^2}}{2}, \quad x = \frac{A_0 D}{R_0}, \quad \theta = \frac{D - 1}{R_0},$$

$$c = x + \theta, \quad \frac{1 - 4\theta + \theta^2}{2(1 - 2\theta)} \leq x \leq \frac{1}{2} - \theta.$$

Let $M = M_c^{\text{sup}}$. The equation $s(1 - M) = c$, where $s(z) = z(2 - z)/(1 + z)$, and

$$2x^2 + (\theta - 1)x + \theta \geq 0$$

give

$$A_0 \leq x \leq 1 - M, \quad \frac{A_0}{A_0 + 1 - u} = x \leq 1 - M.$$

Finally $A_0 + 1 - u = R_0/D \leq 1$. These are exactly the hypotheses of theorem D.28. $\square$

D.15. Alternative one-Vd radial-separation calculations. The calculations in this subsection are retained quantitative alternatives. They assume the original reduced CE2 one-Vd placement, its boundary-cover and strict Vd-cap hypotheses, and all the corresponding residual selectors. They are not deductions from the bare CE2 signed domain, and are not used by the active B/C fixed-point route.

For radial separation along r_i , define

$$d_i^C := \max\{x \in [0, 1] : xV_i \in T_C\},$$

$$c_{\text{loc}} := \sup\{s \in [0, 1] : V_i + s(O - V_i) \in U_{i-1} \cup U_i \cup U_{i+1}\}.$$

Then

$$T_C \cap r_i = \{xV_i : 0 \leq x \leq d_i^C\},$$

$$(U_{i-1} \cup U_i \cup U_{i+1}) \cap r_i \subseteq \{xV_i : 1 - c_{\text{loc}} \leq x \leq 1\}.$$

Thus the exact strict inequality

$$c_{\text{loc}} < 1 - d_i^C$$

leaves the nonempty interval

$$\{xV_i : d_i^C < x < 1 - c_{\text{loc}}\} \subseteq S \setminus \left(U_C \cup \bigcup_{j=0}^5 U_j \right).$$

For the nonadjacent adapter, if $I \subseteq [0, 1]$ is empty or a closed interval, define

$$\mathcal{R}_I(p) := 1 - \sup\{x \in [0, 1] : [0, x] \subseteq [0, p] \cup I\}.$$

Put

$$I_R := \{x \in [0, 1] : X_0(x) \in T_C\}, \quad I_L := \{x \in [0, 1] : X_5(1 - x) \in T_C\},$$

$$\rho_R := \mathcal{R}_{I_R}(B_0), \quad \rho_L := \mathcal{R}_{I_L}(A_0), \quad \rho := \min\{\rho_R, \rho_L\}, \quad D_\tau := \rho V_\tau.$$

Appendix D proves the exact separation inequalities for both adapters.

Lemma D.30 (Solved adjacent Vd separation). *Under the adjacent placement*

$$\sigma = 0, \quad \tau \in \{1, 5\},$$

the radial-separation inequality $c_{\text{loc}} < 1 - d_2^C$ holds on the routed arm r_2 .

Proof. In the adjacent placement, the residuals satisfy

$$\rho_R + \rho_L < \frac{1}{2}.$$

The signed CE2 domain gives

$$\alpha + \delta < \frac{1}{6} \min\{R, W\}.$$

If $\rho_L = W - \alpha$, then

$$4\delta + \alpha \leq 4(\alpha + \delta) < \frac{2W}{3} < W,$$

so $4\delta < \rho_L$.

Otherwise $\rho_L = 1 - A_0$. The opposite residual must be

$$\rho_R = 1 - (W + \delta), \quad B_0 \geq \frac{k}{R}.$$

The boundary-endpoint diameter inequality gives

$$\rho_L > \frac{k}{2R} > \frac{\eta}{2R}.$$

Combining this with the preceding residual-sum inequality gives

$$\delta > R - \frac{1}{2} + \frac{\eta}{2R} =: J(R).$$

The function J is increasing, $J(3/8) = 1/24$, and the displayed center-slack inequality gives $\delta < 1/24$. Thus $R < 3/8$. Then

$$(2 - 3R)^2 - E^2 = (3 - 8R)(1 - R) > 0,$$

so $\eta/R > 1/3$, and the preceding endpoint-distance inequality yields

$$\rho_L > \frac{1}{6} > 4\delta.$$

This proves the residual estimate used by the adjacent Vd terminal. The new one-third radial envelope then gives

$$C_2 < 1 - \frac{\rho_L}{3} < 1 - \delta.$$

□

Lemma D.31 (Solved nonadjacent Vd separation). *Under the nonadjacent placement*

$$\sigma = 0, \quad \tau \in \{2, 3, 4\},$$

the point $D_\tau = \min\{\rho_R, \rho_L\}V_\tau$ lies beyond both the C-triangle exit and the actual Vd radial endpoint.

Proof. Let $x = k/W$ and $y = k/R$ be the two center near endpoints, and let B, U be the farthest already-covered extents. The endpoint-distance inequality and monotonicity of M_0 give

$$B \leq M_0(x), \quad U \leq M_0(y).$$

Since

$$1 - M_0(z) > \frac{z}{2} \quad (z > 0),$$

the residual tails satisfy

$$\rho_R > \frac{x}{2}, \quad \rho_L > \frac{y}{2}.$$

The signed center-slack estimate

$$\alpha + \delta < \frac{1}{2} \min\{x, y\}$$

therefore gives

$$\alpha + \delta < \min\{\rho_R, \rho_L\}.$$

The Vd own-radial margin then places $\min\{\rho_R, \rho_L\}V_\tau$ beyond the Vd trace, while the last strict residual inequality places it beyond the C exit. $\square$

Lemma D.32 (Solved Vd1 supported-tail adapter). *When the unique Vd1 role is based at T_0 , its actual corner data imply the two strict-supercritical envelope inequalities required by theorem D.28.*

Proof. Use the Vd1 corner parameter $t \geq 1$, let $d = \sqrt{t^2 + t + 1}$, and write the supported interval as $[c, u]$. Then

$$c = \frac{t(1-b)}{t+1}, \quad u = \frac{d-a-tb-1}{t}.$$

The midpoint condition gives

$$a + tb \leq d - 1 - \frac{t}{2}.$$

Eliminating $tb = t - c(t+1)$ gives

$$a \leq F(t, c) := d - 1 - \frac{3t}{2} + c(t+1).$$

For fixed $c \leq 1/2$, the function F decreases with t , so

$$a \leq F(1, c) = \sqrt{3} - \frac{5}{2} + 2c =: L(c).$$

The inequality $2L(c) \leq 1 - M_c^{\text{sup}}$ reduces to

$$20c^2 + (18\sqrt{3} - 52)c + 47 - 24\sqrt{3} \geq 0,$$

which holds on $[0, 1/2]$. Hence

$$a \leq 1 - M_c^{\text{sup}}.$$

Put $\varepsilon = 1 - u$. Direct substitution gives

$$\varepsilon = \varepsilon_0 + \frac{a}{t}, \quad \varepsilon_0 = \frac{2t+1-c(t+1)-d}{t} > 0.$$

The map

$$z \mapsto \frac{z}{z + \varepsilon_0 + z/t}$$

is increasing, while $\varepsilon_0 + F(t, c)/t = 1/2$. Therefore

$$\frac{a}{a + \varepsilon} \leq \frac{F(t, c)}{F(t, c) + 1/2} \leq 2L(c) \leq 1 - M_c^{\text{sup}}.$$

The last two estimates are the local hypotheses of the common rescuer-tail lemma. $\square$

APPENDIX E. ZERO-GAP NINE-POINT OPTIMIZATION

This appendix supplies exact frontier and radial calculations, point exclusions, Newton reduction, and the four-contact enclosure proof for the type-independent zero-gap obstruction. The radial step reuses the common C_+, C_- forcing interface; the three asymmetric witnesses and the exact certificate are unchanged. It uses the public parameters and witness $K_{\text{wit}}(a, b)$ from section 5.6. The symbols A, B, C introduced below are appendix-only Newton inner points required by the exact certificate; they do not replace the public frontier points Q_-, Q_0, Q_+ .

The finite caliper criterion used here is the shared result theorem A.5 in Appendix A. It is not rederived in this appendix.

E.1. Corner-clipped AB -unions. Apply the rigid motion that sends the supercritical corner V_4 to the local origin and its two incident edge directions to e_A, e_B . Within this calculation O, A, B, C denote the local origin, anchors, and corner cone; the feasible union keeps the public symbol $\mathcal{R}_{AB}(a, b)$.

Let e_A, e_B be unit vectors with angle 120° , put

$$O = 0, \quad A = ae_A, \quad B = be_B, \quad C = \{ue_A + ve_B : u, v \geq 0\},$$

and define

(87)

$$\mathcal{R}_{AB}(a, b) = C \cap \bigcup \{T : T \text{ is a closed unit equilateral triangle and } O, A, B \in T\}.$$

The finite description in Theorem A.5 applies to the four-point set $\{O, A, B, x\}$ and makes membership in (87) an exact finite semialgebraic condition of degree at most four.

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

Theorem E.1 (Strict supercritical AB -union). *Assume*

$$a + b > 1, \quad \rho = a^2 + ab + b^2 < 1,$$

put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, and define

$$\begin{aligned} \alpha &= \frac{h(a + 2b - aD)}{2\rho}, & \beta &= \frac{h(a - b + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-a + b + (a + b)D)}{2\rho}, & \delta &= \frac{h(2a + b - bD)}{2\rho}. \end{aligned}$$

Then all four coefficients are positive and

$$(88) \quad \mathcal{R}_{AB}(a, b) = (C \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (C \cap \mathcal{D}_B \cap \mathcal{H}_B),$$

where

$$\begin{aligned} \mathcal{D}_A &= \{(u, v) : (u - a)^2 + v^2 - (u - a)v \leq 1\}, \\ \mathcal{D}_B &= \{(u, v) : u^2 + (v - b)^2 - u(v - b) \leq 1\}, \\ \mathcal{H}_A &= \{(u, v) : \alpha(u - a) + \beta v \leq 0\}, \\ \mathcal{H}_B &= \{(u, v) : \gamma u + \delta(v - b) \leq 0\}. \end{aligned}$$

Its non-axis frontier consists, in counterclockwise order, of the A -circle, the A -line, the B -line, and the B -circle.

Proof. The identities

$$4\rho - 3(a + b)^2 = (a - b)^2, \quad (a + b)^2 D^2 - (a - b)^2 = 4\rho((a + b)^2 - 1)$$

give $0 < D < 1$ and positivity of the four coefficients.

We apply the caliper criterion of Theorem A.5. For a point $x = (u, v)$ in the relative interior of C but outside $\triangle OAB$, the four hull vertices occur in the order O, B, x, A . The two cone-axis hull edges cannot certify a unit triangle: their support sums, divided by h , are respectively

$$\max\{a, u\} + \max\{b, v - u\}, \quad \max\{b, v\} + \max\{a, u - v\},$$

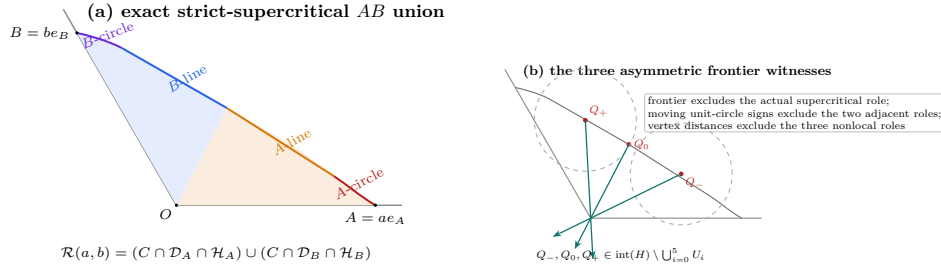


FIGURE 13. The strict-supercritical AB frontier and the three asymmetric witnesses. The source-conditioned union has, in counterclockwise order, an A -circle arc, an A -line segment, a B -line segment, and a B -circle arc. The points Q_-, Q_0, Q_+ are selected on this exact frontier and are then excluded from the six actual V roles by frontier, moving-circle, and vertex-distance arguments. The drawing is schematic; the labelled frontier order and witness roles are exact.

and are at least $a + b > 1$. It remains to test the Ax and Bx calipers.

For the Ax caliper put $p = u - a$ and $r^2 = p^2 + v^2 - pv$. Exact evaluation of the three supports gives

$$(89) \quad \frac{G_A}{h} = \frac{\max\{r^2, -ap + (a+b)v\} + N_A}{r}, \quad N_A = \begin{cases} 0, & p \leq 0, \\ ap, & 0 \leq p \leq v, \\ (a+b)p - bv, & p \geq v. \end{cases}$$

No support label is omitted in these three sectors. If $p \leq 0$, $G_A \leq h$ is equivalent to

$$r \leq 1, \quad -ap + (a+b)v \leq r.$$

The second inequality is the half-plane condition in (88), because

$$r^2 - (-ap + (a+b)v)^2 = \frac{(cp + dv)(c'p + d'v)}{4\rho},$$

where

$$c = a + 2b - aD, \quad d = a - b + (a+b)D, \quad c' = a + 2b + aD, \quad d' = a - b - (a+b)D,$$

and $c, c', d > 0 > d'$. Thus the low sector is exactly $\mathcal{D}_A \cap \mathcal{H}_A$. In the middle sector $0 < p \leq v$ one has $r \leq v$, whereas the numerator in (89) is at least $(a+b)v > r$, so no fit is possible.

In the high sector $p \geq v$, the inequalities $G_A \leq h$ imply

$$(90) \quad r^2 + (a+b)p - bv \leq r, \quad bp + av \leq r.$$

Write $x - A = ry(\theta)$, $0 \leq \theta \leq 60^\circ$, with cone coordinates

$$y(\theta) = te_A + we_B, \quad t = \frac{2}{\sqrt{3}} \cos(\theta - 30^\circ), \quad w = \frac{2}{\sqrt{3}} \sin \theta.$$

Then (90) gives

$$r \leq f(\theta) = 1 - (a+b)t + bw, \quad S(\theta) = bt + aw \leq 1.$$

Let n_B be the unit vector with dual coordinates (γ, δ) . The identities

$$b\gamma + (a+b)\delta = h, \quad (a+b)\gamma + a\delta = \frac{h}{2}(1+D), \quad b\delta - a\gamma = \frac{h}{2}(1-D)$$

show that the unique angle θ_B of this normal satisfies $S(\theta_B) = 1$ and $f(\theta_B) = (1-D)/2$. The function S has at most one critical point, a maximum, so $S(\theta) \leq 1$ implies $\theta \leq \theta_B$. On the feasible part $f, f' \geq 0$; hence $f(\theta) \cos(\theta_B - \theta)$ is increasing. Consequently

$$\langle n_B, x - B \rangle \leq a\gamma - b\delta + \frac{h}{2}(1-D) = 0.$$

The diameter bound also gives $\|x - B\| \leq 1$. Thus every high sector fit belongs to $\mathcal{D}_B \cap \mathcal{H}_B$. Reflection exchanges a, u with b, v and supplies the corresponding result for the Bx caliper. This proves (88) in the cone interior; the axes follow by taking limits, since both sides are closed.

For completeness, the line-circle junctions are obtained by solving the two displayed line and circle equations. The line normals satisfy

$$\alpha^2 + \alpha\beta + \beta^2 = h^2, \quad \gamma^2 + \gamma\delta + \delta^2 = h^2,$$

and the line determinant is

$$\alpha\delta - \gamma\beta = \frac{3(1-D)(D+3)}{16\rho} > 0.$$

The successive cone slopes are

$$+\infty, \quad \frac{2}{1-D}, \quad \frac{a(1-D) + 2b}{2a + b(1-D)}, \quad \frac{1-D}{2}, \quad 0;$$

the two middle comparisons reduce to $a(4 - (1-D)^2) > 0$ and $b(4 - (1-D)^2) > 0$. This proves the asserted four-piece order. $\square$

E.2. Exact-one handoffs and common reach bounds.

Lemma E.2 (Exact-one handoff chain). *Assume the six V triangles cover ∂H and exactly one actual V triangle is supercritical. Rotate its index to 4. The strict handoffs supplied by Proposition 2.11 satisfy*

$$\xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

With

$$a = 1 - \xi_3, \quad b = \xi_4, \quad p = 1 - b, \quad q = 1 - a,$$

one has

$$(91) \quad 0 < a, b < 1, \quad a + b > 1, \quad a^2 + ab + b^2 < 1,$$

and every selected V triangle obeys

$$a_i \geq p, \quad b_i \geq q.$$

Proof. Every selected V triangle other than 4 is strictly nonsupercritical, so $\xi_i < \xi_{i-1}$ for $i \neq 4$, which is precisely (E.2). In particular ξ_3 is the global minimum and ξ_4 the global maximum. The two anchors of V triangle 4 lie in its open triangle; their mutual squared distance is $a^2 + ab + b^2$, strictly below the diameter squared of the closed unit triangle. This proves (91). Finally every $\xi_j \in [\xi_3, \xi_4]$, so $1 - \xi_{i-1} \geq 1 - \xi_4 = p$ and $\xi_i \geq \xi_3 = q$. $\square$

E.3. Six directly forced radial points. Put $h = \sqrt{3}/2$ and

$$s = p + q, \quad m = \min(p, q), \quad M = \max(p, q), \quad \chi = s^4 - s^2 + pq.$$

The strict domain (91) gives

$$(92) \quad pq > (1-s)(2-s), \quad m > 1-s, \quad s > 1-m.$$

Let $c_* = c_{\max}(p, q)$ be the exact radial envelope of Proposition A.9. On the present subcritical domain, equation (37) becomes

$$(93) \quad c_* = \begin{cases} C_L(m), & \chi \leq 0, \\ \frac{2M}{1 + \sqrt{4s^2 - 3}}, & \chi > 0, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal s .

Lemma E.3 (Direct radial forcing). *One has*

$$(94) \quad 0 < c_* < 1, \quad c_* > 1 - m.$$

Assume that $U_C, U_0, \dots, U_5$ cover H , with no restriction on the normalized V types. Then, for every i ,

$$D_i = (1 - c_*)V_i \in U_C.$$

Consequently U_C contains the centered closed disk

$$(95) \quad \mathcal{D}_\eta = \{x : \|x\| \leq \eta\}, \quad \eta = h(1 - c_*).$$

Proof. On the C_L branch, $C_L(m) < 1$ because the defining polynomial is positive at 1; if $s < \sqrt{3}/2$, then $c_* \geq \sqrt{3}/2 > s > 1 - m$, while if $s \geq \sqrt{3}/2$ its value at s is $\chi \leq 0$, so $c_* \geq s > 1 - m$. On the other branch put $E = \sqrt{4s^2 - 3}$. The selector gives $sE > M - m$, whence $c_* < s < 1$. Moreover, with

$$A_* = \frac{2M}{1 - m} - 1,$$

one has

$$A_*^2 - E^2 = \frac{4(1-s)\{(1-m)(1-2m) + m(2-m)(1-s)\}}{(1-m)^2} > 0.$$

Thus $A_* > E$ and $c_* > 1 - m$. This proves (94).

By theorem E.2, every selected pair dominates (p, q) . The common-pair forcing corollary theorem 5.6 therefore forces the same $D_i = (1 - c_*)V_i$ for every V -type pattern. To make the adjacent-role step explicit, theorem 5.5 and (94) give

$$C_+(p, q), C_-(p, q) \leq 1 - m < c_*.$$

A neighboring open V triangle containing D_i would have positive trace on r_i and its closure would realize neighboring coordinate c_* with the common anchors, contradicting this inequality. If its type forbids that trace, openness already gives a contradiction. The own-role and nonlocal exclusions are the shared type-aware forcing lemma. Thus no V triangle contains D_i , and full coverage puts it in U_C .

The six points D_i form a regular hexagon of circumradius $1 - c_*$. Their convex hull contains its centered incircle of radius $h(1 - c_*)$, and convexity of U_C gives (95). $\square$

We next define the three asymmetric witnesses. At V_4 use local coordinates

$$P = V_4 + u(V_5 - V_4) + v(V_3 - V_4), \quad \mathbf{q}(u, v) = u^2 + v^2 - uv.$$

The forward-reach lower bound is b and the backward-reach lower bound is a . Accordingly, set

$$\begin{aligned} \alpha &= \frac{h(b + 2a - bD)}{2\rho}, & \beta &= \frac{h(b - a + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-b + a + (a + b)D)}{2\rho}, & \delta &= \frac{h(2b + a - aD)}{2\rho}, \end{aligned}$$

where now $\rho = a^2 + ab + b^2$ and $D = \sqrt{4\rho - 3}$. The two line pieces of Theorem E.1 are

$$L_-(\lambda) = (b - \beta\lambda, \alpha\lambda), \quad L_+(\mu) = (\delta\mu, a - \gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D + 3))$ in

$$(96) \quad J = \left(\frac{2b + a - aD}{D + 3}, \frac{b + 2a - bD}{D + 3} \right).$$

Put

$$E_- = (1 - b, 2 - b), \quad E_+ = (2 - a, 1 - a).$$

The actual adjacent handoffs have local centers

$$Z_5(t) = (1 + t, t), \quad 1 - a \leq t \leq b, \quad \text{and} \quad Z_3(y) = (y, 1 + y), \quad 1 - b \leq y \leq a.$$

The restrictions of the two unit-circle equations are

$$(97) \quad \begin{aligned} g_-(\lambda) &= \frac{3}{4}\lambda^2 - 3(\alpha + b\beta)\lambda + 3b^2 - 3b + 2, \\ g_+(\mu) &= \frac{3}{4}\mu^2 - 3(\delta + a\gamma)\mu + 3a^2 - 3a + 2. \end{aligned}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\begin{aligned} \lambda_- &= 2(\alpha + b\beta) - \frac{2}{3}\sqrt{9(\alpha + b\beta)^2 - 9b^2 + 9b - 6}, \\ \mu_+ &= 2(\delta + a\gamma) - \frac{2}{3}\sqrt{9(\delta + a\gamma)^2 - 9a^2 + 9a - 6}. \end{aligned}$$

Lemma E.4 (Moving adjacent-triangle signs). *Throughout (91),*

$$(98) \quad \lambda_* < \lambda_- < h^{-1}, \quad \mu_* < \mu_+ < h^{-1}.$$

Moreover $J_u, J_v < 1/2$. The point $L_+(\mu_+)$ satisfies

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3 - 2a,$$

and the reflected bound holds for $L_-(\lambda_-)$. Writing $F(P, t) = \mathbf{q}(P - Z_5(t)) - 1$, one has, for every $t \in [1 - a, b]$,

$$(99) \quad F(L_+(\mu_+), t) \geq 0, \quad F(J, t) > 0, \quad F(L_-(\lambda_-), t) > 0.$$

Under the reflection $(u, v, a, b) \leftrightarrow (v, u, b, a)$, the analogous signs hold for every $y \in [1 - b, a]$: the first-root point $L_-(\lambda_-)$ has nonnegative sign against $Z_3(y)$, and the other two witnesses have strictly positive sign.

Proof. Put

$$U = a + b - 1, \quad z = a - b, \quad D^2 = z^2 + 6U + 3U^2.$$

Then $0 < U < 1/6$ and $z^2 < \Omega := 1 - 6U - 3U^2$. We give the fixed-line calculation for the circle centered at $E_+ = (2 - a, 1 - a)$; reflection gives the calculation for E_- . Put

$$F(P, t) = \mathbf{q}(P - (1 + t, t)) - 1, \quad t_0 = 1 - a,$$

so this circle is $F(P, t_0) = 0$. At the junction, exact expansion gives

$$(D + 3)^2 F(J, t_0) = 3D(z^2 + Uz + 1 - 3U) + P_J,$$

where

$$P_J = z^4 + 5z^2 + 6Uz^2 + 3U^2z^2 + 9Uz + 3U + 15U^2 > 0;$$

indeed $z^2 + Uz + 1 - 3U \geq 1 - 3U - U^2/4 > 0$, and $5z^2 + 9Uz + 15U^2$ is positive definite. Thus $F(J, t_0) > 0$. Also

$$J_u + J_v = \frac{(1 + U)(3 - D)}{3 + D} < 1.$$

Indeed this inequality is $3U < D(2 + U)$, which follows from $D^2 \geq 3U(2 + U)$ and $(2 + U)^3 > 3U$. Since

$$\frac{\partial F}{\partial t}(P, t) = 1 + 2t - P_u - P_v,$$

$F(J, t)$ is strictly increasing for $t \geq 0$, and therefore is positive throughout $[1 - a, b]$.

On the opposite line L_- , the derivative at the junction, after multiplication by a positive factor, has at $t = 1 - a$ the numerator

$$N_0 = D(9 + 3z^2 + 6Uz - 9U^2) + P_0,$$

where

$$P_0 = 18U^3 + 6U^2z^2 + 63U^2 + 18Uz^2 + 18Uz + 54U + 2z^4 + 9z^2 + 9.$$

The coefficient of D is at least $9 - 6U - 9U^2 > 0$, while $P_0 \geq 9 + 54U - 18U > 0$. Hence $N_0 > 0$. At the other endpoint $t = b$, the corresponding numerator is

$$N_1 = D(9 + 3z^2 + 6U - 3U^2) + P_1,$$

where

$$P_1 = 9U^4 - 3U^3z + 45U^3 + 9U^2z^2 - 6U^2z + 72U^2 - Uz^3 + 21Uz^2 + 9Uz + 45U + 2z^4 + 9z^2 + 9.$$

Its D -coefficient is positive and, using $|z| < 1$,

$$P_1 \geq 9 + 45U - 9U - U - 6U^2 - 3U^3 > 0.$$

The junction derivative is affine in t , so it is positive throughout $[1 - a, b]$. Since the restriction of F to L_- is a convex quadratic, is positive at the junction, and is increasing there, it has no intersection on the opposite line side for any actual adjacent handoff.

On the correct line L_+ , the value at its outer endpoint h^{-1} has, after multiplication by a positive factor, numerator

$$A_* = P_2 - 4DU(1 + U + z),$$

where

$$P_2 = 3U^4 + 6U^3z + 6U^3 + 4U^2z^2 + 12U^2z + 12U^2 \\ + 2Uz^3 + 6Uz^2 + 2Uz + 6U + z^4 + 2z^2 - 3.$$

For fixed U , replace the odd powers of z by $x = |z|$. The resulting polynomial $P_2^+(U, x)$ satisfies

$$\partial_x P_2^+ = 2(3U^3 + 4U^2x + 6U^2 + 3Ux^2 + 6Ux + U + 2x^3 + 2x) > 0,$$

and therefore its supremum for $z^2 < \Omega$ occurs at $x = \sqrt{\Omega}$. There

$$P_2^+(U, \sqrt{\Omega}) = 4U(U + \sqrt{\Omega} - 3) < 0.$$

As $1 + U + z = 2a > 0$, this gives $A_* < 0$. The positive junction value and negative endpoint value force the first root strictly between them, proving the L_+ half of (98); reflection proves the other half.

For the coordinate bounds, the exact identities

$$D^2 - (3U - z)^2 = 6U(1 + z - U), \quad D^2 - (3U + z)^2 = 6U(1 - z - U)$$

give $J_u, J_v < 1/2$. It remains to control the coordinate sum along L_+ . For $E = L_+(h^{-1})$, the sign of $3 - 2a - E_u - E_v$ is the sign of

$$D(6U + 2z + 6) + B(U, z),$$

where

$$B(U, z) = -9U^3 - 9U^2z - 9U^2 - 3Uz^2 - 18Uz + 3U \\ - 3z^3 + 3z^2 - 3z + 3.$$

Here

$$B_z = -3(3z^2 + 2Uz + 6U - 2z + 3U^2 + 1) < 0,$$

because the quadratic in parentheses has discriminant $-32U^2 - 80U - 8 < 0$. Hence its minimum on $z^2 < \Omega$ is at $z = \sqrt{\Omega}$, where $B(U, \sqrt{\Omega}) = 6(1 - 3U - \sqrt{\Omega}) > 0$ because $(1 - 3U)^2 - \Omega = 12U^2$. Thus $E_u + E_v < 3 - 2a$. Also $J_u + J_v < 1 < 3 - 2a$, and the coordinate sum is affine on the line segment, so the same strict bound holds at the first root. Now

$$\frac{\partial F}{\partial t}(P, t) = 1 + 2t - P_u - P_v.$$

Since $F(L_+(\mu_+), 1 - a) = 0$, the coordinate-sum bound makes its t -derivative strictly positive for $t \geq 1 - a$. Together with the no-opposite-line result, this proves (99). Reflection gives both the bound on L_- and the complete $Z_3(y)$ statement. $\square$

Define the local and global witnesses by

$$(100) \quad Q_-^{\text{loc}} = L_-(\lambda_-), \quad Q_0^{\text{loc}} = J, \quad Q_+^{\text{loc}} = L_+(\mu_+),$$

and by the displayed conversion from (u, v) to the plane.

Lemma E.5 (Direct asymmetric forcing). *Assume that $U_C, U_0, \dots, U_5$ cover H . Then*

$$Q_-, Q_0, Q_+ \in \text{int}(H) \setminus \bigcup_{i=0}^5 U_i,$$

and consequently all three witnesses belong to U_C .

Proof. The root order and positivity of the line coordinates give $0 < u, v < 1$ for all three points. Each belongs to $\mathcal{R}_{AB}(b, a)$ and hence lies in a closed unit triangle containing V_4 , so $\mathbf{q}(u, v) \leq 1$; hence $|u - v| < 1$ and all three points lie in $\text{int}(H)$.

The closure of U_4 is one of the triangles represented by $\mathcal{R}_{AB}(b, a)$, since it contains V_4, X_3, X_4 . All three witnesses lie on its exact non-axis frontier. If one belonged to the open triangle U_4 , a sufficiently small plane ball about it would lie both in U_4 and in the interior of the corner cone, making the witness an interior point of $\mathcal{R}_{AB}(b, a)$. This contradiction excludes U_4 .

For the triangles U_0, U_1, U_2 , their contained vertices have local coordinates

$$\begin{array}{c|ccc} i & 0 & 1 & 2 \\ \hline V_i & (2, 1) & (2, 2) & (1, 2). \end{array}$$

For $P = (u, v)$,

$$\|P - (2, 1)\|^2 - 1 = \mathbf{q}(u, v) + 2 - 3u, \quad \|P - (1, 2)\|^2 - 1 = \mathbf{q}(u, v) + 2 - 3v.$$

The inequalities $J_u, J_v < 1/2$ and the line order give distance greater than one from V_0 for Q_-, Q_0 and from V_2 for Q_0, Q_+ . For the remaining pair, the first-root circle and the sum bound in Lemma E.4 give

$$\|Q_+ - V_0\|^2 - 1 = a(3 - a - (Q_+)_u - (Q_+)_v) > a^2,$$

and, by reflection, $\|Q_- - V_2\|^2 - 1 > b^2$. All three points are farther than one from $V_1 = (2, 2)$, because writing $u = 1 - \xi$, $v = 1 - \eta$ gives

$$\|P - V_1\|^2 = 1 + \xi + \eta + \xi^2 + \eta^2 - \xi\eta > 1.$$

Since two points in one open unit equilateral triangle have distance strictly less than one, these inequalities exclude U_0, U_1, U_2 .

The actual handoff $X_5 = Z_5(\xi_5)$ belongs to U_5 , and $1 - a < \xi_5 < b$ by (E.2). The three signs in (99) therefore put every witness at distance at least one from X_5 , excluding U_5 . Similarly, if $y = 1 - \xi_2$, then $1 - b < y < a$, $X_2 = Z_3(y) \in U_3$, and the reflected signs exclude U_3 . Thus no V triangle contains any witness. The assumed cover now forces all three interior witnesses into U_C . $\square$

The nine directly forced points are $D_0, \dots, D_5, Q_-, Q_0, Q_+$. Under a putative cover, Lemmas E.3 and E.5 therefore force the compact set

$$(101) \quad K_{\text{wit}}(a, b) = \mathcal{D}_\eta \cup \{Q_-, Q_0, Q_+\} \subset U_C.$$

E.4. Newton inner reduction. For the nontrivial regime $c_* > 2/3$, we first remove the square radicals in the first-root witnesses. Normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D+3)}.$$

Let g_-, g_+ be the rescaled forms of (97), and take one Newton step at the junction:

$$\hat{\xi}_- = \xi_* - \frac{g_-(\xi_*)}{g'_-(\xi_*)}, \quad \hat{\zeta}_+ = \zeta_* - \frac{g_+(\zeta_*)}{g'_+(\zeta_*)}.$$

Define, in local coordinates,

$$\begin{aligned} A &= \left(b - \frac{3}{4}\widehat{\beta\xi}_-, \frac{3}{4}\widehat{\alpha\xi}_- \right), \\ B &= J, \\ C &= \left(\frac{3}{4}\widehat{\delta\zeta}_+, a - \frac{3}{4}\widehat{\gamma\zeta}_+ \right), \end{aligned}$$

and convert them to global vectors.

Lemma E.6 (Newton inner reduction). *The points above have coordinates rational in a, b, D and satisfy*

$$A \in (Q_0, Q_-), \quad C \in (Q_0, Q_+).$$

Consequently

$$\widehat{K} = \text{conv}(\mathcal{D}_\eta \cup \{A, B, C\}),$$

obeys $\widehat{K} \subseteq \text{conv}(K_{\text{wit}})$.

Proof. Each quadratic g_-, g_+ is strictly convex, positive, and decreasing at the junction. Its tangent zero therefore lies strictly between the junction and its first root. This gives the two open-segment inclusions. All displayed coefficients and Newton quotients are rational in a, b, D , and convexity gives the asserted containment. $\square$

E.5. Ordered convex chain and supporting-line bounds.

Proposition E.7 (Four-contact witness geometry). *Assume (91) and $c_* > 2/3$. The rays A, B, C occur in that order in one open 120° cone, and*

$$\text{cross}(B - A, C - B) > 0, \quad \|A\|, \|C\| > \eta.$$

Moreover the outward distances of the lines AB, BC satisfy

$$(102) \quad d_{AB} := \frac{\text{cross}(A, B)}{\|B - A\|} \geq h - 2\eta, \quad d_{BC} := \frac{\text{cross}(B, C)}{\|C - B\|} \geq h - 2\eta.$$

Proof. The local coordinates of all three Newton points lie in $(0, 1)^2$. Centering at O gives coefficient vectors $(u - 1, v - 1)$ in the two incident edge directions, so their rays lie in one open 120° cone. The two line identities

$$d_{AB} = \alpha(1 - b) + \beta > 0, \quad d_{BC} = \gamma + \delta(1 - a) > 0$$

give their order A, B, C . Put $s_A = h\widehat{\xi}_- - \lambda_* > 0$ and $s_C = h\widehat{\zeta}_+ - \mu_* > 0$. The frontier directions give

$$\text{cross}(B - A, C - B) = hs_A s_C (\alpha\delta - \beta\gamma) > 0,$$

since $\alpha\delta - \beta\gamma = 3(1 - D)(D + 3)/(16\rho) > 0$. For $A = (-X, -Y)$ in the centered cone basis, $X > 1/2$, and

$$\|A\|^2 = (Y - X/2)^2 + 3X^2/4 > 3/16.$$

The reflected calculation gives

$$(103) \quad \|A\|, \|C\| > h/2 > \eta.$$

It remains to prove the two supporting-line bounds. Put $\tau = h - 2\eta = h(2c_* - 1)$. We now give the exact analytic verification. By reflection take $a \leq b$ and set

$m = 1 - b$, $M = 1 - a$, $U = a + b - 1$, $s = m + M$, and $w = M - m$. The smaller of the two normalized left sides in (102) is

$$(104) \quad d = \frac{\mathcal{A} + U(2m - 1) + D(\mathcal{A} + U)}{2\rho}, \quad \mathcal{A} = 1 - m + m^2.$$

On the C_L sheet, write $F_L(z) = z^4 - z^2 + mz - m^2$. This defining polynomial evaluated at $r = 1 - m/2 + m^2/2$ is

$$F_L(r) = \frac{m^2(m-1)}{16}(m^5 - 3m^4 + 11m^3 - 17m^2 + 28m - 12) > 0.$$

Indeed the polynomial in parentheses is increasing on $(0, 1/2)$ and has value $-33/32$ at $1/2$; its derivative is $28 - 34m + m^2(5m^2 - 12m + 33) > 0$. Since $F_L(h) = -(m - h/2)^2 \leq 0$ and the selected root is unique, $2c_* - 1 \leq \mathcal{A}$. Put

$$X_0 = \mathcal{A} + U, \quad Y_0 = 2\rho\mathcal{A} - \mathcal{A} - U(2m - 1).$$

Then

$$d - \mathcal{A} = \frac{D(\mathcal{A} + U) - Y_0}{2\rho},$$

and exact expansion gives

$$\begin{aligned} Y_0 &= 2(m^2 - m + 1)U^2 + (2m^3 - 2m + 3)U \\ &\quad + (m^2 - m + 1)(2m^2 - 2m + 1) > 0. \end{aligned}$$

Furthermore

$$D^2 X_0^2 - Y_0^2 = 4m\rho\Phi(U, m),$$

where

$$\begin{aligned} \Phi(U, m) &= (1 - m)(m^2 - m + 2)U^2 \\ &\quad + (2 - m^2)(m^2 - m + 1)U \\ &\quad - m(1 - m)^2(m^2 - m + 1). \end{aligned}$$

In these variables

$$\chi = U^4 - 4U^3 + 5U^2 - (m + 2)U + m(1 - m).$$

Since $U^2(U^2 - 4U + 5) > 0$, the selector $\chi \leq 0$ implies $U > m(1 - m)/(m + 2)$; Φ is increasing in U and at that lower endpoint is

$$\frac{m^2(1 - m)(m^4 - 2m^3 + 6m^2 - 6m + 4)}{(m + 2)^2} > 0.$$

The last quartic equals $(m^2 - m)^2 + 5m^2 - 6m + 4 > 0$. Thus $d > \mathcal{A} \geq 2c_* - 1$.

On the other radial sheet let $E = \sqrt{4s^2 - 3}$. Then $0 \leq w < sE$ and $c_* = (s + w)/(1 + E)$. At fixed U , differentiation of (104) can be checked without an implicit estimate. Namely,

$$d = \frac{1 + D}{2} - UB(D, w), \quad B(D, w) = \frac{(1 + U)(3 + D) + w(1 - D)}{D^2 + 3}.$$

Here $D^2 = w^2 + 6U + 3U^2$, so $0 \leq D' = w/D \leq 1$, and

$$B_w = \frac{1 - D}{D^2 + 3} \geq 0.$$

Writing $N = (1 + U)(3 + D) + w(1 - D)$, direct differentiation gives

$$B_D = \frac{(1 + U - w)(D^2 + 3) - 2DN}{(D^2 + 3)^2} > -\frac{34}{27}.$$

Indeed the first term in the numerator is positive, while $1 + U - w = 2a > 0$, $N < (7/6)4 + 1 = 17/3$, $D < 1$, and $D^2 + 3 \geq 3$. If $B_D < 0$, then $B_D D' \geq B_D$; otherwise $B_D D' \geq 0$. Hence $B' = B_w + B_D D' > -34/27$, and $U < 1/6$ yields

$$d' < \frac{1}{2} + \frac{1}{6} \frac{34}{27} = \frac{115}{162} < 1,$$

whereas $(2c_* - 1)' = 2/(1 + E) \geq 1$. Hence their difference decreases with w . At the boundary $w = sE$, where $\chi = 0$ and $c_* = s$, the preceding sheet applies: this is an admissible point because $h < s < 1$ and $D^2 = 4s^4 - 12s + 9 < 1$. Indeed $s^4 - 3s + 2 = (s - 1)(s^3 + s^2 + s - 2) < 0$ on this interval; the second factor is increasing and already equals $(7h - 5)/4 > 0$ at $s = h$. The preceding sheet gives $d \geq 2s - 1$. Therefore (102) holds on both sheets, and reflection supplies the other line.

This proves both supporting-line bounds. Since $0 < \eta < h/3$, their common lower bound $h - 2\eta$ is strictly greater than η , as required for the exposed-edge geometry used below. $\square$

E.6. Newton reduction in the body notation. The containment established by theorem E.6 is recorded under the public routing label

$$(105) \quad \widehat{K} = \text{conv}(\mathbb{B}(O, \eta) \cup \{A, B, C\}) \subseteq \text{conv}(K_{\text{wit}}(a, b)), \quad \Lambda(K_{\text{wit}}(a, b)) \geq \Lambda(\widehat{K}).$$

E.7. Four supporting contacts and paired radius transfer. Let R and J denote counterclockwise rotations through $2\pi/3$ and $\pi/2$, respectively. For the inner witness define

$$H(n) = \max\{\eta, \langle A, n \rangle, \langle B, n \rangle, \langle C, n \rangle\}, \quad \Psi(n) = \sum_{j=0}^2 H(R^j n).$$

The support formula gives $\Lambda(\widehat{K}) = h^{-1} \min_{\|n\|=1} \Psi(n)$. The minimum can be located without introducing another convex body.

Lemma E.8 (Four-contact enclosure formula). *For the Newton inner witness in the range $c_* > 2/3$, put*

$$n_{AB} = -\frac{J(B - A)}{\|B - A\|}, \quad n_{BC} = -\frac{J(C - B)}{\|C - B\|},$$

$$t_A = \frac{\eta A - \sqrt{\|A\|^2 - \eta^2} JA}{\|A\|^2}, \quad t_C = \frac{\eta C + \sqrt{\|C\|^2 - \eta^2} JC}{\|C\|^2}.$$

Then

$$(106) \quad \Lambda(\widehat{K}) = \frac{1}{h} \min\{\Psi(n_{AB}), \Psi(n_{BC}), \Psi(t_A), \Psi(t_C)\}.$$

Proof. By theorem E.7, the rays A, B, C are ordered in one open 120° cone, the chain turns strictly counterclockwise, and the supporting-line distances exceed η . The third point and the disk are strictly on the inward side of each of AB, BC , so both segments are exposed edges. Moreover

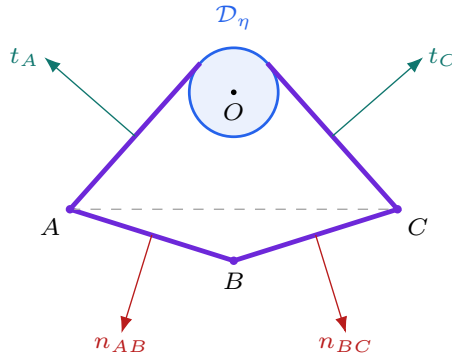
$$\text{cross}(C - A, B - A) < 0, \quad \text{cross}(C - A, -A) > 0.$$

Thus B and the origin are on opposite sides of AC , which is not an exposed edge. The other two straight boundary pieces are the exposed disk tangents through A and C . There is no exposed disk tangent through B . The indicated signs select

the tangent before A and the tangent after C in the counterclockwise boundary traversal.

The support-cell rotation lemma theorem D.11 places a minimum at these exposed ties or on a disk-only cell. In the latter case the same value occurs at a boundary tie, since $\|A\| > \eta$. Rotation through 120° leaves just the four displayed normals. The exposed-hull argument, rather than the general caliper theorem alone, is what discards the other point and disk ties. $\square$

(a) the four exposed contacts



(b) four support sums

Two consecutive-point contacts:

$$\Psi(n_{AB}) \geq d_{AB} + 2\eta \geq h,$$

$$\Psi(n_{BC}) \geq d_{BC} + 2\eta \geq h.$$

Two exposed disk tangencies:

$$P_A(\eta), P_C(\eta) \geq 0$$

$$\implies \Psi(t_A), \Psi(t_C) \geq h.$$

$$\text{Hence } \Lambda(\hat{K}) \geq 1.$$

FIGURE 14. The four-contact terminal for the zero-gap witness. The supporting lines through AB and BC and the two outer disk tangencies exhaust the active changes of support source. The dashed segment AC is not exposed: B and the disk center lie on opposite sides. The drawing is schematic; the exact witness geometry and the paired tangent residuals are proved in the text. No auxiliary Minkowski sum or cap covering is needed.

At the first two contacts the disk supplies the other two supports, so

$$(107) \quad \Psi(n_{AB}) \geq d_{AB} + 2\eta \geq h, \quad \Psi(n_{BC}) \geq d_{BC} + 2\eta \geq h.$$

For the two tangencies, put

$$\Delta = \text{cross}(C, RA) > 0, \quad \nu = \langle C, RA \rangle, \quad N_X = \|X\|^2 \quad (X = A, C).$$

Choosing C as a support source at Rt_A and A at R^2t_C , with the disk in the other directions, gives

$$(108) \quad \Psi(t_A) \geq 2\eta + \frac{\eta\nu + \Delta\sqrt{N_A - \eta^2}}{N_A}, \quad \Psi(t_C) \geq 2\eta + \frac{\eta\nu + \Delta\sqrt{N_C - \eta^2}}{N_C}.$$

Consequently the only remaining targets are the paired residual signs

$$(109) \quad P_X(\eta) := (N_X - \eta^2)\Delta^2 - ((h - 2\eta)N_X - \eta\nu)^2 \geq 0 \quad (X = A, C).$$

Indeed $P_X(\eta) \geq 0$ gives $\Delta\sqrt{N_X - \eta^2} \geq |(h - 2\eta)N_X - \eta\nu|$, which implies the required lower bound even when the expression inside the absolute value is negative. No sign is lost by squaring.

The exact certificate uses a smaller rationally specified radius. The following elementary transfer is the reason those existing signs suffice for the actual-radius tangencies.

Lemma E.9 (Simultaneous radius transfer). *Let X, Y be vectors, $d \geq 0$, and $0 \leq e_0 \leq 1/3$. If*

$$\|(1 - 2e_0)X - e_0Y\| \leq d, \quad \|(1 - 2e_0)Y - e_0X\| \leq d,$$

then both inequalities hold at every $e \in [e_0, 1/3]$.

Proof. The difference of the two vectors at e_0 is $(1 - e_0)(X - Y)$, so

$$\|X - Y\| \leq \frac{2d}{1 - e_0} \leq 3d.$$

At $e = 1/3$ the vectors become $(X - Y)/3$ and $(Y - X)/3$, both in the closed radius- d ball. Each vector depends affinely on e , and the ball is convex. Interpolation proves both inequalities, including the endpoint case. This transfers the pair together; it does not assert monotonicity of either squared residual separately. $\square$

Lemma E.10 (Solved witness-enclosure optimization). *For every parameter pair in equation (14),*

$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

Proof. If $c_* \leq 2/3$, the centered disk gives $\Lambda(K_{\text{wit}}) \geq 3(1 - c_*) \geq 1$. Otherwise $0 < \eta < h/3$, and it suffices to enclose $\hat{K}$ by equation (105). The line contacts satisfy equation (107).

theorem F.3 proves the unchanged eight integer-polynomial signs at the rational radius $0 < \bar{\eta} = h(1 - \bar{c}) \leq \eta$. The Gram factors and paired transfer in theorem F.2 turn those signs into $P_A(\eta), P_C(\eta) \geq 0$. Thus both tangent sums in equation (108) are at least h as well. All four contacts in equation (106) have support sum at least h . Therefore

$$\Lambda(K_{\text{wit}}) \geq \Lambda(\hat{K}) \geq 1.$$

The Newton inner points and exact algebraic certificate are retained; only the terminal geometry has been replaced by supporting contacts. $\square$

APPENDIX F. EXACT DISK-TANGENT RESIDUAL CERTIFICATE

The two disk-tangent contact bounds in the zero-gap nine-point theorem are certified by exact symbolic arithmetic. This section gives the mathematical reduction and verification algorithms; the repository provenance manifests identify the accompanying code and sparse integer data. No floating-point arithmetic, interval arithmetic, adaptive subdivision, branch-and-bound, or unrecorded numerical tolerance is used.

F.1. Certificate manifest. The machine-readable provenance manifest identifies the six sparse-data shards and both exact verifiers by full Git blob and records their roles. They decode to one canonical sparse-polynomial transcript with SHA-256 digest

`dc46aaf263655d5159ecd3a81db72ee82477951d06172f4743b248df37209485`

The derivation verifier reconstructs the transcript from the witness formulas and compares every coefficient with the authenticated object. The positivity verifier derives the Bernstein coefficients from that object and checks their signs exactly.

F.2. Rational radial upper envelopes. Retain the notation of Section E. Thus

$$p = 1 - b, \quad q = 1 - a, \quad s = p + q, \quad m = \min\{p, q\}, \quad M = \max\{p, q\},$$

$$\chi = s^4 - s^2 + mM, \quad h = \frac{\sqrt{3}}{2}, \quad c_* = c_{\max}(p, q).$$

The strict witness domain gives

$$(110) \quad mM > (1 - s)(2 - s), \quad m > 1 - s, \quad s > \frac{2}{3}.$$

Put

$$F_m(x) = x^4 - x^2 + mx - m^2, \quad \kappa = F'_m(s) = 4s^3 - 2s + m.$$

Then

$$\kappa > 4s^3 - 3s + 1 = (s + 1)(2s - 1)^2 > 0.$$

Define

$$(111) \quad \bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \leq 0), \quad \bar{c}_T = s - \frac{\chi}{1 - m} \quad (\chi \geq 0).$$

Lemma F.1 (Branchwise radial upper bounds). *On the corresponding radial cells,*

$$c_* \leq \bar{c}_L < 1, \quad c_* \leq \bar{c}_T < 1.$$

At $\chi = 0$ both upper bounds equal $s = c_*$.

Proof. On the L cell, $F_m(s) = \chi \leq 0$ and $F_m(c_*) = 0$. For $x \geq s > 2/3$, $F''_m(x) > 0$, so convexity gives

$$0 \geq F_m(s) + F'_m(s)(c_* - s),$$

and hence $c_* \leq \bar{c}_L$. If $U = 1 - s$, then

$$\chi + U\kappa > U\{m + 1 - 4U + 8U^2 - 3U^3\} > 0$$

by (110); therefore $\bar{c}_L < 1$.

On the T cell put

$$E = \sqrt{4s^2 - 3}, \quad m_- = \frac{s(1 - E)}{2}, \quad m_+ = \frac{s(1 + E)}{2}.$$

Then

$$\chi = (m - m_-)(m_+ - m), \quad s - c_* = \frac{2(m - m_-)}{1 + E}.$$

Consequently

$$\frac{\chi}{s - c_*} = \frac{1 + E}{2}(m_+ - m) \leq 1 - m,$$

which gives $c_* \leq \bar{c}_T$. The strict inequality $\chi > 0$ gives $\bar{c}_T < s < 1$. $\square$

Reflect so that $z = a - b \geq 0$ and put $U = a + b - 1$. Then

$$s = 1 - U, \quad m = \frac{1 - U - z}{2}, \quad D^2 = z^2 + 6U + 3U^2 =: K(U, z),$$

and

$$\Xi(U, z) = 4\chi = 1 - 10U + 21U^2 - 16U^3 + 4U^4 - z^2,$$

$$G(U, z) = 2\kappa = 5 - 21U + 24U^2 - 8U^3 - z.$$

Thus

$$(112) \quad \bar{c}_L = 1 - U - \frac{\Xi(U, z)}{2G(U, z)}, \quad \bar{c}_T = 1 - U - \frac{\Xi(U, z)}{2(1 + U + z)}.$$

F.3. Gram reduction and the geometric implication. On the active cell let $\bar{c}$ denote the corresponding expression in (112), and put

$$\bar{\eta} = h(1 - \bar{c}), \quad e = 1 - \bar{c}, \quad t = 2\bar{c} - 1.$$

Lemma F.1 gives $0 < \bar{\eta} \leq \eta = h(1 - c_*)$. The algebra below is evaluated at $\bar{\eta}$; theorem E.9 will transfer both residual signs to the actual radius η .

Represent a global vector as (x, hy) and put $C' = R^{-1}C$. Define

$$N_A = \|A\|^2, \quad N_C = \|C'\|^2, \quad \nu = \langle A, C' \rangle, \\ \Delta_0 = x_{C'}y_A - y_{C'}x_A > 0.$$

Thus $h\Delta_0 = \text{cross}(C', A) = \text{cross}(C, RA) > 0$. The authenticated code uses the opposite signed determinant, but every occurrence is squared; its integer polynomials and digest are unchanged. The Gram identity is

$$(113) \quad h^2\Delta_0^2 = N_A N_C - \nu^2.$$

Define

$$(114) \quad Q_A = \Delta_0^2 - \|tA - eC'\|^2, \quad Q_C = \Delta_0^2 - \|tC' - eA\|^2.$$

Lemma F.2 (Residual signs imply the actual-radius tangent bounds). *If $Q_A, Q_C \geq 0$, then the two actual-radius residuals $P_A(\eta), P_C(\eta)$ in equation (109) are nonnegative. Consequently both tangent contact sums in equation (108) are at least h .*

Proof. Let $\bar{\tau} = h - 2\bar{\eta} = ht$. Expanding (114) and using (113) gives

$$(115) \quad \bar{P}_A := (N_A - \bar{\eta}^2)h^2\Delta_0^2 - (\bar{\tau}N_A - \bar{\eta}\nu)^2 = h^2N_AQ_A,$$

$$(116) \quad \bar{P}_C := (N_C - \bar{\eta}^2)h^2\Delta_0^2 - (\bar{\tau}N_C - \bar{\eta}\nu)^2 = h^2N_CQ_C.$$

The signs $Q_A, Q_C \geq 0$ are exactly

$$\|(1 - 2e)A - eC'\| \leq \Delta_0, \quad \|(1 - 2e)C' - eA\| \leq \Delta_0.$$

Since $0 < e = \bar{\eta}/h \leq \eta/h < 1/3$, theorem E.9 gives the same inequalities at η/h . The identical Gram factors at that radius prove $P_A(\eta), P_C(\eta) \geq 0$. Since $h\Delta_0 > 0$, the unsquared bounds following equation (109) show that both tangent contact sums are at least h . Transferring the pair avoids any assumption that a single squared residual is monotone in the disk radius. $\square$

F.4. Eight exact integer-polynomial signs. The Newton witnesses A, C are rational functions of U, z, D . Substituting (112) into (114), clearing denominators, and reducing by

$$D^2 - K(U, z) = 0$$

gives, for $B \in \{L, T\}$ and $X \in \{A, C\}$,

$$(117) \quad \text{num}(Q_{B,X}) = 32(K + 3)^2\{A_{B,X}(U, z) + DB_{B,X}(U, z)\},$$

where all eight core polynomials lie in $\mathbb{Z}[U, z]$. Every omitted denominator is strictly positive on the strict domain: its factors are positive constants, $(D+3)^4$, $G(U, z)^2$ or $(1+U+z)^2$, and squares of the two Newton-derivative numerators. Here $G = 2\kappa > 0$, and the fixed-line sign theorem proves both Newton derivatives strictly negative.

The derivation verifier constructs A, C , both rational envelopes, and all four residuals directly in the exact field $\mathbb{Q}(U, z, D)$. It performs the reduction modulo $D^2 - K$, removes the common positive factor, primitive-normalizes the eight integer polynomials, and compares their sparse term lists coefficient by coefficient with the

authenticated transcript. Thus the data are checked against the geometric formulas rather than trusted as an independent precomputation.

F.5. Three global positive-basis certificates. Set

$$\mathcal{G}(U) = 1 - 6U - 3U^2, \quad \mathcal{P}(U) = 1 - 10U + 21U^2 - 16U^3 + 4U^4,$$

$$U_{\Delta} = 1 - \frac{\sqrt{3}}{2}, \quad U_{\max} = \frac{2}{\sqrt{3}} - 1.$$

The complete reflected domain is the union of the three closed cells

cell	U -range	z^2 -range
T	$0 \leq U \leq U_{\Delta}$	$0 \leq z^2 \leq \mathcal{P}(U)$
L_0	$0 \leq U \leq U_{\Delta}$	$\mathcal{P}(U) \leq z^2 \leq \mathcal{G}(U)$
L_1	$U_{\Delta} \leq U \leq U_{\max}$	$0 \leq z^2 \leq \mathcal{G}(U)$

For the T cell use

$$U = \frac{(x-1)(3-x)}{4x}, \quad z = y \frac{9-x^4}{8x^2}, \quad 1 \leq x \leq \sqrt{3}, \quad 0 \leq y \leq 1,$$

and then $x = 1 + (\sqrt{3} - 1)r$. This maps $[0, 1]^2$ onto the complete cell. After multiplication by positive powers of 2 and x , the four transformed core polynomials have global tensor Bernstein expansions with data

	bidegree	coefficient count	zero count
$A_{T,A}$	$(88, 22)$	2047	2
$B_{T,A}$	$(80, 20)$	1701	0
$A_{T,C}$	$(88, 22)$	2047	2
$B_{T,C}$	$(80, 20)$	1701	0.

Every nonzero coefficient is strictly positive.

For an L -cell polynomial write

$$F(U, z) = E_F(U, w) + zO_F(U, w), \quad w = z^2,$$

and

$$R_F(U, w) = E_F(U, w)^2 - wO_F(U, w)^2.$$

The implications $E_F \geq 0$ and $R_F \geq 0$ give $E_F \geq \sqrt{w}|O_F|$, hence $F \geq 0$ for $z \geq 0$. Use the exact charts

$$U = U_{\Delta}r, \quad w = \mathcal{P}(U) + s\{\mathcal{G}(U) - \mathcal{P}(U)\}$$

for L_0 , and

$$U = U_{\Delta} + (U_{\max} - U_{\Delta})r, \quad w = s\mathcal{G}(U)$$

for L_1 . The positivity verifier forms one global Bernstein expansion for each of E_F, R_F on each square. The authenticated transcript records the bidegrees and coefficient counts of all sixteen expansions; every coefficient is strictly positive.

The verifier implements polynomial addition, multiplication, substitution, and power-to-Bernstein conversion over exact integers. Coefficients in $\mathbb{Q}(\sqrt{3})$ are represented as rational pairs and signed by exact rational comparisons. If $P(r, s) = \sum a_{ij}r^i s^j$ has bidegree (m, n) , its exact tensor Bernstein coefficients are

$$b_{k\ell} = \sum_{i \leq k} \sum_{j \leq \ell} a_{ij} \frac{\binom{k}{i}}{\binom{m}{i}} \frac{\binom{\ell}{j}}{\binom{n}{j}}.$$

This is precisely the conversion implemented by the verifier.

Theorem F.3 (Exact tangent-residual certificate). *On the complete strict witness domain with $c_* > 2/3$, all eight polynomials in (117) are nonnegative. Hence $Q_A, Q_C \geq 0$, and the two actual-radius tangent contact bounds in equation (108) hold.*

Proof. The T chart proves the four T -cell core signs by nonnegative Bernstein coefficients. On each L chart the positive Bernstein certificates for E_F and R_F prove the four L -cell signs. Since $D \geq 0$, (117) gives $Q_A, Q_C \geq 0$ for $z \geq 0$; reflection supplies $z \leq 0$. Lemma F.2 then gives the two tangent contact bounds. $\square$

Remark F.4 (Reproduction). The authenticated data and the two verifiers constitute the exact electronic certificate used by the proof. Reproduce it from its package directory by executing

```
verify_mixed_overlap_core_derivation.py
verify_global_core_positivity.py
```

Both commands use exact arithmetic and must reproduce the transcript digest above and the asserted sign checks.